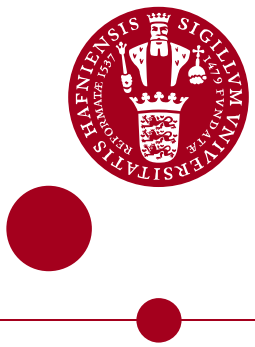
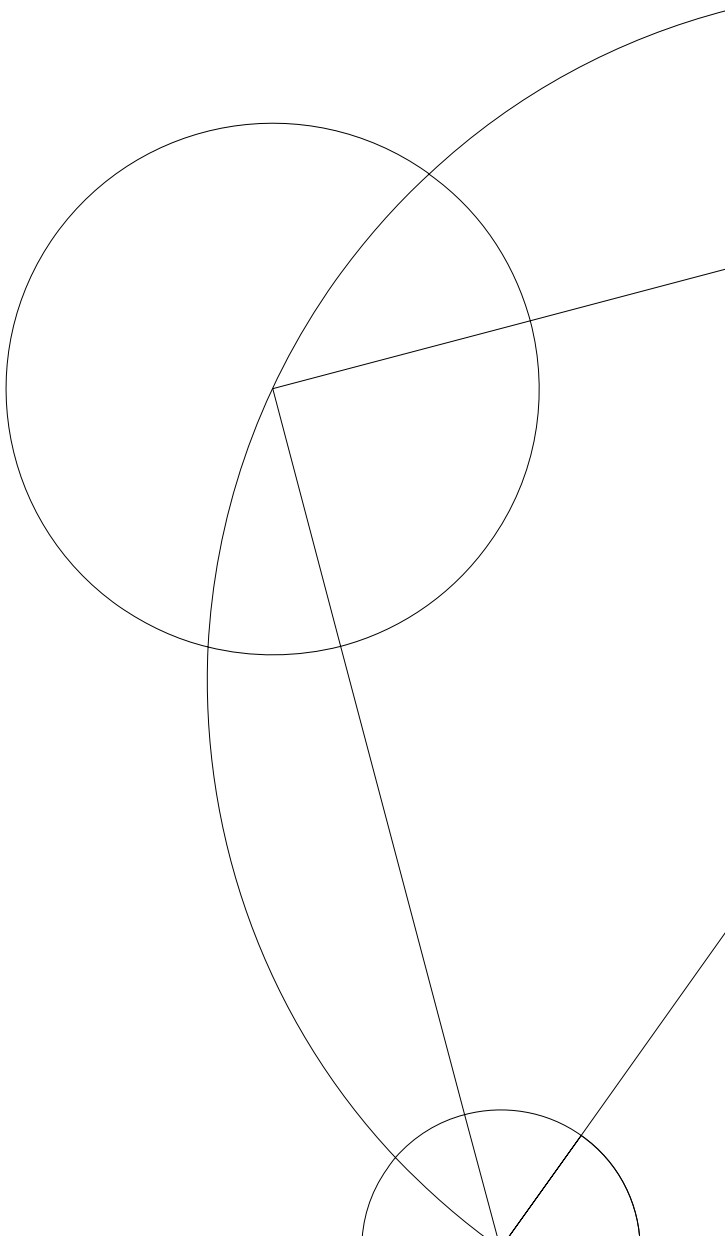
UNIVERSITY OF COPENHAGEN
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Bayesian Optimal Prediction

A Bayesian approach to Optimal Prediction of the ultimate maximum with asymmetric loss functions



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Abstract

In this thesis, we study prediction of the ultimate maximum of a Brownian motion on a finite time horizon with sequential observation of the process. The problem is twofold in the sense that we study prediction of the time at which the ultimate maximum is achieved and prediction of the value of the maximum. We employ a Bayesian decision-theoretic view to find optimal estimators for both the time and space formulation of the problem.

In particular, we focus on asymmetric loss functions such as weighted linear, entropy, and LINEX loss. We find Bayes estimators by using methods from the field of optimal stopping. In many cases, the Bayes estimator can be described by an explicit stopping time, while in some cases we resort to investigating free-boundary equations in the form of nonlinear Volterra equations of the second kind.

Finally, we compare the estimators from a mean squared error perspective and by investigating the admissibility of estimators. We find that even though the space and time problems are closely related one has to be careful of using rules from one formulation in another since inadmissibility is possible.

To our knowledge, the explicit Bayes estimator of the time problem under entropy loss and the proof of an optimal estimator of the maximal value based on the running maximum is completely new. Furthermore, the use of LINEX loss and comparison of estimators are important new contributions to the literature.

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Chapter 1

Introduction

In the financial industry, one typically encounters phrases like "Cut your losses and let the winners run". Although such phrases may seem intuitive it poses the question of how to determine winning and losing assets. And even more important, when to cut or let run. In the field of financial mathematics, we impose a certain rigor, by assuming no-arbitrage conditions which ensures that assets be priced such that rational investors can agree on the price. The motivation for this thesis is to study a setting where an investor wants to sell an asset within some finite timeframe. We impose that the investor should act rationally in that he wants to sell as close as possible to the peak of the asset's price within the given timeframe.

In real markets, rational investors have to take into account trade costs, order book dynamics, impact on the price, and so on. We leave out such frictions and look only at the agent's preferences and the impact this has on an optimal selling strategy when accounting for real-time observations. In particular, we look at the statistical properties of these strategies.

The immediate problem in this setting is of course that the value at which and time when the asset attains its ultimate maximum in a finite timeframe is only known at the terminal time. Hence, this problem is called an *optimal prediction* problem — we have to predict, in an optimal way, when the peak will happen. It should also be clear that there are two intuitive ways of looking at the problem, one is to look at minimizing the distance between the selling price and the ultimate maximum price, and the other is to minimize the distance between the time of selling and the time of the ultimate maximum. We refer to these problems as the (probabilistic) space and the time problem, respectively. We will see in the thesis that the two problems are related.

This is by no means the first exposition of optimal prediction problems. The problem was first studied using stochastic calculus in Graversen et al. (2001) where they consider the space problem for a Brownian motion with mean square distance. This paper gave birth to three

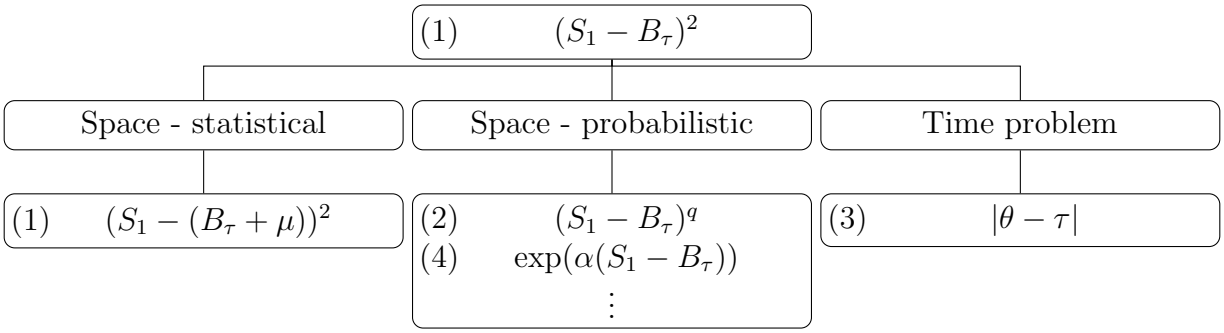


Diagram 1: Overview of literature. B is a Brownian motion, $S_1 = \max_{0 \le t \le 1} B_t$, $\tau \in [0, 1]$ is a stopping time, θ is $B_\theta = S_1$, $\mu, \alpha \in \mathbb{R}$, and $q > 0$. (1) Graversen et al. (2001), (2) Pedersen (2003), (3) Urusov (2005), and (4) du Toit and Peskir (2009).

branches of research, see Diagram 1. By far the most active branch is what we will call the *probabilistic space problems* which are directly motivated by the selling of an asset as close as possible to the ultimate maximum. Here authors like Pedersen (2003) started to consider different metrics like a general q-mean distance, while other authors like du Toit and Peskir (2009) considered different asset models. The *time* branch was introduced by Urusov (2005) where the author shows a direct correspondence between the squared space problem and the linear time problem. The goal of the *statistical space branch* is to estimate the ultimate maximum using sequential observations to produce an estimator at an optimal time. This has applications in many fields such as physics and biology. Specifically in the field of risk management, this produces a real-time estimate of the maximal exposure of a portfolio over a finite horizon.

The goal of this thesis is to expand the time and statistical space branches by employing Bayesian techniques to find optimal estimators of time and value of the ultimate maximum. In particular, we explore different asymmetric loss functions to find Bayes estimators and compare these. Furthermore, we expand the statistical space branch by considering an alternative class of estimators from the running maximum process.

To give an overview of the thesis we proceed in Chapter 2 to give an introduction to *Bayesian decision theory* and formulate a Bayesian interpretation of these problems. In Chapter 3 we give a basic introduction to the field of *optimal stopping*, which will be useful in solving the optimal prediction problems. The main idea is to formulate an auxiliary free-boundary problem that can be solved by standard techniques from differential equations and show that this yields a good candidate solution to the optimal stopping problem. Then in Chapter 4 we first present some useful probabilistic results and go on to solve selected optimal prediction problems, starting with the time formulation and then proceeding to statistical space problems. Finally, in Chapter 5 we make comparisons of the different estimators developed in Chapter 4. Appendix A discusses the numerical procedures used in the thesis and provides a link to the implemented code. Appendix B contains some mathematical results.

Chapter 2

Bayesian Decision Theory

The present chapter will serve as an introduction to Bayesian decision theory. The first section will introduce the general framework of Bayesian inference for decision theory inspired by Berger (1980) and Robert (2007). While the difference between Bayesian and classical statistics is mainly a philosophical matter leading to different practices, we will refrain from this philosophical discussion and only point out differences. Furthermore, we introduce a method for comparing estimators based on admissibility and discuss different loss functions of interest. In the second section, we will see that the Bayesian view is the natural view in our setting.

2.1 Bayesian Inference

In statistical experiments, it is of great importance to ask what the goal of the experiment is. The goals that we consider in the thesis are e.g.:

Goal 1: Should we sell the asset now?

Goal 2: Should we update our view on the maximal exposure of our portfolio?

Note that we do not ask about the true state of nature, but rather when and how to intervene. This is the foundation of *decision theory*. Decision theory is inherently neither Bayesian nor frequentist, however, the principles that are used to answer these questions are different. Furthermore, the goals have a timing component as we want to act in real-time and not wait until we have full information.

Consider a parametric statistical experiment $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\in\mathbb{N}}; (P_\theta)_{\theta\in\Theta})$ where Θ is a (subset) of a finite-dimensional vector space. The filtration consists of $\mathcal{F}_n = \sigma(X_1, X_2, \ldots, X_n)$ where X_i has distribution with some density $f_i(x \mid \theta)$. The functions f_i are assumed to be known and θ

is unknown. In a Bayesian setting, we also assume that θ is a random variable having a *prior distribution* with density π.

The existence of π is the first difference between the Bayesian and classical views. The prior distribution is chosen a priori by the statistician. In classical statistics θ is not considered a random variable, but merely unknown with value in Θ.

The problem at hand is to draw inference on the true value of θ as observations reveal. Since θ is considered a random variable we can update the density of θ by conditioning on the observed variables to obtain the conditional density $\pi(\theta \mid \mathcal{F}_n)$. Hence, we update our view on the distribution of θ posterior to the observations, and so we call $\pi(\theta \mid \mathcal{F}_n)$ the *posterior distribution*.

Since our goal is to take action upon the inference gathered about θ we need to consider the space of possible actions, $\mathcal{A}$, that a decision can lead to. Letting $\mathcal{H}$ be the sample space we can now define a decision rule:

Definition 2.1.1 (Decision rule)**.** A decision rule δ is a measurable function $\delta : \mathcal{H} \to \mathcal{A}$. We denote the set of all decision rules by $\mathcal{D}$.

To evaluate which decision rule is optimal we need a criterion for ordering decision rules. In some settings, this evaluation criterion is obvious, e.g. in simple zero-sum games. In other settings, e.g. in risk management, a criterion may be given by law. However, in most cases, the statistician must choose a *utility* function to optimize, or equivalently a *loss* function to minimize. In any case, we define the loss as follows:

Definition 2.1.2 (Loss function)**.** A loss function L is a measurable function $L : \Theta \times \mathcal{A} \to \mathbb{R}_+$.

An action $\delta(X)$ is a random variable prior to observation making $L(\theta, \delta(X))$ stochastic given θ. We define the risk function as the expectation of the loss given the parameter.

Definition 2.1.3 (Risk function)**.** The risk function associated with a loss function L and a decision rule δ is for given θ defined by

$$R(\theta, \delta) = \mathbb{E}^{X|\theta}\left[L(\theta, \delta(X))\right]. \tag{2.1.1}$$

It should be obvious that the risks have a partial ordering which enables us to rank decision rules for a given loss function. If a decision rule δ is uniformly dominated in θ by another rule $\tilde{\delta}$ then we say $\tilde{\delta}$ is inadmissible, in the sense that we would never choose $\tilde{\delta}$ since we can get a lower risk by choosing δ. To formalize we make the following definition:

Definition 2.1.4 (Admissibility)**.** A decision rule $\tilde{\delta}$ is said to be inadmissible if there exists a decision rule δ for which

$$R(\theta, \delta) \leq R(\theta, \tilde{\delta}) \tag{2.1.2}$$

for all $\theta \in \Theta$ with strict inequality for at least one θ. If no such δ exists we say $\tilde{\delta}$ is admissible.

Uniqueness of an admissible strategy is by no means ensured. However, inadmissibility can be used in conjunction with other principles to rule out particular decision rules as we will discuss below. One such principle is to prefer the decision which minimizes the risk over the worst possible scenarios of θ. This is called the *minimax principle*.

Definition 2.1.5 (Minimax principle)**.** A decision rule δ_1 is preferred to a rule δ_2 in the minimax setting if

$$\sup_{\theta\in\Theta} R(\theta, \delta_1) < \sup_{\theta\in\Theta} R(\theta, \delta_2). \tag{2.1.3}$$

A decision rule δ_M is called a *minimax decision rule* if

$$\sup_{\theta\in\Theta} R(\theta, \delta_M) = \inf_{\delta\in\mathcal{D}} \sup_{\theta\in\Theta} R(\theta, \delta) \tag{2.1.4}$$

provided the last equality is well-defined.

The minimax principle is a very conservative preference principle since it seeks to minimize the maximal risk a rule can yield. However, since we consider θ a random variable this principle does not take into account that some regions of Θ may be very improbable. So in a Bayesian setting, we can take into account the distribution of θ which leads to the *Bayes principle*.

Definition 2.1.6 (Bayes principle)**.** A decision rule δ_1 is preferred to a rule δ_2 in the Bayesian setting if

$$\mathbb{E}^\pi \left[R(\theta, \delta_1)\right] < \mathbb{E}^\pi \left[R(\theta, \delta_2)\right]. \tag{2.1.5}$$

A decision rule δ_π is called a *Bayes rule* if

$$\mathbb{E}^\pi \left[R(\theta, \delta_\pi)\right] = \inf_{\delta\in\mathcal{D}} \mathbb{E}^\pi \left[R(\theta, \delta)\right] \tag{2.1.6}$$

provided the last equality is well-defined.

Introducing these principles need not make the optimal rules unique, however, it reduces the set of optimal rules. Furthermore, they need not exist. Both of these questions of course depend on the choice of loss function, and in the Bayesian case also the prior distribution. We will discuss the existence in optimal stopping problems in Section 3.1. For a general treatment in this statistical setup, we refer to Berger (1980) and Robert (2007).

Consider a scenario where we have found two optimal rules δ_1 and δ_2 according to e.g. the Bayes principle under the loss functions L_1 and L_2 respectively. One may reasonably ask how the two optimal rules compare with each other. For this it is custom in Bayesian decision theory (see e.g. Basu and Ebrahimi, 1991 and Zellner, 1986) to revert to admissibility and compare $R_1(\theta, \delta_1)$ with $R_1(\theta, \delta_2)$ and $R_2(\theta, \delta_1)$ with $R_2(\theta, \delta_2)$ respectively where R_i is the risk under the loss function L_i for $i = 1, 2$. If $R_1(\theta, \delta_2)$ dominates $R_1(\theta, \delta_1)$ for all $\theta \in \Theta$ while

$R_2(\theta, \delta_1) < R_2(\theta, \delta_2)$ for some $\theta \in \Theta$ and $R_2(\theta, \delta_1) > R_2(\theta, \delta_2)$ for some other $\theta \in \Theta$ then one might reasonably say that the rule δ_1 is preferable to δ_2, since δ_2 is inadmissible under L_1 while δ_1 is not necessarily inadmissible under L_2.

Let τ denote an action, i.e. $\tau = \delta(X)$. Examples of the most standard loss functions which are studied in the literature are as follows:

$$L(\theta, \tau) = \begin{cases} I(\tau = \theta) & \text{(0-1 loss)} \\ |\tau - \theta| & \text{(Linear loss)} \\ (\tau - \theta)^2 & \text{(Square loss).} \end{cases} \tag{2.1.7}$$

The 0-1 loss is of course mostly used for classification. One should note that these loss functions are symmetric, implying that an over and underestimation of the same magnitude should induce the same loss. However, from the introduction in Chapter 1 it should be clear that this behavior is not necessarily wanted. In the risk management example, it can have severe effects if the exposure turns out to be larger than the estimate, while if it is lower the situation is not as severe. Likewise, when selling an asset we may want to penalize acting too early differently than acting too late when we have unknowingly seen the peak. Hence, we will study asymmetric loss functions. Examples of the loss functions studied in the thesis are as follows:

$$L(\theta, \tau) = \begin{cases} \alpha(\tau - \theta)^+ + \beta(\tau - \theta)^- & \text{(Weighted linear loss)} \\ \alpha\left(\log\frac{1-\tau}{1-\theta}\right)^+ + \beta\left(\log\frac{1-\tau}{1-\theta}\right)^- & \text{(Weighted entropy loss)} \\ \alpha I(\tau < \theta) + \beta(\tau - \theta)^+ & \text{(Mixed probability and linear loss)} \\ \beta\left(e^{\alpha(\tau-\theta)} - \alpha(\tau - \theta) - 1\right) & \text{(LINEX loss)} \end{cases} \tag{2.1.8}$$

for $\alpha, \beta > 0$. The entropy loss is a slight variation of the entropy distance, also known as the Kullback-Leibler distance, in e.g. Robert (2007). The mixed probability and linear loss is a loss function that was proposed in Shiryaev (1961, 1963) for the so-called quickest detection problems, where the goal is to detect when a change in drift happens for a stochastic process. This loss function can also be seen as a Lagrangian of a constrained problem as we will see in Chapter 4. The LINEX (linear and exponential) loss function was proposed as a well-behaved asymmetric loss function in Varian (1975) for real estate assessment where the use of asymmetric loss fits well. For this particular loss, we let $\alpha \in \mathbb{R} \setminus \{0\}$ since a sign change in α corresponds to flipping the asymmetry. It should be noted that making a Taylor expansion in $\Delta = \tau - \theta$ around 0 yields

$$\beta^{-1}L(\theta, \tau) = \frac{1}{2}\alpha^2\Delta^2 + \mathcal{O}(\alpha^3\Delta^3) \tag{2.1.9}$$

i.e. for small $|\alpha|$ the LINEX loss is approximately equal to the symmetric squared loss function. Some loss functions are depicted in Figure 2.1.1. We note in passing that the weighted linear loss is also known as the LinLin loss and similarly weighting the squared loss is known as QuadQuad,

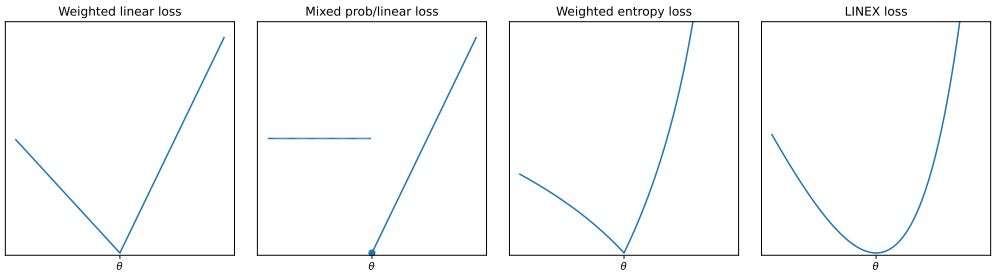


Figure 2.1.1: Examples of loss functions.

however, we do not adopt this terminology. The parametrization with two constants α and β is somewhat unnecessary since considering e.g. the weighted linear loss case we can write

$$V_* = \alpha \mathbb{E}\left[(\tau_* - \theta)^+ + \frac{\beta}{\alpha}(\tau_* - \theta)^-\right]. \tag{2.1.10}$$

So in the following, we will only consider a single weighting parameter.

2.2 Bayesian Formulation of Optimal Prediction Problems

We now turn our attention to the statistical goals presented in Section 2.1. As discussed in the introduction we are motivated by the paper from Graversen et al. (2001) to formulate three branches of problems revolving around prediction of the ultimate maximum on a finite time horizon upon sequential observation of the process. For simplicity, we take the horizon to be 1.

Consider a probability space $(\Omega, \mathcal{F}, P)$. Let $B = (B_t)_{t\in[0,1]}$ be a standard Brownian motion let $(\mathcal{F}_t)_{t\in[0,1]}$ be the natural filtration generated by B, completed and made right continuous such that it obeys the usual conditions, and consider the corresponding filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,1]}, P)$. We denote by $S = (S_t)_{t\in[0,1]}$ the maximal process of B and let θ denote the time of the ultimate maximum, that is $S_t = \max_{s\in[0,t]} B_s$ and $\theta = \inf\{s \in [0,1] \mid B_s = S_1\}$. Note that θ is actually defined as the first time of the ultimate maximum, however the time that B attains its maximum is P-a.s. unique (see Karatzas and Shreve, 1991, Chapter 2, Remark 8.16). We denote the set of stopping times by

$$\mathcal{T} = \{\tau \in [0,1] \mid (\tau < t) \in \mathcal{F}_t \text{ for all } t \in [0,1]\}. \tag{2.2.1}$$

Note in particular that θ is not a stopping time since θ is only $\mathcal{F}_1$ measurable.

We first define the time problem corresponding to Goal 1 which can be considered a binary prediction experiment. We want to answer whether θ has been observed up until time t, i.e. we

want to estimate the process $X = (X_t)_{t\in[0,1]}$ where $X_t = 1_{[\theta,\infty)}(t)$. To estimate X it is natural to match the functional form being a one-jump process i.e. $\hat{X}_t = 1_{[\tau,\infty)}(t)$. Furthermore, the estimator at time t should be produced by observation of X up to time t so $\hat{X}$ should be adapted to the filtration. Hence, τ should be a stopping time. To have $\hat{X}$ be close to X we thus need τ to be close to θ. To this end Urusov (2005) considered $|\theta - \tau|$ which corresponds to a linear loss. However, the Bayesian decision-theoretic view motivates looking at other loss functions as well.

Let the a priori distribution of θ have density π, and denote the posterior density by $\pi_{s|t} = P(\theta \leq s \mid \mathcal{F}_t)$. When $s = t$ we simply denote it by π_t and call it the *posterior process*. We can now formulate the time problem in a Bayesian decision-theoretic sense motivated by Urusov (2005):

Definition 2.2.1 (Bayesian time problems)**.** Let $L : \Theta^2 \to \mathbb{R}_+$ be a loss function and let $\mathcal{T}'$ be the set of acceptable decision rules where $\mathcal{T}' \subseteq \mathcal{T}$. The $(\mathcal{T}', L)$-time problem is to find the Bayes rule τ_*, i.e.

$$\mathbb{E}\left[L(\theta, \tau_*)\right] = \inf_{\tau\in\mathcal{T}'} \mathbb{E}\left[L(\theta, \tau)\right] \tag{2.2.2}$$

if it exists. Furthermore, it is of interest to find the Bayes risk $V_* = \mathbb{E}\left[L(\theta, \tau_*)\right]$. The expectation is taken under P where θ has density π.

Note that since the set of possible decisions coincide with the set of stopping times with respect to the filtration then for $\tau \in \mathcal{T}$

$$\mathbb{E}\left[L(\theta, \tau)\right] = \mathbb{E}\left[\mathbb{E}\left[L(\theta, \tau) \mid \mathcal{F}_\tau\right]\right]. \tag{2.2.3}$$

Under the conditional expectation, we note that θ has the posterior distribution and τ is measurable.

For the prior distribution of θ note that since B is a standard Brownian motion on $[0, 1]$ we can recall the following result from Karatzas and Shreve (1991)

$$\pi(\theta) = \frac{1}{\pi\sqrt{\theta(1-\theta)}}, \qquad P(\theta \leq t) = \frac{2}{\pi}\arcsin\sqrt{t} \tag{2.2.4}$$

i.e. that θ has Arcsine law, or equivalently is Beta distributed with shape parameters $\alpha = \beta = 1/2$.

For the spatial problems, we note that Graversen et al. (2001) gives rise to two different problems namely the probabilistic and statistical space problems. The goal in the statistical space problem is to estimate S_1 corresponding to Goal 2. In a static statistical experiment, one might estimate S_1 by the properties of the underlying process B. However, since we want to develop a dynamic estimator, taking into account sequential observation of the underlying, we consider the class of estimators on the form $\hat{S}_1 = B_\tau + \mu$ where $\tau \in \mathcal{T}$ and $\mu \in \mathbb{R}$ motivated again by the article. Thus, we define the statistical space problem as follows:

Definition 2.2.2 (Statistical space problems)**.** Let $L : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}_+$ be a loss function, Θ' be some subset $\Theta' \subseteq \mathbb{R}$, and let $\mathcal{T}' \subseteq \mathcal{T}$ be the set of acceptable decisions. The $(\Theta', \mathcal{T}', L)$-statistical space problem is defined as

$$V_* = \inf_{\tau \in \mathcal{T}', \mu \in \Theta'} \mathbb{E}\left[L(S_1, B_\tau + \mu)\right]. \tag{2.2.5}$$

If a solution $(\tau_*, \mu_*) \in (\mathcal{T}', \Theta')$ exists such that

$$V_* = \mathbb{E}\left[L(S_1, B_{\tau_*} + \mu_*)\right] \tag{2.2.6}$$

then τ_* and μ_* are called the optimal prediction rule and the optimal offset, respectively.

The main result of Graversen et al. (2001) was not to produce an estimator of S_1 but rather to combat Goal 1 by considering the statistical space problem with $\mu \equiv 0$ under a squared loss, i.e. B_τ is the action corresponding to the price at which we sell the asset. Thus, the authors considered a different spatial problem which we will call the probabilistic space problem.

Definition 2.2.3 (Probabilistic space problems)**.** Let $L : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}_+$ be a loss function and let $\mathcal{T}' \subseteq \mathcal{T}$ be the set of acceptable decisions. The $(\mathcal{T}', L)$-probabilistic space problem is defined as the $(\{0\}, \mathcal{T}', L)$-statistical space problem.

It should be clear that the probabilistic space problems and the time problems are closely related since they both consider Goal 1.

The class of estimators considered in the statistical space problem may seem somewhat unnatural since we are just adding a constant based on the loss function to adjust the mean of our estimator. An alternative class of estimators that will be considered is $\hat{S}_1 = S_\tau$ for some $\tau \in \mathcal{T}$. However, taking $\tau = 1$ will yield an exact estimator, so we will add a time penalty to the loss. Thus, we define an alternative statistical space problem as follows:

Definition 2.2.4 (Alternative statistical space problem)**.** Let $f : [0, 1] \to \mathbb{R}_+$ be a non-decreasing measurable function and let $\mathcal{T}' \subseteq \mathcal{T}$ be the set of acceptable decisions. The $(\mathcal{T}', f)$-alternative statistical space problem is defined as

$$V_* = \inf_{\tau \in \mathcal{T}'} \mathbb{E}\left[S_1 - S_\tau + f(\tau)\right]. \tag{2.2.7}$$

If a solution $\tau_* \in \mathcal{T}'$ exists such that

$$V_* = \mathbb{E}\left[S_1 - S_{\tau_*} + f(\tau_*)\right] \tag{2.2.8}$$

then τ_* is called the optimal prediction rule.

In the proceeding sections, we drop the decision-theoretic lingo of actions and decisions for the general terms of estimators or predictions. Furthermore, the Bayes risk and Bayes rule will be used interchangeably with optimal value and optimal stopping rule or optimal prediction as is custom in the field of optimal stopping.

Chapter 3

Optimal Stopping and Free-Boundary Problems

In the present chapter, we give an introduction to optimal stopping theory. Specifically, we follow the book Peskir and Shiryaev (2006) where the method of reducing to auxiliary classical free-boundary problems is explained. Furthermore, the authors give a great number of solution methods and concrete examples. Although interesting we omit discussions about the discrete setting since this thesis only considers a continuous time setting. We leave out the proofs, all of which can be found in Peskir and Shiryaev (2006).

Optimal stopping problems are classically stated as optimization problems, stemming from utility functions. However, as discussed above our focus is on minimizing loss hence we have adapted the results for our purpose without loss of generality.

3.1 The Markovian Approach

Consider a strong Markov process $X = (X_t)_{t\geq 0}$ taking values in $(E, \mathcal{B})$ and defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, P_x)$. For simplicity, we assume the state space to be $E = \mathbb{R}^d$ for some $d \geq 1$ and $\mathcal{B} = \mathcal{B}(\mathbb{R}^d)$. We assume the filtration satisfies the usual conditions, hence specifically it is right-continuous which makes first entry time τ to open and closed subsets of E have $\mathcal{F}_t$ measurable events $(\tau \leq t)$ for all $t \geq 0$. The measure P_x is such that $X_0 = x \in E$ and we denote expectation under this measure by $\mathbb{E}_x$. We assume the paths of X are continuous over stopping times. We want to view optimal stopping problems as a function of the starting point x hence we also assume $x \mapsto P_x(F)$ is measurable for $F \in \mathcal{F}$. The strong Markov property of X means

$$P_x(X_{\tau+s} \in B \mid \mathcal{F}_\tau) = P_{X_\tau}(X_s \in B) \tag{3.1.1}$$

for any stopping time τ and any $s \geq 0$.

Let $G : E \to \mathbb{R}$ be the *gain function* and assume

$$\mathbb{E}_x \left(\sup_{0 \leq t \leq T} |G(X_t)| \right) < \infty \tag{3.1.2}$$

for all $x \in E$, where $0 \leq T \leq \infty$. In the case $T = \infty$ we simply set $G(X_T) = 0$. The optimal stopping problem is then defined as

$$V_*(x) = \inf_{\tau \in \mathcal{T}^T} \mathbb{E}_x \left[G(X_\tau) \right] \tag{3.1.3}$$

where $\mathcal{T}^T$ is the set of stopping times with respect to $(\mathcal{F}_t)_{t \geq 0}$ where $\tau \in \mathcal{T}^T$ has $\tau \leq T$. The goal is to find the *value function* V_* and the *optimal stopping time* τ_* at which the infimum is obtained if it exists.

One could expect that the finite horizon case is less troublesome than handling infinite horizons since we have a terminal boundary. However, if X is a one-dimensional time-homogeneous process then the optimal strategy becomes dependent on the remaining time. Thus, to solve finite horizon problems one has to extend the dimensionality of the problem by considering the two-dimensional Markov process $Z = (Z_t)_{t \geq 0}$ where $Z_t = (t, X_t)$ and formulate the problem

$$V_*(t, x) = \inf_{\tau \in \mathcal{T}^{T-t}} \mathbb{E}_{t,x}[G(Z_{\tau + t})] \tag{3.1.4}$$

for $x \in E$ and $t \in [0, T]$. However, in some cases, it is, as we will see, possible to circumvent this dimensionality increasing e.g. by doing a suitable transformation of time as done in Section 3.3.

Heuristically since a Markov process only depends on the current state, say $\overline{x} \in E$, we should be able to calculate $G(\overline{x})$ and evaluate whether it is optimal to stop or continue. Furthermore, by shifting the time of the Markov process we can just evaluate under the measures $(P_x)_{x \in E}$. Thus, we are led to split the state space E up in a *continuation set* $C \subseteq E$ and a *stopping set* $D = E \setminus C$ where if $X_t(\omega) \in C$ it is optimal to continue observing, while on $X_t(\omega) \in D$ it is optimal to stop and take the gain.

We now introduce formal definitions for these sets and the concept of a *subharmonic* function. Let

$$C = \{x \in E \mid V_*(x) < G(x)\}, \qquad D = \{x \in E \mid V_*(x) = G(x)\} \tag{3.1.5}$$

be the continuation set and stopping set respectively. The candidate for an optimal stopping time will then be the first entry time

$$\tau_D = \inf\{t \geq 0 \mid X_t \in D\}. \tag{3.1.6}$$

As of now, we do not know if τ_D is in $\mathcal{T}^T$ or not.

Definition 3.1.1. A measurable function $F : E \to \mathbb{R}$ is *subharmonic* with respect to X if

$$\mathbb{E}_x\left[F(X_\tau)\right] \geq F(x) \tag{3.1.7}$$

for all stopping times τ and $x \in E$.

Furthermore, recall that a function H is called lower(upper) semicontinuous if

$$\underset{(\limsup)}{\liminf_{x \to x_0}} H(x) \underset{(\leq)}{\geq} H(x_0) \tag{3.1.8}$$

for all $x_0 \in E$. As of now, we do not know if the problem (3.1.3) is solvable. The following nontrivial theorems and corollary show necessary and sufficient conditions:

Theorem 3.1.2 (Necessary condition)**.** *Assume there exists an optimal stopping time τ_* in (3.1.3), i.e.*

$$V_*(x) = \mathbb{E}_x\left[G(X_{\tau_*})\right] \tag{3.1.9}$$

for all $x \in E$. Then:

The value function V_* is the largest subharmonic function dominated by G on E.

If we furthermore assume V_* is upper semicontinuous and G is lower semicontinuous then:

The first entry time τ_D is a stopping time, $\tau_D \leq \tau_*$ P-a.s. for all $x \in E$ and is optimal.

The stopped process $(V_*(X_{t\wedge\tau_D}))$ is a right-continuous martingale under P_x for all $x \in E$.

Theorem 3.1.3 (Sufficient condition)**.** *Consider the problem (3.1.3) under the mentioned conditions. Assume existence of a largest subharmonic function V dominated by G on E, and assume V is upper semicontinuous and G is lower semicontinuous. Let $D = \{x \in E \mid V(x) = G(x)\}$ and τ_D be the first entry time on that set. Then:*

If $\tau_D < \infty$ P_x-a.s. for all $x \in E$ then $V = V_*$ and τ_D is optimal.

If $P_x(\tau_D < \infty) < 1$ for some $x \in E$ there is no optimal stopping time with probability 1.

Corollary 3.1.4. *Consider the problem (3.1.3) under the mentioned conditions. Assume V_* is upper semicontinuous and G is lower semicontinuous. If $\tau_D < \infty$ P_x-a.s. for all $x \in E$ then τ_D is optimal. If however $P_x(\tau_D < \infty) < 1$ for some $x \in E$ there is no optimal stopping time with probability 1.*

Verification of the assumptions of Corollary 3.1.4 is straightforward for nice gain functions and underlying processes where the initial starting point is well-behaved, see the beginning of the proof in Appendix B.3 for an example.

3.2 Free-Boundary Problems

An alternative method to the Markovian approach is the so-called Martingale method which considers the gain as a process, $G = (G_t)_{t\geq 0}$ with $G_t = G(X_t)$, and seeks to determine the largest submartingale dominated by the gain process.

These are similar however the Martingale method is a probabilistic problem while the Markovian approach is analytic.

For motivational purposes let us briefly assume $F \in C^2$ and X is an Itô process stated in $X_0 = x_0 \in \mathbb{R}$ and given by $\mathrm{d}X_t = \alpha(X_t)\,\mathrm{d}t + \eta(X_t)\,\mathrm{d}B_t$ for a standard Brownian motion $B = (B_t)_{t\geq 0}$ and suitable scalar functions α and η. Then by Itô's lemma

$$\mathrm{d}F(X_t) = \mathbb{L}_X F(X_t)\,\mathrm{d}t + \eta(X_t)\frac{\mathrm{d}F(X_t)}{\mathrm{d}x}\,\mathrm{d}B_t \tag{3.2.1}$$

where $\mathbb{L}$ denotes the infinitesimal generator, in this case $\mathbb{L}_X = \alpha(x)\frac{\mathrm{d}}{\mathrm{d}x} + \frac{1}{2}\eta(x)^2\frac{\mathrm{d}^2}{\mathrm{d}x^2}$. The second term is a (local) martingale so if $\mathbb{L}_X F(x) \geq 0$ then $F(X_t)$ is a (local) submartingale.

The general idea to solve optimal stopping problems by Theorems 3.1.2 and 3.1.3 and the reasoning above is thus to find a function $V : E \to \mathbb{R}$ which solves the *free-boundary* problem

$$\mathbb{L}_X V = 0, \qquad (\text{on } C) \tag{3.2.2}$$
$$V < G, \qquad (\text{on } C) \tag{3.2.3}$$
$$V = G, \qquad (\text{on } D). \tag{3.2.4}$$

By heuristic arguments, one can typically make a good guess for the shape of C and use properties of the process X to infer what should happen at the boundary ∂C. For example, one might guess the solution is continuous and even smooth at the boundary namely

$$V = G\ (\textit{Continuous-fit}), \qquad \frac{\partial V}{\partial x} = \frac{\partial G}{\partial x}\ (\textit{Smooth-fit}), \qquad \textit{on } \partial C. \tag{3.2.5}$$

Dependent on the functional form of G different free-boundary problems might be valuable to study. The usual functional form in optimal stopping problems consists of two measurable functions $M, L : E \to \mathbb{R}$ satisfying the integrability condition (3.1.2) and the problem is

$$V_* = \inf_\tau \mathbb{E}\left[M(X_\tau) + \int_0^\tau L(X_t)\,\mathrm{d}t + \sup_{0\leq t\leq \tau} X_t\right]. \tag{3.2.6}$$

This class of problems is called the *MLS* formulation (Mayer-Lagrange-Supremum). Of course, integrability conditions should also be imposed on the supremum functional, however, this term does not arise in any of the problems studied in the thesis. We now present the auxiliary problems that are studied in this thesis.

When both the Mayer and Lagrangian functionals are present then the auxiliary problem to study is the *Bolza problem*:

$$\mathbb{L}_X V(x) = -L(x), \qquad \text{for } x \in C \tag{3.2.7}$$
$$V(x) = M(x), \qquad \text{for } x \in \partial C \tag{3.2.8}$$
$$V_x(x) = M_x(x), \qquad \text{for } x \in \partial C. \tag{3.2.9}$$

When there is no integral term i.e. no Lagrangian functional then the right-hand side of the differential equation is set to zero. This problem is called the *Dirichlet problem*. In the case where M vanishes the right-hand side of the smooth- and continuous-fit conditions are set to zero. This problem is called the *Dirichlet/Poisson problem*.

For the boundary, if the process is one-dimensional then a reasonable guess for ∂C is a one-dimensional ball since the stopping boundary only depends on x. Thus, in the free-boundary problem, we have to solve for the boundary points $\partial C = \{x_*, x^*\}$ where $x_* \leq x^*$ are points on the extended real line. For higher dimensions, the boundary will generally be some curve that has to be studied case by case.

3.3 Transformation Methods for Reduction

When solving optimal stopping problems one usually has to perform one or more transformations in order to formulate a solvable problem. These transformations include but are not limited to changes of time, space, and measure. The general idea is to find a transformation method that transforms a complicated process X into a simple process $\tilde{X}$. In the following, we present two transformations that will be used throughout the thesis.

Transformation by killing

Consider a one-dimensional Markov process X with values in $\mathbb{R}$. In many contexts, the MLS formulation will be a killed (discounted) version, namely

$$V_* = \inf_\tau \mathbb{E}\left[\tilde{M}(\tau, X_\tau) + \int_0^\tau \tilde{L}(t, X_t)\,\mathrm{d}t\right] \tag{3.3.1}$$
$$= \inf_\tau \mathbb{E}\left[e^{-\int_0^\tau \lambda(X_s)\,\mathrm{d}s} M(X_\tau) + \int_0^\tau e^{-\int_0^t \lambda(X_s)\,\mathrm{d}s} L(X_t)\,\mathrm{d}t\right] \tag{3.3.2}$$

where λ is some measurable function called the killing rate. Since the gain is a function of both X and time the problem is two-dimensional hence we would have to consider the two-dimensional Markov process (t, X). However, if λ is constant then we can circumvent this dimensionality increase. Consider a process $\tilde{X} = (\tilde{X}_t)_{t\geq 0}$ with transition function

$$P_x(\tilde{X}_t \in A) = \mathbb{E}_x\left[e^{-\lambda t} I_A(X_t)\right] \tag{3.3.3}$$

for $x \in \mathbb{R}$ where A is a measurable set. We call $\tilde{X}$ the killed process, which is one-dimensional and Markovian. The equivalent problem when considering the killed process is

$$V_* = \inf_\tau \mathbb{E}\left[M(\tilde{X}_\tau) + \int_0^\tau L(\tilde{X}_t)\,\mathrm{d}t\right], \tag{3.3.4}$$

and it can be shown that the infinitesimal generator of $\tilde{X}$ is

$$\mathbb{L}_{\tilde{X}} = \mathbb{L}_X - \lambda I \tag{3.3.5}$$

where I is the identity operator. Hence, we have transformed the two-dimensional process (t, X) into a one-dimensional process $\tilde{X}$. Thus, when faced with a killed MLS problem one is led to consider auxiliary problems as in the previous subsection with an added linear term in the differential equation. For details, we again refer to Peskir and Shiryaev (2006, Section 5.4).

Method of deterministic time-change

We first introduce the method following Pedersen and Peskir (2000) and then a particular example to be used later following Graversen et al. (2001). Let X be a one-dimensional time-homogeneous Markov process with values in $\mathbb{R}$. Specifically we assume $X_0 = x$ under P_x and X solves

$$\mathrm{d}X_t = \alpha(X_t)\,\mathrm{d}t + \eta(X_t)\,\mathrm{d}B_t \tag{3.3.6}$$

for some positive and continuous function η, a continuous function α, and $B = (B_t)_{t\geq 0}$ being a standard Brownian motion. Thus, X has infinitesimal generator

$$\mathbb{L}_X = \alpha(x)\frac{\partial}{\partial x} + \frac{1}{2}\eta^2(x)\frac{\partial^2}{\partial x^2}. \tag{3.3.7}$$

Often one encounters functionals in the MLS formulation where terms of the form $\gamma(t)X_t$ appear for some nonlinear but smooth $t \mapsto \gamma(t)$. As mentioned this will usually imply that we have to consider the two-dimensional Markov process (t, X). However, from the method of transformation by killing we know that for exponential γ the problem can be kept one-dimensional. Hence, we seek to find a time transformation $s \mapsto T_s$ which is continuous and strictly increasing such that there exists a one-dimensional time-homogeneous Markov process $Z = (Z_s)_{s\geq 0}$ where

$$\gamma(T_s)X_{T_s} = e^{-\lambda s}Z_s \tag{3.3.8}$$

for some $\lambda \in \mathbb{R}$. Consider the process $Y = (Y_t)_{t\geq 0}$ with $Y_t = \gamma(t)X_t$. Since γ is smooth it is in particular a function of finite variation. Hence, by the product rule

$$Y_t = Y_0 + \int_0^t (\gamma'(u)X_u + \gamma(u)\alpha(X_u))\,\mathrm{d}u + \int_0^t \gamma(u)\eta(X_u)\,\mathrm{d}B_u \tag{3.3.9}$$

$$= Y_0 + \int_0^t \left(\frac{\gamma'(u)}{\gamma(u)}Y_u + \gamma(u)\alpha\left(\frac{Y_u}{\gamma(u)}\right)\right)\mathrm{d}u + \int_0^t \gamma(u)\eta\left(\frac{Y_u}{\gamma(u)}\right)\mathrm{d}B_u. \tag{3.3.10}$$

Thus, Y has infinitesimal generator

$$\mathbb{L}_Y = \left(\frac{\gamma'(t)}{\gamma(t)} y + \gamma(t)\alpha\left(\frac{y}{\gamma(t)}\right)\right)\frac{\partial}{\partial y} + \frac{1}{2}\gamma^2(t)\eta^2\left(\frac{y}{\gamma(t)}\right)\frac{\partial^2}{\partial y^2}. \tag{3.3.11}$$

If we consider a time-change on the form

$$T_s = \inf\left\{t > 0 \;\middle|\; \int_0^t \rho(u)\,\mathrm{d}u > s\right\} \tag{3.3.12}$$

for some positive and measurable ρ where we assume $\int_0^t \rho(u)\,\mathrm{d}u = \infty$ for $t = \infty$ and possibly for finite t. Then by Rogers and Williams (1987, chapter V.26) the infinitesimal generator of $Z = (Y_{T_s})_{s\geq 0}$ is given by

$$\mathbb{L}_Z = \frac{1}{\rho(t)}\mathbb{L}_Y. \tag{3.3.13}$$

In order for Z to be time-homogeneous ρ must be chosen such that there exists functions G_1 and G_2 of y such that

$$\begin{aligned} \frac{\gamma'(t)}{\gamma(t)} y + \gamma(t)\alpha\left(\frac{y}{\gamma(t)}\right) &= \rho(t)G_1(y) \\ \frac{1}{2}\gamma^2(t)\eta^2\left(\frac{y}{\gamma(t)}\right) &= \rho(t)G_2(y), \end{aligned} \tag{3.3.14}$$

i.e. the drift and diffusion terms of Y should be separable into time and space. In most cases, this is rather difficult. However, for particular γ, α, and η the construction can be surprisingly simple. As of now, we have not considered the goal of an exponential term in (3.3.8). If the terms are separable with a ρ that integrates to a logarithmic function, then taking the inverse to get the time-change in (3.3.12) we will get such an exponential term.

In the optimal prediction problems studied in Chapter 4, a process on the form

$$\frac{B_t}{\sqrt{1-t}} \tag{3.3.15}$$

for $t \in (0, 1)$ appears. This is a complicated process on a finite time horizon. The following lemma shows that we can formulate an equivalent problem with a simple one-dimensional time-homogeneous process with an infinite time horizon:

Lemma 3.3.1. *Consider a Brownian motion $B = (B_t)_{t\in[0,1]}$. The process $Z = (Z_s)_{s\geq 0}$ defined by*

$$Z_s = \frac{B_{T_s}}{\sqrt{1-T_s}}, \tag{3.3.16}$$

where $T_s = 1 - e^{-2s}$, is an Ornstein-Uhlenbeck process with infinitesimal generator

$$\mathbb{L}_Z = z\frac{\mathrm{d}}{\mathrm{d}z} + \frac{\mathrm{d}^2}{\mathrm{d}z^2}. \tag{3.3.17}$$

Furthermore, σ is a stopping time for Z if and only if $\tau = 1 - e^{-2\sigma}$ is a stopping time for B.

Proof. Let $B = (B_t)_{t\in[0,1]}$ be a Brownian motion. The system (3.3.14) reduces significantly since $\alpha(t) = 0$ and $\eta(t) = 1$ for all $t \in [0, 1]$. Inserting $\gamma(t) = \frac{1}{\sqrt{1-t}}$ we get the system

$$\begin{aligned} \frac{(2(1-t)^{3/2})^{-1}}{(1-t)^{-1/2}} y &= \rho(t)G_1(y) \\ (2(1-t))^{-1} &= \rho(t)G_2(y). \end{aligned} \tag{3.3.18}$$

Thus, we find $G_1(y) = y, G_2(y) = 1$ and $\rho(t) = (2(1-t))^{-1}$ to solve the system. Hence, by (3.3.12) we consider the time-change $T_s = 1 - e^{-2s}$ and the process

$$Z_s = \frac{B_{T_s}}{\sqrt{1-T_s}} = e^s B_{1-e^{-2s}}. \tag{3.3.19}$$

By Itô's lemma, we have

$$\mathrm{d}Z_s = Z_s\,\mathrm{d}s + e^s\,\mathrm{d}B_{1-e^{-2s}} = Z_s\,\mathrm{d}s + \sqrt{2}\,\mathrm{d}\beta_s \tag{3.3.20}$$

where the process $\beta = (\beta_s)_{s\geq 0}$ is given by

$$\beta_s = \frac{1}{\sqrt{2}}\int_0^s e^u\,\mathrm{d}B_{1-e^{-2u}} = \frac{1}{\sqrt{2}}\int_0^{T_s} \frac{1}{\sqrt{1-u}}\,\mathrm{d}B_u. \tag{3.3.21}$$

Note that β is a continuous centered Gaussian martingale and by the Itô isometry we have

$$\mathbb{E}\left[\beta_s^2\right] = \frac{1}{2}\int_0^{1-e^{-2s}} \frac{1}{1-u}\,\mathrm{d}u = \frac{1}{2}\left[\log x\right]_{e^{-2s}}^1 = s. \tag{3.3.22}$$

Thus, by Lévy's characterization theorem (see e.g. Revuz and Yor, 1999), β is a standard Brownian motion. Hence, by (3.3.20) Z is an Ornstein-Uhlenbeck process with an infinitesimal generator given by

$$\mathbb{L}_Z = z\frac{\mathrm{d}}{\mathrm{d}z} + \frac{\mathrm{d}^2}{\mathrm{d}z^2}. \tag{3.3.23}$$

The time-change can trivially be extended to a bijection between $[0, \infty]$ and $[0, 1]$ thus letting $(\mathcal{F}_s^Z)_{s\geq 0}$ be the filtration generated by Z we see that $\mathcal{F}_s^Z = \mathcal{F}_{T_s}^B$. A stopping time σ with respect to $(\mathcal{F}_s^Z)_{s\geq 0}$ will hence yield a stopping time $\tau = T_\sigma$ with respect to $(\mathcal{F}_t^B)_{t\in[0,1]}$ and vice versa. □

3.4 Verification

Solving a free-boundary problem need not give the right solution to the particular optimal stopping problem. This has to be verified. In the case where an explicit solution to the free-boundary problem can be found, then verification can usually be done by an application of an extension of Itô's lemma and the optional sampling theorem. As we will see in Theorem 4.2.1, this formally usually also involves the application of Fatou's lemma and the dominated convergence theorem over a localizing sequence of bounded stopping times, e.g. taking $\tau_n \wedge n$ where $(\tau_n)_{n\in\mathbb{N}}$ is a localizing sequence.

In case no explicit solution exists, which often is the case when the dimensionality is greater than one, the verification becomes tedious. In such cases, we will make use of the following Assumptions 3.4.1 for an optimal stopping problem on the form

$$V_*(t,x) = \inf_{\tau \le 1-t} \mathbb{E}_{t,x}\left[M(t+\tau, X_{t+\tau}) + \int_0^\tau L(t+s, X_{t+s})\,\mathrm{d}s\right] \tag{3.4.1}$$

for $(t,x) \in [0,1]\times\mathbb{R}_+$ where $X = (X_t)_{t\in[0,1]}$ is a one-dimensional continuous and time-homogeneous process and the measure is such that $X_t = x$ under $P_{t,x}$:

Assumptions 3.4.1. *Assume:*

1. *Existence of an optimal stopping time τ_* being the first hitting time of the stopping set D.*
2. *The stopping set is on the form*

$$D = \{(t,x) \in [0,1]\times\mathbb{R} \mid |x| \ge b(t)\} \tag{3.4.2}$$

   for some continuous and real function b satisfying $b(1) = 0$.
3. *The mapping $(t,x) \mapsto V_*(t,x)$ is continuous.*
4. *The mapping $x \mapsto \frac{\partial V_*}{\partial x}(t,x)$ is continuous for any $t \in [0,1]$.*

Under these assumptions, we can formulate a free-boundary equation as we will discuss in the following section.

3.5 Free-Boundary Equation

In the case of a non-explicit solution then we have to resort to numerical methods for computing the value function and the shape of ∂C. For the present section let us assume that the underlying process is two-dimensional $Z = (Z_t)_{t\ge0}$ with $Z_t = (t, X_t)$ for a one-dimensional time-homogeneous diffusion $X = (X_t)_{t\ge0}$, and that the horizon of the problem is some finite T. We furthermore assume that the value function of the optimal stopping problem solves the

Bolza problem and the stopping set can be characterized by a continuous function $b : [0, T] \mapsto \mathbb{R}$ in the following way:

$$D = \{(t, x) \in [0, T] \times \mathbb{R} \mid X_t \geq b(t)\} \tag{3.5.1}$$

hence ∂C is described by the curve b. By the Markov approach, we can write the value function as a function of Z, and for convenience, we drop the subscript on the value function. Utilizing the change of variable formula (see Peskir and Shiryaev, 2006, Chapter II.3.5) under $P_{t,x}$ we have

$$\begin{aligned} V(t+u, X_{t+u}) =& V(t,x) + \int_0^u \mathbb{L}_Z V(t+s, X_{t+s}) I(X_{t+s} \neq b(t+s))\, \mathrm{d}s && (3.5.2)\\ &+ \int_0^u V_x(t+s, X_{t+s}) \eta(X_{t+s}) I(X_{t+s} \neq b(t+s))\, \mathrm{d}B_s && (3.5.3)\\ &+ \int_0^u \Delta_x V_x(t+s, X_{t+s}) I(X_{t+s} = b(t+s))\, \mathrm{d}\ell_s^b(X) && (3.5.4) \end{aligned}$$

where the operator Δ_x is such that $\Delta_x V_x(t,x) = V_x(t,x+) - V_x(t,x-)$ and $\ell^b(X)$ is the local time of X at b (see e.g. Karatzas and Shreve, 1991). Since we assume the smooth-fit condition the local time integral term vanishes. Furthermore, the process $N = (N_u)_{u \geq 0}$ given by

$$N_u = \int_0^u V_x(t+s, X_{t+s}) \eta(X_{t+s}) I(X_{t+s} \neq b(t+s))\, \mathrm{d}B_s \tag{3.5.5}$$

is a local martingale. Taking expectation under $P_{t,x}$ and evaluating in $u = T - t$ we thus get

$$\mathbb{E}_{t,x}\left[V(T, X_T)\right] = V(t,x) + \int_0^{T-t} \mathbb{E}_{t,x}\left[\mathbb{L}_Z V(t+s, X_{t+s}) I(X_{t+s} \neq b(t+s))\right] \mathrm{d}s. \tag{3.5.6}$$

We can now split up the indicator using (3.2.4) and (3.2.7) to arrive at the equation

$$\begin{aligned} \mathbb{E}_{t,x}\left[V(T, X_T)\right] =& V(t,x) && (3.5.7)\\ &+ \int_0^{T-t} \mathbb{E}_{t,x}\left[\mathbb{L}_Z M(t+s, X_{t+s}) I(X_{t+s} > b(t+s))\right] \mathrm{d}s && (3.5.8)\\ &- \int_0^{T-t} \mathbb{E}_{t,x}\left[L(t+s, X_{t+s}) I(X_{t+s} < b(t+s))\right] \mathrm{d}s. && (3.5.9) \end{aligned}$$

The equation has both V and b as unknowns, however evaluating in the specific measure $P_{t,b(t)}$ we can utilize the continuous-fit condition to get

$$\begin{aligned} \mathbb{E}_{t,b(t)}\left[M(T, X_T)\right] =& M(t, b(t)) && (3.5.10)\\ &+ \int_0^{T-t} \mathbb{E}_{t,b(t)}\left[\mathbb{L}_Z M(t+s, X_{t+s}) I(X_{t+s} > b(t+s))\right] \mathrm{d}s && (3.5.11)\\ &- \int_0^{T-t} \mathbb{E}_{t,b(t)}\left[L(t+s, X_{t+s}) I(X_{t+s} < b(t+s))\right] \mathrm{d}s && (3.5.12) \end{aligned}$$

for $t \in [0, T]$. This is a nonlinear Volterra equation of the function b. Given that we can find a terminal condition for b then this can be solved iteratively, see Appendix A.1.

Chapter 4

Solutions to Optimal Prediction Problems

With the tools from Chapter 3 we are now ready to solve a range of optimal prediction problems on the form of Definition 2.2.1, 2.2.2 and 2.2.4. We start by considering some probabilistic results gathered from the literature. Then from the outset of the linear time problem which is solved in Urusov (2005) we set out to solve more intricate problems.

4.1 Probabilistic Results

As discussed in Chapter 3 we need some preliminary probabilistic results in order to solve optimal stopping problems. The main problem is that the time and space problems are not on the usual *MLS* form since θ and S_1 are not adapted to the filtration.

Let Φ and φ be the distribution and density functions of a standard normal, respectively. Recall the reflection theorem (see e.g. Revuz and Yor, 1999, Chapter III, Proposition 3.7) giving $P(S_t \geq x) = 2P(B_t \geq x)$, hence

$$P(S_t \leq z) = 2\Phi\left(\frac{z}{\sqrt{t}}\right) - 1. \tag{4.1.1}$$

By differentiating we get the density

$$P(S_t \in \mathrm{d}z) = \frac{2}{\sqrt{t}}\varphi\left(\frac{z}{\sqrt{t}}\right)\mathrm{d}z. \tag{4.1.2}$$

Furthermore, a process that will be of interest is the *CUSUM*-type process $S-B = (S_t - B_t)_{t\in[0,1]}$. By Lévy's distributional theorem (see e.g. Revuz and Yor, 1999, Chapter VI, Theorem 2.3) we

have that

$$S_t - B_t \overset{d}{=} |B_t| \tag{4.1.3}$$

for all $t \in [0,1]$.

We need an expression for the posterior distribution of θ, specifically the posterior process π_t. The following result is from Peskir and Shiryaev (2006, Chapter VIII section 30.1.4) however there is a typo, so we give the proof for completeness.

Lemma 4.1.1. *Let $t \in (0,1)$. The posterior process $\pi_t = P(\theta \leq t \mid \mathcal{F}_t)$ satisfies*

$$\pi_t = 2\Phi\left(\frac{S_t - B_t}{\sqrt{1-t}}\right) - 1. \tag{4.1.4}$$

Proof. Note that

$$P(\theta \leq t \mid \mathcal{F}_t) = P\left(S_t \geq \max_{u\in[t,1]} B_u \mid \mathcal{F}_t\right). \tag{4.1.5}$$

So by (4.1.1) it is easily seen that

$$\begin{aligned} 1 - \pi_t &= P\left(S_t < \max_{u\in[t,1]} B_u \mid \mathcal{F}_t\right) = P\left(S_t - B_t < \max_{u\in[t,1]}(B_u - B_t) \mid \mathcal{F}_t\right) && (4.1.6)\\ &= P(s - x < S_{1-t})|_{s=S_t, x=B_t} = 1 - F_{1-t}(S_t - B_t) && (4.1.7)\\ &= 2 - 2\Phi\left(\frac{S_t - B_t}{\sqrt{1-t}}\right) && (4.1.8) \end{aligned}$$

where the third equality holds by stationary and independent increments of B and $F_t(z) = P(S_t \leq z)$. □

The following result comes from Pedersen (2003) and is useful for reducing the dimensionality by one since we know $S - B \overset{d}{=} |B|$. The proof is somewhat similar to the above hence we omit it.

Lemma 4.1.2. *Let $G : \mathbb{R}_+ \to \mathbb{R}$ be a measurable function and fix $t \in (0,1)$. Then*

$$\mathbb{E}\left[G(S_1 - B_t) \mid \mathcal{F}_t\right] = G(S_t - B_t)P(S_{1-t} \leq S_t - B_t) + \int_{S_t - B_t}^{\infty} G(z)P(S_{1-t} \in \mathrm{d}z). \tag{4.1.9}$$

Note that we can extend the results in Lemma 4.1.1 and 4.1.2 to stopping times τ since τ is finite so a standard argument based on a decreasing sequence of simple stopping times will yield the wanted.

Lastly, we also need the transition density of the reflected Brownian motion $|B|$. Referring to Borodin and Salminen (1996) it is given by

$$p(t;x,y) = \frac{1}{\sqrt{2\pi t}}\left(e^{-\frac{(x-y)^2}{2t}} + e^{-\frac{(x+y)^2}{2t}}\right). \tag{4.1.10}$$

4.2 Time Problems

We are now ready to turn our attention to optimal prediction problems specifically the time formulation from Definition 2.2.1. We start by considering different linear time problems.

4.2.1 Linear Time Problems

Weighted linear loss

The linear time problem was first proposed in Urusov (2005). As discussed earlier the events $(\tau < \theta)$ and $(\tau > \theta)$ have different interpretations, so we want to consider asymmetric loss functions. We start by considering the weighted linear loss $L(\tau,\theta) = c(\tau-\theta)^+ + (\tau-\theta)^-$ for some $c \geq 0$. This leads to the following theorem:

Theorem 4.2.1. *Consider the family of optimal prediction problems with weighted linear loss*

$$V_*^{(c)} = \inf_{\tau\in\mathcal{T}} \mathbb{E}\left[c(\tau-\theta)^+ + (\tau-\theta)^-\right] \tag{4.2.1}$$

for $c \geq 0$. Then the value $V_*^{(c)}$ is given by

$$V_*^{(c)} = \frac{c}{2} + (c+1)\left(\Phi\left(z_*^{(c)}\right) - 1\right) \tag{4.2.2}$$

with $z_*^{(c)}$ being the unique strictly positive root of the equation

$$4\Phi\left(z_*^{(c)}\right) - 2z_*^{(c)}\varphi\left(z_*^{(c)}\right) - 3 + \frac{c-1}{c+1} = 0 \tag{4.2.3}$$

and the following stopping time is optimal:

$$\tau_*^{(c)} = \inf\{t \in [0,1] \mid S_t - B_t \geq z_*^{(c)}\sqrt{1-t}\}. \tag{4.2.4}$$

Proof. As in Urusov (2005) we first rewrite the problem to an optimal stopping problem. We fix a $c \geq 0$ and drop the superscript. Note that

$$c(\tau-\theta)^+ + (\tau-\theta)^- = c(\tau-\theta)^+ + \theta - \tau\wedge\theta \tag{4.2.5}$$

$$= \int_0^\tau cI(\theta \leq t)\,\mathrm{d}t + \theta - \int_0^\tau I(\theta > t)\,\mathrm{d}t \tag{4.2.6}$$

$$= \theta + \int_0^\tau \big((c+1)I(\theta \leq t) - 1\big)\,\mathrm{d}t. \tag{4.2.7}$$

Now taking expectation and noting that τ is a stopping time we get by the tower property that

$$\mathbb{E}\left[c(\tau-\theta)^+ + (\tau-\theta)^-\right] = \mathbb{E}[\theta] + \mathbb{E}\left[\int_0^\infty \big((c+1)I(\theta \le t) - 1\big) I(t<\tau)\,\mathrm{d}t\right] \tag{4.2.8}$$

$$= \frac{1}{2} + \mathbb{E}\left[\int_0^\infty \big((c+1)\mathbb{E}\left[I(\theta \le t) \mid \mathcal{F}_t\right] - 1\big) I(t<\tau)\,\mathrm{d}t\right] \tag{4.2.9}$$

$$= \frac{1}{2} + \mathbb{E}\left[\int_0^\tau \big((c+1)\pi_t - 1\big)\,\mathrm{d}t\right] \tag{4.2.10}$$

where π_t is the posterior process. Thus, by Lemma 4.1.1 we may write

$$\mathbb{E}\left[c(\tau-\theta)^+ + (\tau-\theta)^-\right] = \frac{1}{2} + \mathbb{E}\left[\int_0^\tau \left((c+1)\left(2\Phi\left(\frac{S_t - B_t}{\sqrt{1-t}}\right) - 1\right) - 1\right)\mathrm{d}t\right] \tag{4.2.11}$$

$$= \frac{1}{2} + \frac{c+1}{2}\mathbb{E}\left[\int_0^\tau \left(4\Phi\left(\frac{S_t - B_t}{\sqrt{1-t}}\right) - 2 - \frac{2}{c+1}\right)\mathrm{d}t\right] \tag{4.2.12}$$

$$= \frac{1}{2} + \frac{c+1}{2}\mathbb{E}\left[\int_0^\tau \left(4\Phi\left(\frac{S_t - B_t}{\sqrt{1-t}}\right) - 3 + \frac{c-1}{c+1}\right)\mathrm{d}t\right]. \tag{4.2.13}$$

We will denote $F(z) = 4\Phi(z) - 3 + \frac{c-1}{c+1}$. For $c = 1$ this is exactly the optimal stopping problem considered in Graversen et al. (2001), hence we proceed following their approach. Since $S - B \overset{d}{=} |B|$ we will study the optimal stopping problem

$$V_* = \frac{1}{2} + \frac{c+1}{2}\inf_{\tau\in\mathcal{T}}\mathbb{E}\left[\int_0^\tau F\left(\frac{|B_t|}{\sqrt{1-t}}\right)\mathrm{d}t\right]. \tag{4.2.14}$$

However, this problem is of a finite time horizon which makes the problem inherently two-dimensional. To circumvent this we utilize Lemma 3.3.1, to get an Ornstein-Uhlenbeck process $Z = (Z_s)_{s\ge 0}$ where

$$Z_s = \frac{B_{T_s}}{\sqrt{1-T_s}} \tag{4.2.15}$$

for the time-change $T_s = 1 - e^{-2s}$. The process Z has infinitesimal generator given by (3.3.17). Inserting the time-change into (4.2.14) by a substitution we get

$$V_* = \frac{1}{2} + \frac{c+1}{2}\inf_{\tau\in\mathcal{T}}\mathbb{E}\left[\int_0^{\sigma_\tau} 2e^{-2s}F\left(|Z_s|\right)\mathrm{d}s\right] \tag{4.2.16}$$

where $\sigma_\tau : \tau \mapsto -\log\sqrt{1-\tau}$. Recall that there is a one-to-one correspondence between stopping times for Z and B. Furthermore, since $|Z_s| \to \infty$ P-a.s. for $s \to \infty$, which can be realized by using the scale function of Z and (Karatzas and Shreve, 1991, Proposition 5.5.22), then any first hitting time is a stopping time for $|Z|$. We denote the set of stopping times for Z by $\mathcal{S}$.

We now consider the following optimal stopping problem:

$$W_*(z) = \inf_{\sigma \in \mathcal{S}} \mathbb{E}_z \left[\int_0^\sigma e^{-2s} F\left(|Z_s|\right) \mathrm{d}s \right] \tag{4.2.17}$$

for $z \in \mathbb{R}$ with $Z_0 = z$ under P_z. The problem is a killed Dirichlet/Poisson problem, see Section 3.2. Note that since we consider $|Z|$ we argue that the value function should be even. Furthermore, since F is continuous and strictly increasing we guess that the optimal stopping time is on the form

$$\sigma_* = \inf\left\{ s \geq 0 \mid |Z_s| > z_* \right\} \tag{4.2.18}$$

for some $z_* > 0$. That is we argue that the stopping boundary is $\partial C = \{-z_*, z_*\}$. We also impose smooth- and continuous-fit conditions as mentioned above. The free-boundary problem to study is thus

$$\begin{aligned}
\mathbb{L}_Z W(z) - 2W(z) &= -F(|z|), && z \in (-z_*, z_*) && (4.2.19)\\
W(\pm z_*) &= 0, && \text{(Instantaneous-stopping/Cont. fit)} && (4.2.20)\\
W'(\pm z_*) &= 0, && \text{(Smooth-fit)} && (4.2.21)\\
W'(0) &= 0, && \text{(Even).} && (4.2.22)
\end{aligned}$$

Since we impose an even value function we consider the ODE (4.2.19) for $z \in [0, z_*)$, that is

$$W''(z) + zW'(z) - 2W(z) = -F(z). \tag{4.2.23}$$

By general techniques (see e.g. Polyanin and Zaitsev, 2017) we have the general solution

$$W(z) = K_1(z^2+1) + K_2(z\varphi(z) + (z^2+1)\Phi(z)) + \frac{1}{2}F(z) \tag{4.2.24}$$

where K_1 and K_2 are to be determined. Note by the even and smooth-fit properties we have

$$0 = W'(0) = 2\varphi(0)(K_2 + 1) \tag{4.2.25}$$

yielding $K_2 = -1$ and

$$0 = W'(z_*) = 2z_*(K_1 - \Phi(z_*)) \tag{4.2.26}$$

yielding $K_1 = \Phi(z_*)$, where we use $z_* > 0$. Finally, the continuous-fit implies

$$0 = W(z_*) = \Phi(z_*)(z_*^2+1) - (z_*\varphi(z_*) + (z_*^2+1)\Phi(z_*)) + 2\Phi(z_*) - \frac{3}{2} + \frac{c-1}{2(c+1)}. \tag{4.2.27}$$

So we can determine z_* as the solution to

$$4\Phi(z_*) - 2z_*\varphi(z_*) - 3 + \frac{c-1}{c+1} = 0. \tag{4.2.28}$$

We note that $f(z) := 4\Phi(z) - 2z\varphi(z) - 3 + \frac{c-1}{c+1}$ has a unique root $z_* > 0$ since $\frac{c-1}{c+1} \in (-1, 1)$ for $c > 0$, the derivative is $f'(z) = 2\varphi(z) + 2z^2\varphi(z) > 0$, the initial point is $f(0) = \frac{c-1}{c+1} - 1 < 0$ and the limit is $\lim_{z\to\infty} f(z) = 1 + \frac{c-1}{c+1} > 0$.

We need to verify that our proposed solution from the auxiliary problem W, given by (4.2.24) and trivially extended to $z \in \mathbb{R}$, is equal to W_*. Note first that $W \in C^2(\mathbb{R} \setminus \{\pm z_*\})$ and $W \in C^1(\mathbb{R})$. So by the Itô-Tanaka formula, we have

$$e^{-2s}W(Z_s) = W(Z_0) + \int_0^s e^{-2u} \left(\mathbb{L}_Z W(Z_u) - 2W(Z_u)\right) \mathrm{d}u + M_s \tag{4.2.29}$$

where $M_s = \sqrt{2}\int_0^s e^{-2u} W'(Z_u)\, \mathrm{d}\beta_u$ is a continuous and centered local martingale. We specified that $W(z) = 0$ on $z \notin [-z_*, z_*]$ so $\mathbb{L}_Z W(Z_u) - 2W(Z_u) = 0$ furthermore on this set

$$F(|z|) = 4\Phi(|z|) - 3 + \frac{c-1}{c+1} \geq f(|z|) \geq f(|z_*|) = 0 \tag{4.2.30}$$

since $2\,|z|\,\varphi(|z|)$ is positive and f is increasing. Using that W solves the ODE (4.2.19) on $z \in (-z_*, z_*)$ we get the following bound for all $z \in \mathbb{R}$:

$$e^{-2s}W(Z_s) \geq W(Z_0) - \int_0^s e^{-2u} F(|Z_u|)\, \mathrm{d}u + M_s \tag{4.2.31}$$

where we use that $(Z_s = \pm z_*)$ is a null set since Z is a diffusion. Note that $W(z) \leq 0$ for all z since for $z \in (-z_*, z_*)$ we have that

$$W'(z) = 2z(\Phi(z_*) - \Phi(|z|)) \tag{4.2.32}$$

and $W(\pm z_*) = 0$. So specifically also $W(Z_\sigma) \leq 0$ for any stopping time σ. Let $(\overline{\sigma}_n)_{n\in\mathbb{N}}$ be a localizing sequence of bounded stopping times for M. Then taking expectation in the stopped version of (4.2.31) we get

$$\begin{aligned}
\mathbb{E}_z\left[\int_0^{\sigma\wedge\overline{\sigma}_n} e^{-2u} F(|Z_u|)\, \mathrm{d}u\right] &\geq W(z) + \mathbb{E}_z[M_{\sigma\wedge\overline{\sigma}_n}] - \mathbb{E}_z\left[e^{-2(\sigma\wedge\overline{\sigma}_n)} W(Z_{\sigma\wedge\overline{\sigma}_n})\right] && (4.2.33)\\
&\geq W(z) + \mathbb{E}_z[M_{\sigma\wedge\overline{\sigma}_n}] = W(z) && (4.2.34)
\end{aligned}$$

by optional sampling on the martingale $M_{\cdot\wedge\overline{\sigma}_n}$. Thus, by Fatou's lemma, we get

$$\begin{aligned}
W(z) &\leq \lim_{n\to\infty} \mathbb{E}_z\left[\int_0^{\sigma\wedge\overline{\sigma}_n} e^{-2u} F(|Z_u|)\, \mathrm{d}u\right] \leq \limsup_{n\to\infty} \mathbb{E}_z\left[\int_0^{\sigma\wedge\overline{\sigma}_n} e^{-2u} F(|Z_u|)\, \mathrm{d}u\right] && (4.2.35)\\
&\leq \mathbb{E}_z\left[\int_0^{\sigma} e^{-2u} F(|Z_u|)\, \mathrm{d}u\right]. && (4.2.36)
\end{aligned}$$

So taking infimum over $\mathcal{S}$ we get $W(z) \leq W_*(z)$.

For the other inequality note that since σ_* is a first hitting time we have $\sigma_* < \infty$ P_z-a.s. for any $|z| \leq z_*$ as argued above. For $0 \leq s \leq \sigma_*$ it holds that

$$\mathbb{L}_Z W(Z_s) - 2W(Z_s) = -F(|Z_s|) \tag{4.2.37}$$

so for such s it holds that

$$e^{-2s} W(Z_s) = W(Z_0) - \int_0^s e^{-2u} F(|Z_u|)\,\mathrm{d}u + M_s. \tag{4.2.38}$$

Specifically on $\sigma_* \wedge \overline{\sigma}_n$ we have by taking expectation

$$W(Z) = \mathbb{E}_z\left[\int_0^{\sigma_* \wedge \overline{\sigma}_n} e^{-2u} F(|Z_u|)\,\mathrm{d}u\right] - \mathbb{E}_z\left[M_{\sigma_* \wedge \overline{\sigma}_n}\right] + \mathbb{E}_z\left[e^{-2(\sigma_* \wedge \overline{\sigma}_n)} W(Z_{\sigma_* \wedge \overline{\sigma}_n})\right]. \tag{4.2.39}$$

Here the second term vanishes due to optional sampling. Hence, letting $n \to \infty$ we get by the dominated convergence theorem and (4.2.20) that

$$W(z) = \mathbb{E}_z\left[\int_0^{\sigma_*} e^{-2u} F(|Z_u|)\,\mathrm{d}u\right] \geq \inf_{\sigma \in \mathcal{S}} \mathbb{E}_z\left[\int_0^{\sigma} e^{-2u} F(|Z_u|)\,\mathrm{d}u\right] = W_*(z) \tag{4.2.40}$$

thus $W(z) = W_*(z)$.

By backtracking we see that

$$V_* = \frac{1}{2} + \frac{c+1}{2} 2W_*(0) = \frac{c}{2} + (c+1)\left(\Phi(z_*) - 1\right) \tag{4.2.41}$$

and transforming σ_* back to τ_* we get the optimal strategy as stated. Since $c > 0$ was arbitrary we are done. □

In Graversen et al. (2001) the authors show that

$$\mathbb{E}\left[(S_1 - B_\tau)^2\right] = \mathbb{E}\left[\int_0^\tau F\left(\frac{S_t - B_t}{\sqrt{1-t}}\right)\mathrm{d}t\right] + 1 \tag{4.2.42}$$

were $F(x) = 4\Phi(x) - 3$ by using a particular representation of the maximum process. In Urusov (2005) the author utilizes (4.2.13) with $c = 1$ to imply that the squared spatial problem is equivalent to the linear time loss problem in the following way:

$$\mathbb{E}\left[|\tau - \theta|\right] = \mathbb{E}\left[(B_\tau - S_1)^2\right] - \frac{1}{2} \tag{4.2.43}$$

for any stopping time $\tau \in \mathcal{T}$. So an optimal stopping time in the linear time problem is also optimal in the squared space problem. However, we can extend the relation between the linear

time loss and the squared spatial loss further by using (4.2.43) and (4.2.13) for a general $c > 0$ to get

$$\mathbb{E}\left[c(\tau-\theta)^+ + (\tau-\theta)^-\right] = \frac{c+1}{2}\mathbb{E}\left[(B_\tau - S_1)^2\right] - \frac{c}{2} + \frac{c-1}{2}\mathbb{E}\left[\tau\right]. \tag{4.2.44}$$

Thus, the relation, that the optimal prediction rule for the linear time loss is equal to the optimal rule for the squared space loss does not carry over to the weighted linear time loss since the right-hand side of the above equation also has a preference for τ being large when $c < 1$ and vice versa.

However, since we have solved a whole family of loss functions where we weight the loss between $(\tau < \theta)$ and $(\tau > \theta)$, we can use the free parameter c to get a similar relation by choosing a particular loss function.

Corollary 4.2.2. *Let $\mu \in (0,1)$ be the wanted expectation of the stopping rule. The following relation holds:*

$$\inf_{\tau\in\mathcal{T}'}\mathbb{E}\left[\overline{c}(\tau-\theta)^+ + (\tau-\theta)^-\right] = \frac{\overline{c}+1}{2}\inf_{\tau\in\mathcal{T}'}\left(\mathbb{E}\left[(B_\tau - S_1)^2\right]\right) - \frac{\overline{c}}{2} + \frac{\overline{c}-1}{2}\mu \tag{4.2.45}$$

where $\mathcal{T}' = \{\tau \in \mathcal{T} \mid \mathbb{E}[\tau] = \mu\}$. Here $\overline{c} > 0$ is given by

$$\overline{c} = -\frac{2\Phi\left(z_*^{(\mu)}\right) - \varphi\left(z_*^{(\mu)}\right) - 2}{2\Phi\left(z_*^{(\mu)}\right) - \varphi\left(z_*^{(\mu)}\right) - 1} \tag{4.2.46}$$

and $z_*^{(\mu)}$ is given by

$$z_*^{(\mu)} = \sqrt{\frac{\mu}{1-\mu}} \tag{4.2.47}$$

and the stopping time $\tau_*^{(\mu)} = \inf\left\{t \in [0,1] \;\middle|\; S_t - B_t \geq z_*^{(\mu)}\sqrt{1-t}\right\}$ is optimal.

Proof. The optimal rule in Theorem 4.2.1 is given by (4.2.4). Note that $S - B \overset{d}{=} |B|$ so since the process has continuous paths we may write

$$\tau_*^{(c)} = \inf\{t \in [0,1] \mid S_t - B_t = z_*^{(c)}\sqrt{1-t}\} \overset{d}{=} \tilde{\tau}_*^{(c)} := \inf\{t \in [0,1] \mid |B_t| = z_*^{(c)}\sqrt{1-t}\}. \tag{4.2.48}$$

But note since $\tilde{\tau}_*^{(c)} \leq 1$ then $\mathbb{E}[\tilde{\tau}_*^{(c)}] < \infty$ thus by Wald's identities (see e.g. Peskir and Shiryaev, 2006, Chapter II, Section 3.2) we have $\mathbb{E}[B^2_{\tilde{\tau}_*^{(c)}}] = \mathbb{E}[\tilde{\tau}_*^{(c)}]$ which yields

$$\mathbb{E}[\tau_*^{(c)}] = \mathbb{E}[\tilde{\tau}_*^{(c)}] = \mathbb{E}[\left|B_{\tilde{\tau}_*^{(c)}}\right|^2] = \left(z_*^{(c)}\right)^2\mathbb{E}[1-\tilde{\tau}_*^{(c)}] = \left(z_*^{(c)}\right)^2(1-\mathbb{E}[\tau_*^{(c)}]). \tag{4.2.49}$$

Hence,

$$\mathbb{E}[\tau_*^{(c)}] = \frac{\left(z_*^{(c)}\right)^2}{1+\left(z_*^{(c)}\right)^2}. \tag{4.2.50}$$

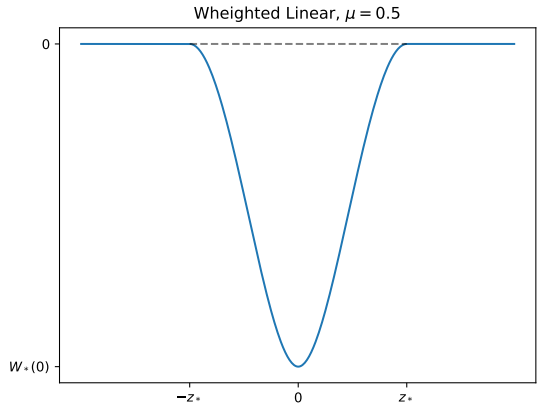


Figure 4.2.1: Value function from (4.2.17).

So choosing $z_*^{(\mu)} > 0$ by (4.2.47), then to have Theorem 4.2.1 hold we must let $\bar{c}$ solve (4.2.3) for $z_*^{(\mu)}$. Since $c \mapsto \frac{c-1}{c+1}$ is a bijection from $\mathbb{R}_+$ to $(-1, 1)$ and

$$4\Phi(z) - 2z\varphi(z) - 3 \begin{cases} = -1 & z = 0 \\ \to 1 & z \to \infty \end{cases} \tag{4.2.51}$$

then there is a unique c for each μ. Rewriting (4.2.3) we get (4.2.46). □

Remark 4.2.3. For the case of $c = 1$ we see that (4.2.50) yields $\mathbb{E}\left[\tau_*^{(1)}\right] \approx 0.55$. In frequentistic statistics, we usually prefer an unbiased estimator. So to make $\tau_*^{(c)}$ an unbiased estimator of θ we set $\mu = \mathbb{E}[\theta] = 1/2$. Using Corollary 4.2.2 we thus get $z_* = 1$ and $c \approx 1.269$. So to get an unbiased estimator we choose the stopping rule provided by the weighted linear loss problem, where we put a larger penalty on the event $(\tau > \theta)$ thus resulting in an earlier stopping rule than for the linear loss case. We have computed the value function from (4.2.17) and depicted it in Figure 4.2.1 for the unbiased optimal rule. Furthermore, we have simulated paths and exercised the optimal rule from Corollary 4.2.2 with $\mu = 1/2$ as depicted in Figure 4.2.2. A further discussion and comparison of the optimal stopping rules are done in Chapter 5. △

Instead of considering (4.2.44) as the Lagrangian of the constrained problem in Corollary 4.2.2, we can consider the problem in its own right. Using that τ is finite so $\mathbb{E}[\tau] = \mathbb{E}[B_\tau^2]$ we can note that

$$\frac{c+1}{2}(B_\tau - S_1)^2 + \frac{c-1}{2}B_\tau^2 = \left(\frac{c+1}{2} + \frac{c-1}{2}\right)B_\tau^2 + \frac{c+1}{2}S_1^2 - 2\frac{c+1}{2}B_\tau S_1 \tag{4.2.52}$$

$$= c\left(B_\tau - \frac{c+1}{2c}S_1\right)^2 + \left(\frac{c+1}{2} - c\left(\frac{c+1}{2c}\right)^2\right)S_1^2. \tag{4.2.53}$$

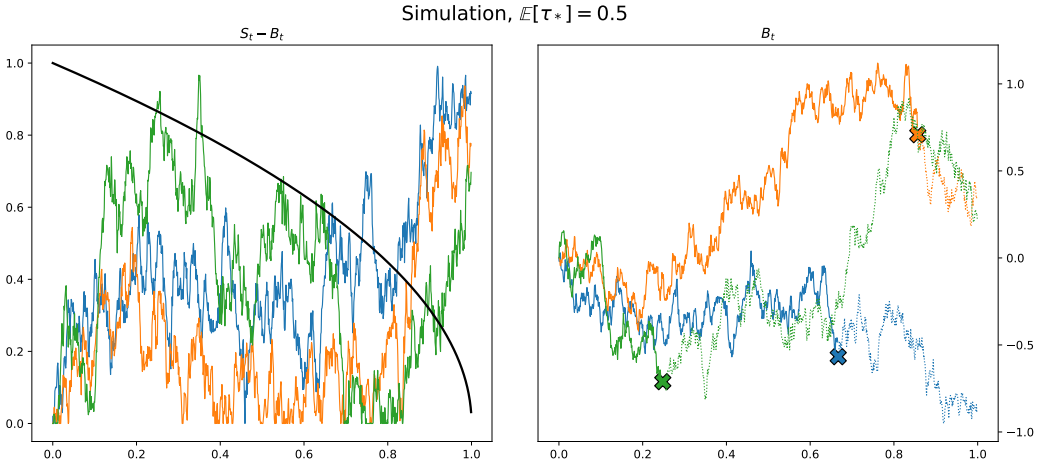


Figure 4.2.2: Simulation of optimal stopping rules when conditioned on $\mu = 1/2$. The left panel depicts simulations of the CUSUM process $S - B$ with $t \mapsto \sqrt{1-t}$ and the right panel is the underlying Brownian paths.

Noting furthermore that $\mathbb{E}[S_1^2] = \mathbb{E}[|B_1|^2] = 1$ we get

$$\begin{aligned}\mathbb{E}\left[c(\tau-\theta)^+ + (\tau-\theta)^-\right] &= c\mathbb{E}\left[\left(B_\tau - \frac{c+1}{2c}S_1\right)^2\right] - \frac{c}{2} + \left(\frac{c+1}{2} - c\left(\frac{c+1}{2c}\right)^2\right)\mathbb{E}[S_1^2] \\ &= c\mathbb{E}\left[\left(B_\tau - \frac{c+1}{2c}S_1\right)^2\right] - \frac{c^2+1}{4c}. \end{aligned} \tag{4.2.54}$$

Note that for $c > 0$ then $\frac{c+1}{2c} \in (1/2, \infty)$. So the weighted linear time problems are equivalent to a squared spatial problem, where instead of considering the maximum of the Brownian we now consider a scaled version.

Corollary 4.2.4. *Consider the family of optimal stopping problems*

$$V_*^{(\alpha)} = \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[(B_\tau - \alpha S_1)^2\right] \tag{4.2.55}$$

for $\alpha > 0$. When $\alpha > 1/2$ then the value $V_*^{(\alpha)}$ is given by

$$V_*^{(\alpha)} = \left(1 + \frac{1}{c}\right)\left(\Phi\left(z_*^{(c)}\right) - 1\right) + \frac{3c^2+1}{4c^2} \tag{4.2.56}$$

with $z_*^{(c)}$ being the unique root of (4.2.3), the optimal stopping time being given by (4.2.4) and $c = (2\alpha - 1)^{-1}$. When $\alpha \leq 1/2$ then $\tau = 0$ is optimal.

Proof. See Appendix B.1. □

Mixed probability and linear loss

As discussed in Section 2.1 the field of quickest detection is interested in the mixed probability and linear loss function. This is from a probabilistic view mainly motivated by being interested in looking at a smaller set of stopping times, namely $\mathcal{T}' = \{\tau \in \mathcal{T} \mid P(\tau < \theta) \leq \alpha\}$, and a loss function $(\tau - \theta)^+$. So we want to condition on a confidence level α that we stop before the event while minimizing the linear loss of stopping too late. Since the loss is minimized when the constraint is optimized this leads to the following optimal prediction problem, by the method of Lagrangian multipliers:

$$W_*^{(\lambda)} = \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[(\tau - \theta)^+ + \lambda \cdot (I(\tau < \theta) - \alpha)\right], \tag{4.2.57}$$

for $\lambda > 0$. Letting $c = \lambda^{-1}$ we can instead solve

$$V_*^{(c)} = \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[c(\tau - \theta)^+ + I(\tau < \theta)\right]. \tag{4.2.58}$$

This is exactly the mixed probability and linear loss. For now, we postpone this problem and as a warm-up, we start by considering the same loss function under the restriction of $\mathcal{T}' = \{\tau \in \mathcal{T} \mid \mathbb{E}[(1-\tau) \cdot I(\tau < \theta)] \leq \alpha\}$. The following theorem solves the Lagrangian problem:

Theorem 4.2.5. *Consider the family of optimal stopping problems*

$$V_*^{(c)} = \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[(1-\tau)I(\tau < \theta) + c(\tau - \theta)^+\right] \tag{4.2.59}$$

for $c \geq 0$. Then the value $V_*^{(c)}$ is given by

$$V_*^{(c)} = c\Phi\left(z_*^{(c)}\right) - \frac{c}{2} - \frac{\varphi\left(z_*^{(c)}\right)}{z_*^{(c)}} \tag{4.2.60}$$

with $z_*^{(c)}$ being the unique strictly positive root of the equation

$$-\varphi\left(z_*^{(c)}\right)\left(1 + (1+c)\left(z_*^{(c)}\right)^2\right) + 2(1+c)z_*^{(c)}\left(\Phi\left(z_*^{(c)}\right) - \frac{1}{2}\right) - z_*^{(c)} = 0 \tag{4.2.61}$$

and the following stopping time is optimal:

$$\tau_*^{(c)} = \inf\{t \in [0,1] \mid S_t - B_t \geq z_*^{(c)}\sqrt{1-t}\}. \tag{4.2.62}$$

Remark 4.2.6. Before the proof note that from a statistical point of view the loss function in (4.2.59) is very non-standard. The loss function is depicted in the left panel of Figure 4.2.3. We have essentially added a loss of $1 - \theta$ on the event $(\tau < \theta)$ in the weighted linear case. This can be realized by noting that

$$(1-\tau)I(\tau < \theta) - (\tau - \theta)^- = (1-\theta)I(\tau < \theta) \tag{4.2.63}$$

hence

$$(1-\tau)I(\tau<\theta)+c(\tau-\theta)^+=(1-\theta)I(\tau<\theta)+(\tau-\theta)^-+c(\tau-\theta)^+. \tag{4.2.64}$$

If θ is large then the added penalty is small and the loss function will be close to the weighted linear loss. △

Proof. As in the proof of Theorem 4.2.1 we fix a $c>0$ and drop the superscript. Likewise,

$$V_*=\inf_{\tau\in\mathcal{T}}\mathbb{E}\left[(1-\tau)I(\tau<\theta)+c(\tau-\theta)^+\right] \tag{4.2.65}$$

$$=\inf_{\tau\in\mathcal{T}}\mathbb{E}\left[(1-\tau)(1-\pi_\tau)+c\int_0^\tau \pi_t\,\mathrm{d}t\right]. \tag{4.2.66}$$

Hence, we will study

$$W_*(z)=\inf_{\sigma\in\mathcal{T}}\mathbb{E}_z\left[e^{-2\sigma}(1-F(|Z_\sigma|))+2c\int_0^\sigma e^{-2s}F(|Z_s|)\,\mathrm{d}s\right] \tag{4.2.67}$$

by utilization of Lemma 3.3.1 as above, where now $F(z)=2\Phi(z)-1$. We now also have a Mayer functional so by Section 3.2 we consider the following Bolza free-boundary problem for some $z_*>0$:

$$\mathbb{L}_Z W(z)-2W(z)=-2cF(|z|), \qquad z\in(-z_*,z_*) \tag{4.2.68}$$

$$W(\pm z_*)=(1-F(z_*)), \qquad \text{(Instantaneous-stopping/Cont. fit)} \tag{4.2.69}$$

$$W'(\pm z_*)=-2\varphi(z_*), \qquad \text{(Smooth-fit)} \tag{4.2.70}$$

$$W'(0)=0, \qquad \text{(Even)}. \tag{4.2.71}$$

Furthermore, we let $W(z)=(1-F(|z|))$ for $|z|>z_*$. The general solution for (4.2.68) on $z\geq 0$ is

$$W(z)=K_1(z^2+1)+K_2\left(z\varphi(z)+\Phi(z)(z^2+1)\right)+\frac{c}{2}F(z)(1-z^2)-cz\varphi(z). \tag{4.2.72}$$

Provided that $z_*>0$ equations (4.2.71) and (4.2.70) yield $K_2=0$ and

$$K_1=\frac{c}{2}F(z_*)-\frac{\varphi(z_*)}{z_*} \tag{4.2.73}$$

respectively. Thus, by continuous-fit we get

$$(1-F(|z_*|))=cF(z_*)-\frac{\varphi(z_*)}{z_*}(z_*^2+1)-cz_*\varphi(z_*) \tag{4.2.74}$$

hence

$$0=-\varphi(z_*)\left(1+(1+c)z_*^2\right)+2(1+c)z_*\left(\Phi(z_*)-\frac{1}{2}\right)-z_*. \tag{4.2.75}$$

Let $f(z) = -\varphi(z)\,(1+(1+c)z^2) + 2(1+c)z\left(\Phi(z) - \frac{1}{2}\right) - z$. It is easily seen that $f(0) = -\varphi(0)$ and $\lim_{z\to\infty} f(z) = \infty$. Studying the monotonicity relations of f further reveals that it has a unique positive root z_*. In particular, we note that $z_* > \Phi^{-1}\left(\frac{c+2}{2(c+1)}\right) =: \underline{z}$ since

$$f(\underline{z}) = -\varphi(\underline{z})\left(1+(1+c)\underline{z}^2\right) + 2(1+c)\underline{z}\left(\frac{c+2}{2(c+1)} - \frac{1}{2}\right) - \underline{z} \tag{4.2.76}$$

$$= -\varphi(\underline{z})\left(1+(1+c)\underline{z}^2\right) \tag{4.2.77}$$

$$< 0, \tag{4.2.78}$$

and $\underline{z} > 0$. The final solution to the free-boundary problem is thus

$$W(|z|) = \begin{cases} 1 - F(|z|), & |z| \geq z_* \\ \frac{c}{2}\left(F(z_*)(z^2+1) + F(|z|)(1-z^2)\right) - \frac{\varphi(z_*)}{z_*}(z^2+1) - c\,|z|\,\varphi(|z|) & |z| < z_*, \end{cases} \tag{4.2.79}$$

which needs to be verified. The Itô-Tanaka formula yields

$$e^{-2s}W(Z_s) = W(Z_0) + \int_0^s e^{-2u}\left(\mathbb{L}_Z W(Z_u) - 2W(Z_u)\right)\mathrm{d}u + M_s, \tag{4.2.80}$$

where M is a particular (local) martingale similar to the process in the proof of Theorem 4.2.1. For $|z| > z_*$ we have

$$\mathbb{L}_Z W(z) - 2W(z) = 4\Phi(|z|) - 4. \tag{4.2.81}$$

As noted above $|z| > z_* > \Phi^{-1}\left(\frac{c+2}{2(c+1)}\right)$, so $(1+c)\Phi(|z|) > 1 + \frac{c}{2}$ hence

$$\mathbb{L}_Z W(z) - 2W(z) > -4c\Phi(|z|) + 2c = -2cF(|z|). \tag{4.2.82}$$

So, by (4.2.68) we have

$$\mathbb{L}_Z W(z) - 2W(z) \geq -2cF(|z|) \tag{4.2.83}$$

for $z \in \mathbb{R} \setminus \{\pm z_*\}$. Note furthermore by construction $W(z) \leq 1 - F(|z|)$ for all z hence from (4.2.80) we get

$$e^{-2s}(1 - F(|Z_s|)) \geq W(Z_0) - 2c\int_0^s e^{-2u}F(|Z_u|)\,\mathrm{d}u + M_s \tag{4.2.84}$$

by an extension to the null set ($Z \in \{\pm z_*\}$). Hence, by the optional sampling theorem, we get

$$\mathbb{E}_z\left[e^{-2s}(1 - F(|Z_s|)) + 2c\int_0^s e^{-2u}F(|Z_u|)\,\mathrm{d}u\right] \geq W(z) \tag{4.2.85}$$

and so replacing s by σ and taking infimum yields $W_*(z) \geq W(z)$. For the converse we define $\sigma_* = \inf\{s \geq 0 \mid |Z_s| \geq z_*\}$ and so analogous to Theorem 4.2.1 we get from (4.2.80) that

$$W_*(z) \leq \mathbb{E}_z\left[e^{-2\sigma_*}(1 - F(|Z_{\sigma_*}|)) + 2c\int_0^{\sigma_*} e^{-2u}F(|Z_u|)\,\mathrm{d}u - M_{\sigma_*}\right] = \mathbb{E}_z\left[W(Z_0)\right] = W(z) \tag{4.2.86}$$

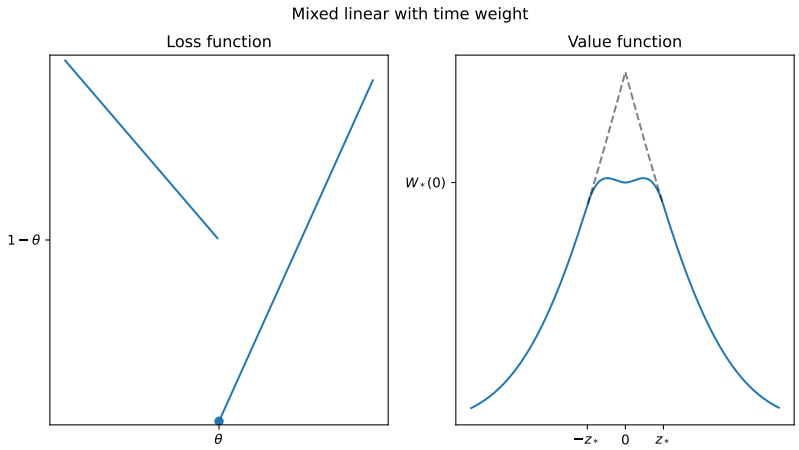


Figure 4.2.3: Left panel: Mixed linear time-weighted probability loss. Right panel: Value function from (4.2.79), when $\alpha = 1/2$. The dashed line is $1 - F(|z|)$.

by continuity of Z, continuous-fit, and (4.2.68). Reversing the time-change and evaluating W in zero we get the wanted. □

It should be noted that in the proof we have disregarded that M is only a local martingale. However, taking a localizing sequence of bounded stopping times, using Fatou's lemma and dominated convergence theorem as in the proof of Theorem 4.2.1, we get the same conclusion. We have computed and depicted the value function in the right panel of Figure 4.2.3.

To solve the original constrained problem we do as in Corollary 4.2.2.

Corollary 4.2.7. *Let $\alpha \in (0,1)$ and consider the optimal prediction problem*

$$V_*^{(\alpha)} = \inf_{\tau \in \mathcal{T}_\alpha} \mathbb{E}[(\tau - \theta)^+] \tag{4.2.87}$$

where $\mathcal{T}_\alpha = \{\tau \in \mathcal{T} \mid \mathbb{E}[(1-\tau)I(\tau < \theta)] \leq \alpha\}$. Then the value $V_*^{(\alpha)}$ is given by

$$V_*^{(\alpha)} = \Phi\left(z_*^{(\alpha)}\right) - \frac{1}{2} - \frac{\varphi\left(z_*^{(\alpha)}\right)}{c z_*^{(\alpha)}} - \frac{\alpha}{c} \tag{4.2.88}$$

with $z_*^{(\alpha)}$ being the unique strictly positive root of

$$0 = \frac{2 - 2\Phi\left(z_*^{(\alpha)}\right)}{1 + \left(z_*^{(\alpha)}\right)^2} - \alpha \tag{4.2.89}$$

and $c > 0$ is given by

$$c = \frac{\varphi\left(z_*^{(\alpha)}\right) + z_*^{(\alpha)}}{2z_*^{(\alpha)}\left(\Phi\left(z_*^{(\alpha)}\right) - \frac{1}{2}\right) - \varphi\left(z_*^{(\alpha)}\right)\left(z_*^{(\alpha)}\right)^2} - 1 \tag{4.2.90}$$

and the following stopping time is optimal:

$$\tau_*^{(\alpha)} = \inf\{t \in [0,1] \mid S_t - B_t \geq z_*^{(\alpha)}\sqrt{1-t}\}. \tag{4.2.91}$$

Proof. Note by the considerations in the introduction of this subsection we have the Lagrangian formulation

$$W_*^{(\lambda)} = \lambda\left(\tilde{V}_*^{(c)} - \alpha\right) \tag{4.2.92}$$

where $c = \lambda^{-1} > 0$ and $\tilde{V}_*^{(c)}$ is the value from Theorem 4.2.5. By the tower property, Lemma 4.1.1, (4.2.62), and (4.2.50) we have that

$$\mathbb{E}[(1-\tau_*^{(c)})I(\tau_*^{(c)} \leq \theta)] = \mathbb{E}\left[(1-\tau_*^{(c)})\left(2 - 2\Phi\left(\frac{S_{\tau_*^{(c)}} - B_{\tau_*^{(c)}}}{\sqrt{1-\tau_*^{(c)}}}\right)\right)\right] \tag{4.2.93}$$

$$= \left(1 - \mathbb{E}\left[\tau_*^{(c)}\right]\right)\left(2 - 2\Phi\left(z_*^{(c)}\right)\right) \tag{4.2.94}$$

$$= \frac{2 - 2\Phi\left(z_*^{(c)}\right)}{1 + \left(z_*^{(c)}\right)^2}. \tag{4.2.95}$$

Letting $f(z) := \frac{2-2\Phi(z)}{1+z^2}$ we see that the initial point is $f(0) = 1$, the limit is $\lim_{z\to\infty} f(z) = 0$ and the derivative obeys

$$f'(z) = -2\left(\frac{\varphi(z)}{1+z^2} + \frac{z(2-2\Phi(z))}{(1+z^2)^2}\right) < 0 \tag{4.2.96}$$

for all $z > 0$. Hence, there is a unique solution $z_*^{(\alpha)} > 0$ to (4.2.89) for any $\alpha \in (0,1)$. Rewriting (4.2.61) in terms of c we get (4.2.90) by noting that

$$g(z) := 2z\left(\Phi(z) - \frac{1}{2}\right) - \varphi(z)\, z^2 > 0 \tag{4.2.97}$$

for $z > 0$ since $g'(z) = 2\Phi(z) - 1 + z^3\varphi(z) > 0$ and $g(0) = 0$. Finally, combining (4.2.92), (4.2.60), and the choice of λ we get (4.2.88) as wanted. □

Now let us consider the problem (4.2.58). As discussed the loss function stems from the field of quickest detection. In that setting, it can be shown that the loss function is equivalent to

a weighted linear loss function due to the distribution of the disorder time and the resulting distribution of the posterior process (see Shiryaev, 2004).

However, in our case of optimal prediction, we are in no such luck. We make a remark on the special case of $\mathcal{T}'$ being deterministic times to see that this problem has some intricacies.

Remark 4.2.8. Recall the density and distribution of θ form (2.2.4). Define for $t \in (0,1)$

$$V(t,c) := P(t < \theta) + c\mathbb{E}\left[(t-\theta)^+\right] \tag{4.2.98}$$

$$= 1 - \frac{2}{\pi}\arcsin\sqrt{t} + c\int_0^t \frac{t-y}{\pi\sqrt{y(1-y)}}\,\mathrm{d}y \tag{4.2.99}$$

$$= 1 - \frac{2}{\pi}\arcsin\sqrt{t} + c\frac{(2t-1)\arcsin(2t-1) + 2\sqrt{t(1-t)}}{2\pi} + c\frac{2t-1}{4} \tag{4.2.100}$$

by a lengthy calculation. Note that for instant stopping we have $V(0,c) := \lim_{t\searrow 0} V(t,c) = 1$ while terminal stopping yields $V(1,c) := \lim_{t\nearrow 1} V(t,c) = \frac{c}{2}$. The monotonicity relations of this function are very ill-behaved in that for some values of c it has only a minimum in $t = 1$, for other values it has global minimum in $t = 1$ but a local minimum in a $t \neq 1$ while the case of a global minimum at $t \neq 1$ is also possible for some c. The interior extremes can be characterized by the first-order condition

$$c^{-1} = \sqrt{t(1-t)}\left(\frac{2}{\pi} + \arcsin(2t-1)\right). \tag{4.2.101}$$

To find the minimal c for which the interior minimum of V is equal to the minimum along the boundary, we can add the condition $V(t,c) = \frac{c}{2}$ and solve the nonlinear system numerically. We find a solution $\bar{c} \approx 1.455$ with $t \approx 0.413$. We have plotted the value function for a range of c parameters in Figure 4.2.4. So in the deterministic case the optimal stopping time is $t = 1$ for $c < \bar{c}$ and for $c \geq \bar{c}$ the optimal stopping time is given by the minimal root with $t \in (0,1)$ of (4.2.101). △

The result from Remark 4.2.8 is somewhat unexpected since a small change in the weight of the loss function can change the optimal deterministic strategy from a large positively biased estimation to a slight negatively biased estimator. Remarkably the $\bar{c}$ constant also seems to play a role in the general stopping time case as we will now discuss.

We first need to formulate the optimal prediction problem (4.2.58) as an optimal stopping problem. With the usual tricks, we have

$$V_*^{(c)} = \inf_{\tau\in\mathcal{T}} \mathbb{E}\left[M(\tau, |B_\tau|) + \int_0^\tau L^{(c)}(s, B_s)\,\mathrm{d}s\right] \tag{4.2.102}$$

where

$$M(t,x) = 2 - 2\Phi\left(\frac{x}{\sqrt{1-t}}\right) \tag{4.2.103}$$

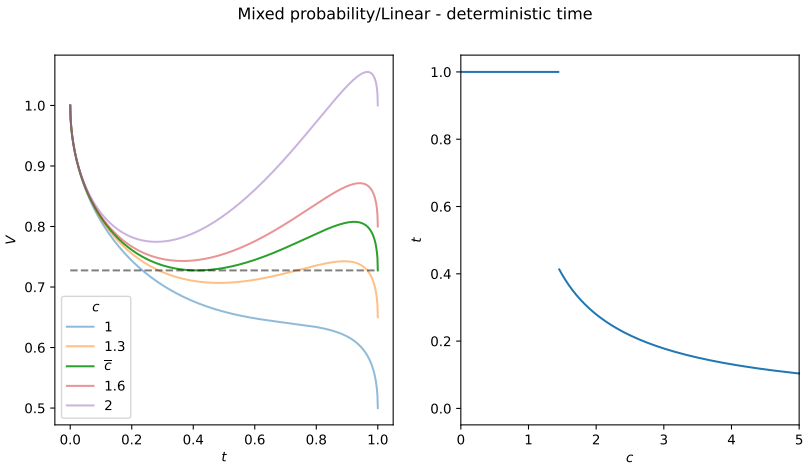


Figure 4.2.4: Value function V from Remark 4.2.8

and

$$L^{(c)}(t,x) = c\left(2\Phi\left(\frac{x}{\sqrt{1-t}}\right) - 1\right). \tag{4.2.104}$$

Since the problem is time-inhomogeneous we can formulate the equivalent optimal stopping problem

$$V_*^{(c)}(t,x) = \inf_{\tau \leq 1-t} \mathbb{E}_{t,x}\left[M(t+\tau, |B_{t+\tau}|) + \int_0^\tau L^{(c)}(t+s, B_{t+s})\,\mathrm{d}s\right] \tag{4.2.105}$$

where $B_t = x$ under $P_{t,x}$ for $(t,x) \in [0,1] \times \mathbb{R}_+$. We want to derive a free-boundary equation, so we take Assumptions 3.4.1 as given.

Under these assumptions, we can derive a free-boundary equation to determine b numerically since we have the following free-boundary problem for $(t,x) \in [0,1] \times \mathbb{R}_+$:

$$\mathbb{L}_Z V_*^{(c)}(t,x) = -L^{(c)}(t,x) \qquad \text{for } x < b(t) \tag{4.2.106}$$

$$V_*^{(c)}(t,x) < M(t,x) \qquad \text{for } x < b(t) \tag{4.2.107}$$

$$V_*^{(c)}(t,x) = M(t,x) \qquad \text{for } x \geq b(t) \tag{4.2.108}$$

$$\frac{\partial V_*^{(c)}}{\partial x}(t,x) = M_x(t,x) \qquad \text{for } x = b(t), \tag{4.2.109}$$

where $\mathbb{L}_Z$ is the infinitesimal generator of the process $Z = (Z_t)_{t\in[0,1]}$ given by $Z_t = (t, |B_t|)$ i.e.

$$\mathbb{L}_Z = \frac{\partial}{\partial t} + \frac{1}{2}\frac{\partial^2}{\partial x^2}. \tag{4.2.110}$$

Applying the results from Section 3.5 we thus get the free-boundary equation

$$\mathbb{E}_{t,b(t)}\left[M(1,|B_1|)\right] = M(t,b(t)) \tag{4.2.111}$$

$$+ \int_0^{1-t} \mathbb{E}_{t,b(t)}\left[\mathbb{L}_Z M(t+s,|B_{t+s}|) I(|B_{t+s}| > b(t+s))\right] \mathrm{d}s \tag{4.2.112}$$

$$- \int_0^{1-t} \mathbb{E}_{t,b(t)}\left[L_*^{(c)}(t+s,|B_{t+s}|) I(|B_{t+s}| < b(t+s))\right] \mathrm{d}s \tag{4.2.113}$$

for $t \in [0,1]$. But note that

$$\mathbb{L}_Z M(t,x) = M_t(t,x) + \frac{1}{2} M_{xx}(t,x) = 0 \tag{4.2.114}$$

and furthermore $M(1,x) = 0$ for all $x \in \mathbb{R}_+$ hence we have

$$M(t,b(t)) = \int_0^{1-t} \mathbb{E}_{t,b(t)}\left[L^{(c)}(t+s,|B_{t+s}|) I(|B_{t+s}| < b(t+s))\right] \mathrm{d}s \tag{4.2.115}$$

or equivalently

$$M(t,b(t)) = \int_0^{1-t} \int_0^{b(t+s)} L^{(c)}(t+s,y) p(s; b(t), y) \,\mathrm{d}y \,\mathrm{d}s \tag{4.2.116}$$

where $p(t,x,y)$ is the transition density of the reflected Brownian motion $|B|$ given in (4.1.10).

We have computed and depicted the numerical optimal boundary in Figure 4.2.5. The weight $\bar{c}$ from Remark 4.2.8 appears to be an important threshold for the general case, since we get that the boundary is strictly decreasing for $c < \bar{c}$ while for $c > \bar{c}$ the boundary appears to have a *hump*. Thus, as in the case of deterministic stopping the problem is to some extent easier for small weights than for large. Having a non-monotone boundary may seem somewhat unintuitive since achieving an optimal exiting value at some time point may be suboptimal at a later time, i.e. even though less time is left the process is more likely to achieve an even more preferable level.

If we again consider (4.2.102) to be the Lagrangian associated with

$$V_*^\alpha = \inf_{\tau \in \mathcal{T}_\alpha} \mathbb{E}\left[(\tau - \theta)^+\right] \tag{4.2.117}$$

for $\mathcal{T}_\alpha = \{\tau \in \mathcal{T} \mid P(\tau < \theta) \leq \alpha\}$ then we see that doing as in Corollary 4.2.7 is not analytically tractable since we have to determine a c such that α equals

$$P(\tau^{(c)} < \theta) = \mathbb{E}\left[2 - 2\Phi\left(\frac{S_{\tau^{(c)}} - B_{\tau^{(c)}}}{\sqrt{1-\tau^{(c)}}}\right)\right] = 2 - 2\mathbb{E}\left[\Phi\left(\frac{b(\tau^{(c)})}{\sqrt{1-\tau^{(c)}}}\right)\right] \tag{4.2.118}$$

by the tower property, Lemma 4.1.2, and definition of the optimal stopping time $\tau^{(c)}$ in (4.2.102). Such a c does not necessarily exist and one would need to continue numerically by estimating the expectation. As an alternative one could consider a loss function under the same constraint as we do in the proceeding section.

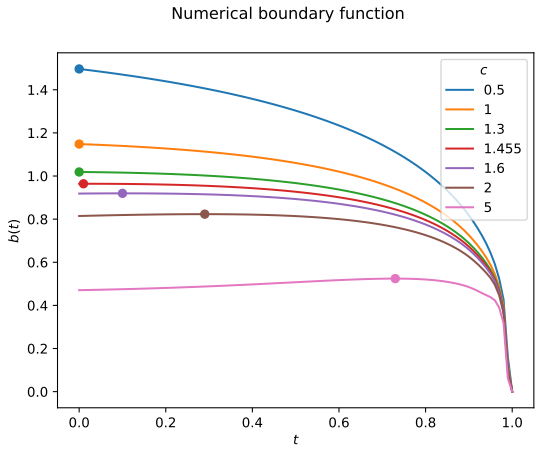


Figure 4.2.5: Numerical solutions to free-boundary equation (4.2.116) for different c parameters. The parameter $1.455 \approx \bar{c}$ is from Remark 4.2.8. The dots are the numerical maxima of the function b.

4.2.2 Logarithmic Time Problem

In the linear time problems, we have a killing factor in the Lagrangian functional. This made the problem in Theorem 4.2.5 one dimensional after the deterministic time-change. However, as noted the set of stopping times where $\mathbb{E}\left[(1-\tau)\cdot I(\tau \leq \theta)\right] \leq \alpha$ is not as easy to interpret as considering a problem where we condition on $P(\tau < \theta) \leq \alpha$. The idea in the following is to remove the killing term in the Lagrangian functional, which can be done by considering an alternative loss function.

Consider the entropy loss function. For convenience, we start with unit weights. Although the loss function is less standard than the linear loss it should be clear that the loss is zero when $\tau = \theta$ and there is a large penalty, tending to infinity, if $\tau \to 1$. Hence, in some sense, this loss function has a higher degree of asymmetric behavior.

As in the linear case, we see that

$$\left|\log\left(\frac{1-\tau}{1-\theta}\right)\right| = I(\theta \leq \tau)\log\left(\frac{1-\theta}{1-\tau}\right) + I(\tau < \theta)\log\left(\frac{1-\tau}{1-\theta}\right) \tag{4.2.119}$$

$$= \int_0^\tau \frac{I(\theta \leq t)}{1-t}\,\mathrm{d}t - \log(1-\theta) - \int_0^\tau \frac{I(\theta > t)}{1-t}\,\mathrm{d}t \tag{4.2.120}$$

$$= -\log(1-\theta) + \int_0^\tau \frac{2I(\theta \leq t) - 1}{1-t}\,\mathrm{d}t. \tag{4.2.121}$$

Taking expectation we get

$$\mathbb{E}\left[\left|\log\left(\frac{1-\tau}{1-\theta}\right)\right|\right] = -\mathbb{E}\left[\log(1-\theta)\right] + \mathbb{E}\left[\int_0^\infty \frac{2\mathbb{E}\left[I(\theta \le t) \mid \mathcal{F}_t\right] - 1}{1-t} I(t \le \tau)\,\mathrm{d}t\right] \tag{4.2.122}$$

$$= -\mathbb{E}\left[\log\theta\right] + \mathbb{E}\left[\int_0^\tau \frac{2\pi_t - 1}{1-t}\,\mathrm{d}t\right] \tag{4.2.123}$$

since $\theta \overset{d}{=} 1-\theta$ and τ is a stopping time. From the density of θ, see (2.2.4), it can be seen that

$$\mathbb{E}\left[\log\theta\right] = \Psi\left(\frac{1}{2}\right) - \Psi\left(1\right) \tag{4.2.124}$$

where Ψ is the digamma function, by using

$$\log(t)(t^{\alpha-1}(1-t)^{\beta-1}) = \frac{\partial}{\partial\alpha}(t^{\alpha-1}(1-t)^{\beta-1}) \tag{4.2.125}$$

and the relations between the beta and digamma functions (see e.g. Polyanin and Zaitsev, 2017). Specifically $\mathbb{E}[\log\theta]$ is finite.

Recall $\pi_{T_s} \overset{d}{=} 2\Phi(|Z_s|) - 1$ where Z is the Ornstein-Uhlenbeck process from (3.3.19) and $T_s = 1 - e^{-2s}$. Making the deterministic time-change we have

$$\mathbb{E}\left[\int_0^\tau \frac{2\pi_t - 1}{1-t}\,\mathrm{d}t\right] = \mathbb{E}\left[\int_0^{\sigma_\tau} \frac{4\Phi(|Z_s|) - 3}{1-(1-e^{-2s})} 2e^{-2s}\,\mathrm{d}s\right] \tag{4.2.126}$$

$$= 2\mathbb{E}\left[\int_0^{\sigma_\tau} \Big(4\Phi(|Z_s|) - 3\Big)\,\mathrm{d}s\right], \tag{4.2.127}$$

where σ_τ is as in Theorem 4.2.1. So by the time-change being a bijection, we are thus let to study the optimal stopping problem

$$W_*(z) = \inf_{\sigma\in\mathcal{S}} \mathbb{E}\left[\int_0^\sigma \Big(4\Phi(|Z_s|) - 3\Big)\,\mathrm{d}s\right]. \tag{4.2.128}$$

If we instead consider the weighted entropy prediction problem then, by a calculation similar to (4.2.11)-(4.2.13), we are led to study the family of optimal stopping problems

$$W_*^{(c)}(z) = \inf_{\sigma\in\mathcal{S}} \mathbb{E}\left[\int_0^\sigma \left(4\Phi(|Z_s|) - 3 + \frac{c-1}{c+1}\right)\mathrm{d}s\right] \tag{4.2.129}$$

where $c > 0$, and we have

$$V_*^{(c)} := \inf_{\tau\in\mathcal{T}} \mathbb{E}\left[c\left(\log\left(\frac{1-\tau}{1-\theta}\right)\right)^- + \left(\log\left(\frac{1-\tau}{1-\theta}\right)\right)^+\right] \tag{4.2.130}$$

$$= -\mathbb{E}\left[\log\theta\right] + (c+1)W_*^{(c)}(0). \tag{4.2.131}$$

Theorem 4.2.9. *Consider the family of optimal prediction problems (4.2.130) for $c > 0$. The value function is given by (4.2.131) where $W_*^{(c)}$ is the solution to the optimal stopping problem in (4.2.129) which is given by*

$$W_*(z) = \begin{cases} -\sqrt{2\pi} \displaystyle\int_0^{|z|} \int_0^x \varphi(x) e^{\frac{y^2}{2}} \left(4\Phi(y) - 3 + \frac{c-1}{c+1} \right) \mathrm{d}x \, \mathrm{d}y + k & |z| < z_* \\ 0 & |z| \geq z_*, \end{cases} \tag{4.2.132}$$

where z_* is the unique strictly positive solution to

$$\frac{3 - \frac{c-1}{c+1}}{4} \int_0^{z_*} e^{\frac{y^2}{2}} \, \mathrm{d}y = \int_0^{z_*} e^{\frac{y^2}{2}} \Phi(y) \, \mathrm{d}y \tag{4.2.133}$$

and k is given by

$$k = \sqrt{2\pi} \int_0^{z_*} \int_0^x \varphi(x) e^{\frac{y^2}{2}} \left(4\Phi(y) - 3 + \frac{c-1}{c+1} \right) \mathrm{d}y \, \mathrm{d}x. \tag{4.2.134}$$

The optimal prediction rule is given by

$$\tau_* = \inf \left\{ t \in [0,1] \,\middle|\, S_t - B_t \geq z_* \sqrt{1-t} \right\}. \tag{4.2.135}$$

Proof. Fix an arbitrary $c > 0$. Considering the optimal stopping problem (4.2.129) we naturally look at the following Dirichlet/Poisson free-boundary problem for some $z_* > 0$:

$$\mathbb{L}_Z W(z) = -F(|z|), \qquad |z| < z_* \tag{4.2.136}$$
$$W(z_*) = 0, \qquad \text{(Instantaneous-stopping/Cont. fit)} \tag{4.2.137}$$
$$W'(z_*) = 0, \qquad \text{(Smooth-fit)} \tag{4.2.138}$$
$$W'(0) = 0, \qquad \text{(Even)} \tag{4.2.139}$$

where $F(z) = 4\Phi(z) - 3 + \frac{c-1}{c+1}$.

The general solution to (4.2.136) when restricting to $z \geq 0$ is

$$W(z) = K_1 \Phi(z) - \sqrt{2\pi} \int_0^z \int_0^x \varphi(x) e^{\frac{y^2}{2}} F(y) \, \mathrm{d}y \, \mathrm{d}x + K_2. \tag{4.2.140}$$

Using (4.2.139) we get $K_1 = 0$. In the following we denote $\alpha = 3 - \frac{c-1}{c+1} \in (2,4)$. Note that

$$W'(z) = -\sqrt{2\pi}\varphi(z) \int_0^z e^{\frac{y^2}{2}} F(y) \, \mathrm{d}y = \sqrt{2\pi}\varphi(z) \left(\alpha \int_0^z e^{\frac{y^2}{2}} \, \mathrm{d}y - 4 \int_0^z e^{\frac{y^2}{2}} \Phi(y) \, \mathrm{d}y \right). \tag{4.2.141}$$

So by (4.2.138) we need to find $z_* > 0$ such that $f(z_*) = 0$ where f is defined by

$$f(z) := \int_0^z e^{\frac{y^2}{2}} \left(\frac{\alpha}{4} - \Phi(y)\right) \mathrm{d}y. \tag{4.2.142}$$

Differentiating we see

$$f'(z) = e^{\frac{z^2}{2}} \left(\frac{\alpha}{4} - \Phi(z)\right) \begin{cases} \geq 0 & z \leq \Phi^{-1}\left(\frac{\alpha}{4}\right) \\ < 0 & z > \Phi^{-1}\left(\frac{\alpha}{4}\right), \end{cases} \tag{4.2.143}$$

where we note $\alpha/4 \in (1/2, 1)$ so specifically by $f(0) = 0$ we know a strictly positive solution to (4.2.133) is unique provided $f < 0$ at some point. For this note that the integrant in the definition of f obeys

$$e^{\frac{y^2}{2}} \left(\frac{\alpha}{4} - \Phi(y)\right) \leq -\varepsilon e^{\frac{y^2}{2}} \tag{4.2.144}$$

for $y \geq \Phi^{-1}(1/4 + \varepsilon)$ which diverges to $-\infty$ implying that $f(z)$ is negative for z large enough. Finally, the determination of $K_2 = k$ from (4.2.134) is thus obvious by the continuous-fit condition.

It only remains to verify the proposed solution after extending W to be symmetric around $z = 0$ and $W(z) = 0$ for $|z| \geq z_*$. Since $W \in C^1(\mathbb{R})$ and $W \in C^2(\mathbb{R} \setminus \{\pm z_*\})$ then Itô-Tanaka yields

$$W(Z_s) = W(Z_0) + \int_0^s \mathbb{L}_Z W(Z_u)\, \mathrm{d}u + \sqrt{2} \int_0^s W'(Z_u)\, \mathrm{d}\beta_u \tag{4.2.145}$$

where the last term is a continuous centered (local) martingale which we denote M. On $z \notin [-z_*, z_*]$ we have $\mathbb{L}_Z W(z) = 0$ and $|z| > z_* > \Phi^{-1}\left(\frac{\alpha}{4}\right)$ hence $F(|z|) \geq 0$. So using that W solves (4.2.136) on $|z| < z_*$ we have

$$W(Z_s) \geq W(Z_0) - \int_0^s F(|Z_u|)\, \mathrm{d}u + M_s. \tag{4.2.146}$$

Noting that $W(Z_\sigma) \leq 0$ for any stopping time σ we get by optional sampling when replacing s by σ in (4.2.146) that

$$W(z) \leq \mathbb{E}\left[\int_0^\sigma F(|Z_u|)\, \mathrm{d}u\right] \tag{4.2.147}$$

and taking infimum that $W(z) \leq W_*(z)$.

For the converse, define $\sigma_* = \inf\{s \in \mathbb{R}_+ \mid |Z| \geq z_*\}$ which is almost surely finite. We get

$$W(Z_0) = W(Z_{\sigma_*}) + \int_0^{\sigma_*} F(|Z_u|)\, \mathrm{d}u - M_{\sigma_*}, \tag{4.2.148}$$

by (4.2.145) and $u \leq \sigma_*$ implying that $|Z_u| < z_*$. Taking expectation, utilizing the optional sampling theorem, using $W(Z_{\sigma_*}) \leq 0$, and taking infimum we get $W(z) \leq W_*(z)$. Thus, $W(z) = W_*(z)$ and σ_* is optimal, so we are done. □

We are now ready to solve the mixed probability and entropy loss case.

Theorem 4.2.10. *Consider the family of optimal prediction problems with mixed probability and entropy loss*

$$W_*^{(c)} = \inf_{\tau \in \mathcal{T}} P(\tau < \theta) + c\mathbb{E}\left[\left(\log\left(\frac{1-\tau}{1-\theta}\right)\right)^-\right] \tag{4.2.149}$$

for $c > 0$. The value is given by

$$W_*^{(c)} = 2 - 2\Phi(z_*) + 2c\sqrt{2\pi}\int_0^{z_*}\int_0^x \varphi(x)e^{\frac{y^2}{2}}\left(2\Phi(y) - 1\right)\mathrm{d}y\,\mathrm{d}x \tag{4.2.150}$$

and z_* is the unique strictly positive solution to

$$\int_0^{z_*} e^{\frac{y^2}{2}}\left(\frac{1}{2} - \Phi(y)\right)\mathrm{d}y = -\frac{1}{2c\sqrt{2\pi}} \tag{4.2.151}$$

and the following stopping time is optimal:

$$\tau_*^{(\alpha)} = \inf\{t \in [0,1] \mid S_t - B_t \geq z_*^{(\alpha)}\sqrt{1-t}\}. \tag{4.2.152}$$

Proof. By a deterministic time-change, we are led to study

$$W_*(z) = \inf_{\sigma \in \mathcal{S}} \mathbb{E}\left[1 - F(|Z_\sigma|) + \int_0^\sigma 2c\left(F(|Z_s|)\,\mathrm{d}s\right)\right] \tag{4.2.153}$$

where $F(z) = 2\Phi(z) - 1$. The reasoning above yields the following natural Bolza free-boundary problem for some $z_* > 0$:

$$\mathbb{L}_Z W(z) = -2cF(|z|), \qquad |z| < z_* \tag{4.2.154}$$
$$W(z_*) = 1 - F(z_*), \qquad \text{(Instantaneous-stopping/Cont. fit)} \tag{4.2.155}$$
$$W'(z_*) = -2\varphi(z_*), \qquad \text{(Smooth-fit)} \tag{4.2.156}$$
$$W'(0) = 0, \qquad \text{(Even)}. \tag{4.2.157}$$

As before considering $z \geq 0$ the solution to (4.2.154) in combination with (4.2.157) yields

$$W(z) = -2c\sqrt{2\pi}\int_0^z\int_0^x \varphi(x)e^{\frac{y^2}{2}}F(y)\,\mathrm{d}y\,\mathrm{d}x + K. \tag{4.2.158}$$

We denote $f(z) := \int_0^z e^{\frac{y^2}{2}}\left(\frac{1}{2} - \Phi(y)\right)\mathrm{d}y$. To have a smooth-fit we must then have

$$f(z_*) = -\frac{1}{2c\sqrt{2\pi}}. \tag{4.2.159}$$

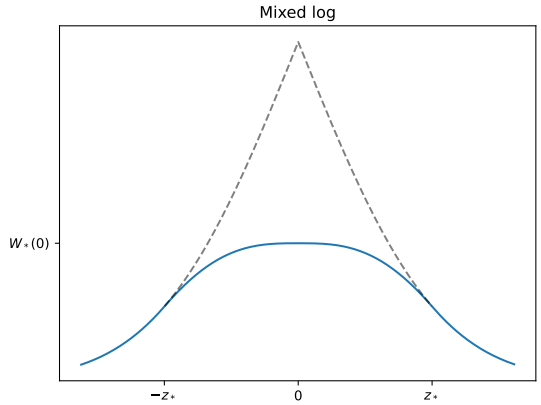


Figure 4.2.6: Value function from (4.2.162) for $c = 1/2$.

An argument similar to that of (4.2.143) and (4.2.144) implies $f(0) = 0$, $f'(z) < 0$ for $z > 0$, and

$$\lim_{z\to\infty} f(z) = -\infty. \tag{4.2.160}$$

Thus, a unique solution to (4.2.159) exists for any $c > 0$ and hence K can be determined by

$$K = 2 - 2\Phi(z_*) + 2c\sqrt{2\pi}\int_0^{z_*}\int_0^x \varphi(x)e^{\frac{y^2}{2}}F(y)\,\mathrm{d}y\,\mathrm{d}x. \tag{4.2.161}$$

The verification is exactly like that of Theorem 4.2.9 except we note that for $|z| > z_*$ then $W(z) = 1 - F(|z|)$ hence $\mathbb{L}_Z W(z) = 0 \geq -2cF(|z|)$. Thus,

$$W_*(z) = \begin{cases} -2c\sqrt{2\pi}\displaystyle\int_0^{|z|}\int_0^x \varphi(x)e^{\frac{y^2}{2}}F(y)\,\mathrm{d}y\,\mathrm{d}x + K & |z| < z_* \\ 2 - 2\Phi(|z|) & |z| \geq z_*. \end{cases} \tag{4.2.162}$$

Hence, we are done. □

We have computed and depicted the value function in Figure 4.2.6. We now easily get the solution to the constrained problem in the entropy loss case.

Corollary 4.2.11. *Consider the family of optimal prediction problems*

$$V_*^{(\alpha)} = \inf_{\tau\in\mathcal{T}_\alpha} \mathbb{E}\left[\left(\log\left(\frac{1-\tau}{1-\theta}\right)\right)^-\right] \tag{4.2.163}$$

for $\alpha \in (0,1)$ where $\mathcal{T}_\alpha = \{\tau \in \mathcal{T} \mid P(\tau < \theta) \leq \alpha\}$. Then the value $V_*^{(\alpha)}$ is given by

$$V_*^{(\alpha)} = \frac{1}{c}\left(2 - 2\Phi(z_*) + 2c\sqrt{2\pi}\int_0^{z_*}\int_0^x \varphi(x)e^{\frac{y^2}{2}}\left(2\Phi(y) - 1\right)\mathrm{d}y\,\mathrm{d}x - \alpha\right) \tag{4.2.164}$$

with $z_*^{(\alpha)} > 0$ given by

$$z_*^{(\alpha)} = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right) \tag{4.2.165}$$

and $c > 0$ is given by

$$c = -\left(\sqrt{2\pi}\left(\int_0^{z_*} e^{\frac{y^2}{2}}\left(1 - 2\Phi(y)\right)\mathrm{d}y\right)\right)^{-1} \tag{4.2.166}$$

and the following stopping time is optimal:

$$\tau_*^{(\alpha)} = \inf\{t \in [0,1] \mid S_t - B_t \geq z_*^{(\alpha)}\sqrt{1-t}\}. \tag{4.2.167}$$

Proof. Recall for $\tau_* = \inf\{t \in [0,1] \mid S_t - B_t \geq z_*\sqrt{1-t}\}$ then

$$P(\tau_* \leq \theta) = \mathbb{E}\left[2 - 2\Phi\left(\frac{S_{\tau_*} - B_{\tau_*}}{\sqrt{1-\tau_*}}\right)\right] = 2 - 2\Phi(z_*). \tag{4.2.168}$$

Hence, by the method of Lagrangian multipliers and Theorem 4.2.10 we are done. □

4.3 Space Problems

We now move on to consider spatial optimal prediction problems from Definition 2.2.2. Recall that since we consider the statistical space problem we are interested in estimation of S_1 and taking action upon an updated estimate of S_1, rather than selling an asset at the optimal time. We start by commenting on the problem in a polynomial loss setting and then go to the asymmetric case of the LINEX loss.

4.3.1 Polynomial Space Problems

We start by considering the squared loss in the statistical space problem. This was also remarked in Graversen et al. (2001).

Theorem 4.3.1. *Consider the squared statistical spatial optimal prediction problem*

$$V_* = \inf_{\tau \in \mathcal{T},\, \mu \in \mathbb{R}} \mathbb{E}\left[\left(S_1 - (B_\tau + \mu)\right)^2\right]. \tag{4.3.1}$$

The value is given by $V_* = W_* - 2/\pi$ where W_* and the optimal prediction rule is given by Theorem 4.2.1 with $c = 1$. The optimal offset for the estimator is $\mu_* = \sqrt{2/\pi}$.

Proof. Note that

$$(S_1 - (B_\tau + \mu))^2 = (S_1 - B_\tau)^2 + \mu^2 + 2\mu(B_\tau - S_1). \tag{4.3.2}$$

Taking expectation we note that the variables separate additively i.e.

$$\begin{aligned} V_* &= \inf_{\tau \in \mathcal{T},\, \mu \in \mathbb{R}} \left(\mathbb{E}\left[(S_1 - B_\tau)^2\right] + \mu^2 - 2\mu\sqrt{\frac{2}{\pi}} \right) \\ &= \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[(S_1 - B_\tau)^2\right] + \inf_{\mu \in \mathbb{R}} \left(\mu^2 - 2\mu\sqrt{\frac{2}{\pi}} \right). \end{aligned} \tag{4.3.3}$$

Thus, the optimal prediction rule coincides with that of Theorem 4.2.1 by (4.2.43) and minimizing $\mu \mapsto \mu^2 - 2\mu\sqrt{2/\pi}$ we see that $\mu_* = \sqrt{2/\pi}$ with a minimal value of $-2/\pi$. □

Thus, the optimal estimator when evaluating under a squared spatial loss is unbiased and the optimal rule is the same as for the probabilistic space problem, i.e. the $\mu \equiv 0$ case. While the proof is relatively straight forward it should be clear that having an unbiased estimator as the optimal estimator with the same prediction rule under a general polynomial loss can not be expected to hold in general. To see this consider the family of polynomials $G_i(x) = x^i$ for $i \in \mathbb{N}$ then by Lemma 4.1.2 we have

$$\begin{aligned} &\mathbb{E}\left[G_n(S_1 - (B_\tau + \mu)) \mid \mathcal{F}_\tau\right] \\ =& \sum_{i=0}^{n} (-1)^{n-i} \binom{n}{i} \mu^{n-i} \mathbb{E}\left[G_i(S_1 - B_\tau) \mid \mathcal{F}_\tau\right] \\ =& \sum_{i=0}^{n} (-1)^{n-i} \binom{n}{i} \mu^{n-i} \left(G_i(S_\tau - B_\tau) F_{1-\tau}(S_\tau - B_\tau) + \int_{S_\tau - B_\tau}^{\infty} G_i(z) f_{1-\tau}(z)\, \mathrm{d}z \right), \end{aligned} \tag{4.3.4}$$

for any $n \in \mathbb{N}$ where F_t and f_t denotes the distribution and density of S_t given in (4.1.1) and (4.1.2), respectively. It should be clear that the squared case is solvable since the decomposition only depends on the square and the linear term, where the latter is trivial when taking expectation. However, for higher order polynomials μ and τ are connected, and it is very unlikely that the prediction rule will be the optimal prediction rule from the $\mu \equiv 0$ case under G_n loss. Before considering the G_n loss we note that the same simplicity can be obtained for higher-order polynomials by adding lower-order polynomials to the loss such that the resulting gain function only depends on the maximal order with $\mu \equiv 0$ and the linear terms.

Consider the function

$$\overline{G}_n(x - \mu) = \sum_{i=2}^{n} \binom{n}{i} \mu^{n-i} G_i(x - \mu) \tag{4.3.5}$$

for $n \geq 2$ and even. When $x - \mu$ is small the function behaves like the squared loss while for large $x - \mu$ it behaves like G_n. Furthermore, for $x = S_1 - B_\tau$ then the function has global minimum when $B_\tau + \mu = S_1$, so it can be used to find an estimator.

Theorem 4.3.2. *Consider the spatial optimal prediction problem*

$$V_* = \inf_{\tau \in \mathcal{T},\, \mu \in \mathbb{R}} \mathbb{E}\left[\overline{G}_n(S_1 - (B_\tau + \mu))\right] \tag{4.3.6}$$

for $n \geq 2$ and even. The value function is given by $V_* = W_* - (2/\pi)^{n/2}$ where W_* is given by

$$W_* = e^{z_n^2/2} \frac{H(z_n)}{M\left(\frac{1+n}{2}, \frac{1}{2}, \frac{1}{2}z_n^2\right)} \tag{4.3.7}$$

where z_n is the unique strictly positive root of

$$\frac{H'(z)}{H(z)} + z = (1+n)z \frac{M\left(\frac{3+n}{2}, \frac{3}{2}, \frac{1}{2}z^2\right)}{M\left(\frac{1+n}{2}, \frac{1}{2}, \frac{1}{2}z^2\right)} \tag{4.3.8}$$

where H is given by

$$H(z) = z^n + 2\int_{z^n}^{\infty} (1 - \Phi(u^{1/n}))\, \mathrm{d}u \tag{4.3.9}$$

for $z \geq 0$ and $M(a,b,x) = 1 + \sum_{k=1}^{\infty} \frac{(a)_k}{(b)_k} \frac{x^k}{k!}$ with $(a)_k = (a)(a+1)\cdot \ldots \cdot (a+k-1)$ and $a, b \in \mathbb{R}$ is the degenerate hypergeometric function (Kummer function) (see e.g. Polyanin and Zaitsev, 2017). The optimal stopping time is given by

$$\tau_* = \inf\{t \in [0,1] \mid S_t - B_t \geq z_n \sqrt{1-t}\} \tag{4.3.10}$$

and the optimal offset is given by $\mu_* = \sqrt{2/\pi}$.

Remark 4.3.3. The proof below heavily relies on Pedersen (2003, Theorem 1). The author proves that the optimal stopping time in the probabilistic space problem under a G_q loss for $q > 0$ is on the form (4.3.10). This thus implies a further connection between the linear and entropy time formulation and the probabilistic space problems as we noted in (4.2.43) and (4.2.44). △

Proof. Note by (4.3.4) we have

$$\mathbb{E}\left[\overline{G}_n(S_1 - (B_\tau + \mu)) \mid \mathcal{F}_\tau\right] \tag{4.3.11}$$

$$= \sum_{i=2}^{n} \binom{n}{i} \mu^{n-i} \sum_{j=0}^{i} (-1)^{i-j} \binom{i}{j} \mu^{i-j} \mathbb{E}\left[G_j(S_1 - B_\tau) \mid \mathcal{F}_\tau\right] \tag{4.3.12}$$

$$= \mathbb{E}\left[G_n(S_1 - B_\tau) \mid \mathcal{F}_\tau\right] - n\mu^{n-1}\mathbb{E}\left[S_1 - B_\tau \mid \mathcal{F}_\tau\right] + n\mu^n - \mu^n, \tag{4.3.13}$$

since the sums telescope. Thus using $\mathbb{E}[S_1 - B_\tau] = \sqrt{2/\pi}$ for any $\tau \in \mathcal{T}$ and Pedersen (2003, Theorem 1) we get the optimal stopping rule as wanted. For the optimal offset note that the function $\mu \mapsto (n-1)\mu^n - n\mu^{n-1}\sqrt{2/\pi}$ has global minimum in $\sqrt{2/\pi}$ for any integer $n \geq 2$ as wanted. □

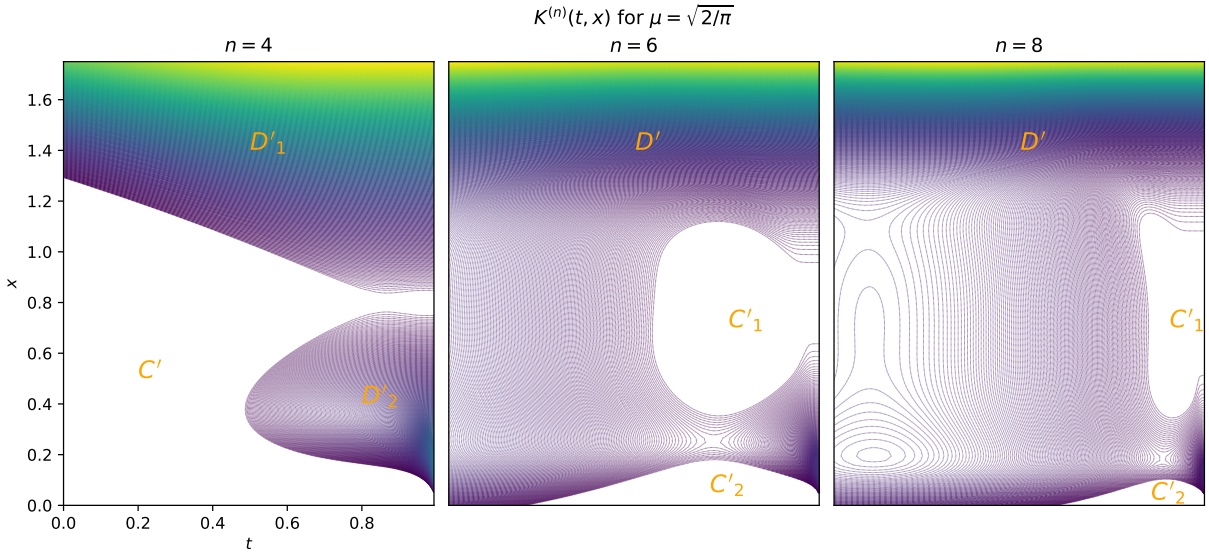


Figure 4.3.1: Plot of $K^{(n)}$ for the offset $\mu = \sqrt{2/\pi}$. The colored regions are $\{K^{(n)} \geq 0\}$ while the white regions are $\{K^{(n)} < 0\}$.

We now move on to consider the $G_n(S_1 - (B_\tau + \mu))$ loss case for $\mu \neq 0$. The problem is not separable, so we need to work with (4.3.4) directly. To explore the continuation and stopping sets of a time-inhomogeneous problem we typically would like to inspect the function $K^{(n)} : [0,1] \times R_+ \to \mathbb{R}$ being the infinitesimal generator applied to the gain function. We know from general theory that $C' = \{K^{(n)} < 0\} \subseteq C$ while $D' = \{K^{(n)} \geq 0\} \supseteq D$. Exploring these candidate sets for $n = 4, 6, 8$ we get the regions depicted in Figure 4.3.1. We see that the candidate sets are very ill-behaved which leaves us with the conclusion that the squared loss case is truly remarkable in its simplicity. For further details about the G_n loss, we refer to Appendix B.2.

4.3.2 Exponential Space Problem

As with the time problem the events $(B_\tau + \mu > S_1)$ and $(B_\tau + \mu < S_1)$ may have drastically different consequences so in general we would like to consider asymmetric loss functions. As an example, we could put an exponential loss on underestimation while letting overestimation be linear. This is somewhat encapsulated by the LINEX loss function defined in (2.1.8). For convenience, we consider the loss function

$$L(S_1, B_\tau + \mu) = e^{\alpha(S_1 - (B_\tau + \mu))} - \alpha(S_1 - (B_\tau + \mu)) - 1. \tag{4.3.14}$$

We start by having $\alpha > 0$. The expected loss is thus

$$\mathbb{E}\left[L(S_1, B_\tau + \mu)\right] = e^{-\alpha\mu}\mathbb{E}\left[e^{\alpha(S_1 - B_\tau)}\right] - \alpha\left(\sqrt{\frac{2}{\pi}} - \mu\right) - 1. \tag{4.3.15}$$

As such we have a preliminary prediction problem

$$W_* = \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[e^{\alpha(S_1 - B_\tau)}\right] \tag{4.3.16}$$

where we note that the exponent is positive for any τ in $\mathcal{T}$.

Consider the process $X = (X_t)_{t\in[0,1]}$ given by

$$X_t = e^{\alpha B_t} = e^{\left(r - \frac{\alpha^2}{2}\right)t + \alpha B_t} \tag{4.3.17}$$

for $r = \frac{\alpha^2}{2}$. Recall that this is the unique strong solution to the stochastic differential equation

$$\mathrm{d}X_t = rX_t\,\mathrm{d}t + \alpha X_t\,\mathrm{d}B_t \tag{4.3.18}$$

with $X_0 = 1$, i.e. X is a geometric Brownian motion. Furthermore, we define $M = (M_t)_{t\in[0,1]}$ by

$$M_t = \sup_{s\in[0,t]} X_s = e^{\alpha \sup_{s\in[0,t]} B_s} = e^{\alpha S_t}. \tag{4.3.19}$$

So the preliminary optimal prediction problem can be written as

$$W_* = \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[\frac{M_1}{X_\tau}\right] \tag{4.3.20}$$

i.e. a ratio loss optimal prediction problem for geometric Brownian motion. This problem was solved in du Toit and Peskir (2009) for general drift. We will now reduce the optimal prediction problem into an optimal stopping problem and present the result for our special case.

Using Lemma 4.1.2 we get

$$\mathbb{E}\left[e^{\alpha(S_1 - B_t)} \mid \mathcal{F}_t\right] = e^{\alpha(S_t - B_t)}\left(2\Phi\left(\frac{S_t - B_t}{\sqrt{1-t}}\right) - 1\right) + \int_{S_t - B_t}^{\infty} e^{\alpha z}\frac{2}{\sqrt{1-t}}\varphi\left(\frac{z}{\sqrt{1-t}}\right)\mathrm{d}z. \tag{4.3.21}$$

We can rewrite the integral by substitution and completion of the square

$$\int_x^\infty e^{\alpha z}\frac{2}{\sqrt{1-t}}\varphi\left(\frac{z}{\sqrt{1-t}}\right)\mathrm{d}z = 2\int_{\frac{x}{\sqrt{1-t}}}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2 - 2\alpha y\sqrt{1-t}}{2}}\,\mathrm{d}y \tag{4.3.22}$$

$$= 2e^{\frac{\alpha^2(1-t)}{2}}\int_{\frac{x}{\sqrt{1-t}}}^\infty \varphi\left(y - \alpha\sqrt{1-t}\right)\mathrm{d}y \tag{4.3.23}$$

$$= 2e^{\frac{\alpha^2(1-t)}{2}}\left(1 - \Phi\left(\frac{x - \alpha(1-t)}{\sqrt{1-t}}\right)\right). \tag{4.3.24}$$

Thus, we have

$$\mathbb{E}\left[e^{\alpha(S_1-B_\tau)}\right] = \mathbb{E}\left[G(\tau, S_\tau - B_\tau)\right] \tag{4.3.25}$$

where

$$G(t,x) = e^{\alpha x}\left(2\Phi\left(\frac{x}{\sqrt{1-t}}\right) - 1\right) + 2e^{\frac{\alpha^2(1-t)}{2}}\left(1 - \Phi\left(\frac{x-\alpha(1-t)}{\sqrt{1-t}}\right)\right) \tag{4.3.26}$$

which is a three-dimensional problem. However, using $S - B \overset{d}{=} |B|$ it can be reduced to a two-dimensional problem of finite horizon. So the function to study using the Markovian method is

$$W_*(t,x) = \inf_{\tau \leq 1-t} \mathbb{E}_{t,x}\left[G(t+\tau, |B_{t+\tau}|)\right] \tag{4.3.27}$$

where $B_t = x$ under $P_{t,x}$ for all $(t,x) \in [0,1] \times \mathbb{R}_+$. Recall the infinitesimal generator of (t,B) is $\frac{\partial}{\partial t} + \frac{1}{2}\frac{\partial^2}{\partial x^2}$ and hence

$$\begin{aligned} H(t,x) = G_t(t,x) + \frac{1}{2}G_x x(t,x) =& \alpha^2 e^{\frac{\alpha^2(1-t)}{2}}\left(\Phi\left(\frac{x-\alpha(1-t)}{\sqrt{1-t}}\right) - 1\right) && (4.3.28) \\ &+ \frac{\alpha^2}{2}e^{\alpha x}\left(2\Phi\left(\frac{x}{\sqrt{1-t}}\right) - 1\right) && (4.3.29) \end{aligned}$$

by a straightforward calculation.

This special case of $\alpha^2/2$ drift has the following solution from du Toit and Peskir (2009):

Theorem 4.3.4. *Consider the optimal stopping problem (4.3.27) for $\alpha > 0$. There exists a continuous decreasing function $b : [0,1] \to \mathbb{R}_+$ with $b(T) = 0$ such that the optimal stopping set is on the form*

$$D = \{(t,x) \in [0,1] \times \mathbb{R}_+ \mid x \geq b(t)\}. \tag{4.3.30}$$

The value function is given by

$$W_*(t,x) = E_{t,x}\left[G(1, |B_1|)\right] - \int_0^{1-t} \mathbb{E}_{t,x}\left[H(t+s, |B_{t+s}|) I\left(|B_{t+s}| > b(t+s)\right)\right] \mathrm{d}s \tag{4.3.31}$$

for all $(t,x) \in [0,1] \times \mathbb{R}_+$. The value in (4.3.16) is given by $W_* = W_*(0,0)$ and the optimal stopping rule is given by the first hitting time on D i.e.

$$\tau_* = \inf\{t \in [0,1] \mid S_t - B_t \geq b(t)\}. \tag{4.3.32}$$

So there exists a solution to the preliminary problem. We can now solve the original problem

Theorem 4.3.5. *Consider the family of spatial optimal prediction problems with the LINEX loss for $\alpha > 0$, i.e.*

$$V_* = \inf_{\tau \in \mathcal{T},\, \mu \in \mathbb{R}} \mathbb{E}\left[e^{\alpha(S_1 - (B_\tau + \mu))} - \alpha(S_1 - (B_\tau + \mu)) - 1\right]. \tag{4.3.33}$$

The value function is given by

$$V_* = \log(W_*) - \alpha\sqrt{\frac{2}{\pi}} \tag{4.3.34}$$

where W_* is given in Theorem 4.3.4. The optimal estimator is $B_{\tau_*} + \frac{\log(W_*)}{\alpha}$ where τ_* is given by (4.3.32).

Proof. For clarity, we first rewrite the expected loss from (4.3.15) to

$$F(\tau, \mu) = e^{-\alpha\mu} f(\tau) - \alpha\left(\sqrt{\frac{2}{\pi}} - \mu\right) - 1 \tag{4.3.35}$$

where $f(\tau) = \mathbb{E}\left[e^{\alpha(S_1 - B_\tau)}\right]$. Note in particular that $f(\tau) \geq 1$.

This is a multivariate function that is to be minimized. Note that for any $\tau \in \mathcal{T}$ then along $\mu \mapsto F(\tau, \mu)$ a unique infimum is obtained at $\mu \in \mathbb{R}$ such that

$$-\alpha e^{-\alpha\mu} f(\tau) + \alpha = 0. \tag{4.3.36}$$

Since the infimum is unique then the infimum of F must lie on the curve $(\tau, \alpha^{-1}\log f(\tau))$. Evaluating F on this curve yields

$$F(\tau, \alpha^{-1}\log f(\tau)) = \log f(\tau) - \alpha\sqrt{\frac{2}{\pi}}. \tag{4.3.37}$$

Thus, $f(\tau) \mapsto F(\tau, \alpha^{-1}\log f(\tau))$ is increasing on $[1, \infty)$ and hence the infimum of F can be found by

$$V_* = \inf_{\mu \in \mathbb{R}} \left(e^{-\alpha\mu} \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[e^{\alpha(S_1 - B_\tau)}\right] - \alpha\left(\sqrt{\frac{2}{\pi}} - \mu\right) - 1 \right). \tag{4.3.38}$$

Hence, recognizing the preliminary problem we get the optimal offset

$$\mu_* = \frac{\log(W_*)}{\alpha} \tag{4.3.39}$$

and the wanted value function. □

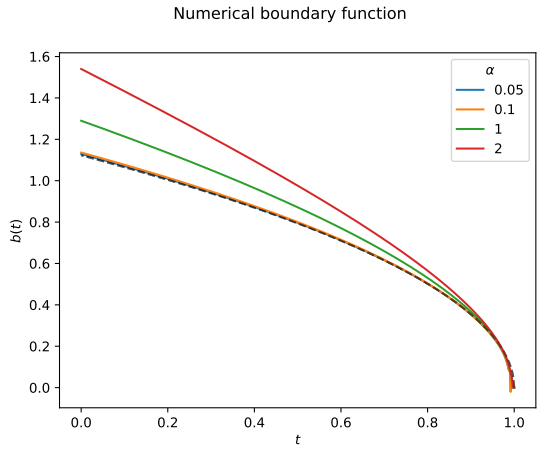


Figure 4.3.2: Solution to (4.3.41) for different α. Transparent curve is $t \mapsto z_* \sqrt{1-t}$ from Theorem 4.2.1 with $c = 1$, see Remark 4.3.6.

Since no explicit solution exists to the preliminary problem then we have to resort to numerical methods for computations of the boundary and value functions. Using that $G(1, x) = e^{\alpha x}$ we have

$$\mathbb{E}_{t,x}\left[G(1, |B_1|)\right] = \int_0^\infty e^{\alpha y} p(1-t; x, y)\, \mathrm{d}y. \tag{4.3.40}$$

As in Section 3.5 we need to rewrite (4.3.31) from an equation of two unknowns into an equation of only b which can be done by considering the points $(t, b(t))$ for $t \in [0, 1]$. Hence, we need to solve

$$0 = -\int_0^\infty e^{\alpha y} p(1-t; b(t), y)\, \mathrm{d}y + G(t, b(t)) + \int_0^{1-t} \int_{b(t+s)}^\infty H(t+s, y) p(s; b(t), y)\, \mathrm{d}y\, \mathrm{d}s \tag{4.3.41}$$

which has been done for a range of α parameters plotted in Figure 4.3.2. We note that the boundary increases in α which also could be expected from the loss function since the loss function increases. This implies that with a larger penalty α it is optimal to continue observations for longer than for small α.

Since we now have the boundary function we can calculate the value function form (4.3.31) by

$$W_*(t, x) = \int_0^\infty e^{\alpha y} p(1-t; x, y)\, \mathrm{d}y - \int_0^{1-t} \int_{b(t+s)}^\infty H(t+s, y) p(s; x, y)\, \mathrm{d}y\, \mathrm{d}s. \tag{4.3.42}$$

Setting $t = 0$ we have depicted this in Figure 4.3.3 for $\alpha = 1$. However, to determine μ_* from Theorem 4.3.5 we only need $W_* = W_*(0, 0)$ The value is recorded for the different α parameters in Table 4.3.1.

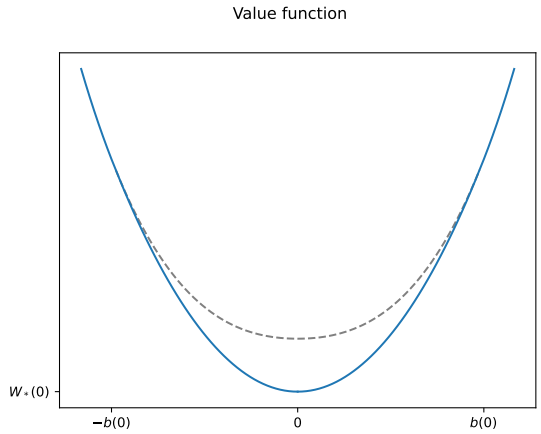


Figure 4.3.3: Numerical calculation of $W_*(0,x)$ for $\alpha = 1$.

α	0.05	0.1	1.0	2.0
W_*	1.041	1.084	2.366	6.703
μ_*	0.8007	0.8035	0.8612	0.9512

Table 4.3.1: Calculated W_* for range of α parameters. The optimal offset μ_* is calculated by (4.3.39).

Remark 4.3.6. As noted in Section 2.1 the LINEX loss is approximately equal to a squared loss for $|\alpha|$ small. Comparing Theorem 4.3.1 with Table 4.3.1 we see that the optimal offset in the LINEX case approaches $\sqrt{2/\pi} \approx 0.7979$ as $|\alpha|$ becomes small. So when considering the asymmetric LINEX loss function that puts an exponential loss on underestimation, we get a positive bias in our optimal estimator $B_\tau + \mu_*$, but as the asymmetry vanishes we recover an unbiased estimator. Furthermore, we have plotted the optimal stopping boundary $t \mapsto z_*\sqrt{1-t}$ from Theorem 4.2.1 with $c = 1$ as the transparent dashed line in Figure 4.3.2. We see that the boundary function b also approaches the squared case for decreasing α. △

Now consider the converse loss function putting exponential weight on overestimation, i.e.

$$L(S_1, B_\tau + \mu) = e^{\beta(B_\tau + \mu - S_1)} - \beta(B_\tau + \mu - S_1) - 1 \tag{4.3.43}$$

for $\beta = -\alpha > 0$. Taking expectation we arrive at the preliminary problem

$$W_* = \inf_{\tau \in \mathcal{T}} \mathbb{E}\left[e^{\beta(B_\tau - S_1)}\right]. \tag{4.3.44}$$

Note that the exponent is negative for any $\tau \in \mathcal{T}$ and since we want to minimize the expectation we now have the goal of B_{τ_*} being far away from S_1.

Since the loss function is simply a sign change of α from the previous loss function then we easily get that the problem is a time-inhomogeneous optimal stopping problem on the form

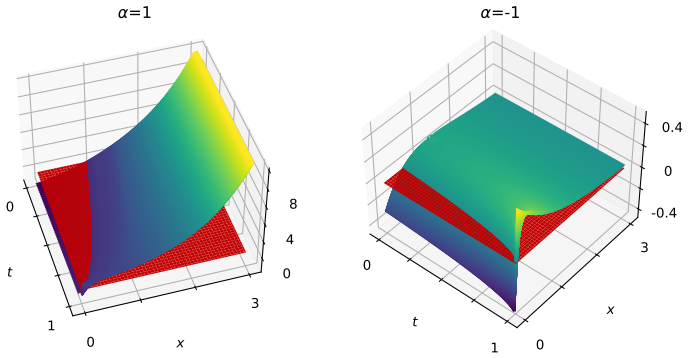


Figure 4.3.4: Plot of H from (4.3.28). Red plain is 0.

(4.3.25) where

$$G(t,x) = e^{-\beta x}\left(2\Phi\left(\frac{x}{\sqrt{1-t}}\right)-1\right) + 2e^{\frac{a^2(1-t)}{2}}\left(1-\Phi\left(\frac{x+a(1-t)}{\sqrt{1-t}}\right)\right). \tag{4.3.45}$$

This problem is rather difficult. To see why note that applying the infinitesimal generator to G we have

$$H(t,x) = \beta^2 e^{\frac{\beta^2(1-t)}{2}}\left(\Phi\left(\frac{x+\beta(1-t)}{\sqrt{1-t}}\right)-1\right) + \frac{\beta^2}{2}e^{-\beta x}\left(2\Phi\left(\frac{x}{\sqrt{1-t}}\right)-1\right). \tag{4.3.46}$$

Note that if $H(t,x) < 0$ for some $(t,x) \in [0,1]\times\mathbb{R}_+$ then it is optimal to continue hence $\{H<0\} \subseteq C$. Inspection of H, see right panel of Figure 4.3.4, yields existence of a smooth function $h : [0,1] \to \mathbb{R}_+$ such that

$$\{H<0\} = \{(t,x) \in [0,1]\times\mathbb{R}_+ \mid x < h(t)\} \tag{4.3.47}$$

with $h(1) = 0$. So parametrizing the stopping set by a function $b : [0,1] \to \mathbb{R}_+$ such that

$$D = \{(t,x) \in [0,1]\times\mathbb{R}_+ \mid x \geq b(t)\} \tag{4.3.48}$$

then $b(t) \geq h(t)$. Since the problem is closely related to Theorem 4.3.4 then one could expect a similar proof of sufficient conditions for deriving a free-boundary equation. However, the proof of Theorem 4.3.4 in du Toit and Peskir (2009) heavily relies on showing that $t \mapsto V(t,x) - G(t,x)$

$-\alpha$	0.05	0.1	1.0	2.0
W_*	0.9610	0.9238	0.4739	0.2588
μ_*	0.7951	0.7924	0.7466	0.6931

Table 4.3.2: Calculated W_* and optimal offset from (4.3.44) for range of α parameters.

is increasing on $[0,1]$. To show this it is sufficient to have H_t be increasing. However, in our case, standard calculations yield

$$H_t(t,x) = \frac{\beta^2}{2}\left(\frac{2x-\beta(1-t)}{(1-t)^{3/2}}\right)\varphi\left(\frac{x}{\sqrt{1-t}}\right)e^{-\beta x} + \frac{\beta^4}{2}e^{\frac{\beta^2(1-t)}{2}}\Phi\left(-\frac{x+\beta(1-t)}{\sqrt{1-t}}\right) \tag{4.3.49}$$

which can be both positive and negative as can also be observed numerically in Figure 4.3.4.

Assume that Assumptions 3.4.1 hold. Then the free-boundary equation is given by

$$\mathbb{E}_{t,b(t)}\left[G(1,|B_1|)\right] = G(t,b(t)) + \int_0^{1-t}\mathbb{E}_{t,b(t)}\left[H(t+s,|B|_{t+s})I(|B|_{t+s} > b(t+s))\right]\mathrm{d}s \tag{4.3.50}$$

and noting that $G(1,x) = e^{-\beta x}$ we need to solve

$$\int_0^\infty e^{-\beta y}p(1-t;b(t),y)\,\mathrm{d}y = G(t,b(t)) + \int_0^{1-t}\int_{b(t+s)}^\infty H(t+s,y)p(s;b(t),y)\,\mathrm{d}y\,\mathrm{d}s. \tag{4.3.51}$$

To get the value function at $t,x=0$ we then need to calculate

$$W_*(0,0) = \int_0^\infty e^{-\beta y}p(1;0,y)\,\mathrm{d}y - \int_0^1\int_{b(s)}^\infty H(s,y)p(s;0,y)\,\mathrm{d}y\,\mathrm{d}s. \tag{4.3.52}$$

Using the method from Theorem 4.3.5 we then get the optimal offset by $\mu_* = -\frac{\log W_*}{\beta}$. We have calculated the optimal stopping boundary and plotted it in Figure 4.3.5. We note that the boundary lies below the squared problem, so we would like to stop earlier. Furthermore, we have calculated the value function and the optimal offset in Table 4.3.2. In accordance with Remark 4.3.6 the estimator now has a negative bias while still approaching an unbiased estimator for $|\alpha|$ small.

Remark 4.3.7. As noted in (4.3.44) the goal in the preliminary optimal stopping problem is to maximize the distance between the ultimate maximum and the stopped process. This may seem odd when wanting to estimate S_1. We could instead consider the worst possible case, i.e.

$$\tilde{W} = \sup_{\tau\in\mathcal{T}}\mathbb{E}\left[e^{\beta(B_\tau - S_1)}\right]. \tag{4.3.53}$$

It is shown in du Toit and Peskir (2009) that $\mathbb{E}\left[e^{\beta(B_\tau-S_1)}\right]$ is bounded by the value at $\tau=0$ and $\tau=1$. Thus, the worst possible decision rules are at $\tau=0$ and $\tau=1$.

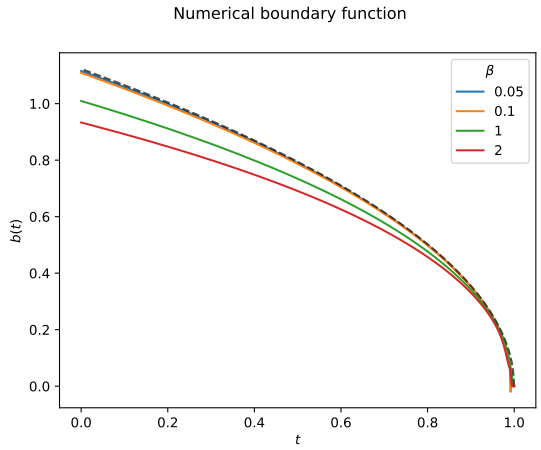


Figure 4.3.5: Solution to (4.3.51) for different β. Transparent curve is $t \mapsto z_* \sqrt{1-t}$ from Theorem 4.2.1 with $c = 1$.

If we take the stopping time $\tau = 0$ from the supremum formulation of the preliminary problem and plug it into our loss then we can still optimize for the optimal offset. To do so note that $\tilde{W}^{(\beta)} = \mathbb{E}\left[e^{-\beta S_1}\right]$ is given by

$$\tilde{W}^{(\beta)} = \int_0^\infty e^{-\beta y} \frac{2}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \, \mathrm{d}y \tag{4.3.54}$$

$$= 2e^{\frac{\beta^2}{2}} \int_\beta^\infty \varphi(y) \, \mathrm{d}y = 2e^{\frac{\beta^2}{2}} \left(1 - \Phi(\beta)\right). \tag{4.3.55}$$

As above the optimal offset is given by $\hat{\mu}^{(\beta)} = \frac{-\log \tilde{W}^{(\beta)}}{\beta}$ i.e.

$$\hat{\mu}^{(\beta)} = -\frac{\log\left(2(1 - \Phi(\beta))\right)}{\beta} - \frac{\beta}{2}. \tag{4.3.56}$$

Note that $\log \tilde{W}^{(\beta)} \to 0$ for $\beta \searrow 0$ however by an application of L'Hospital we get that

$$\lim_{\beta \searrow 0} \hat{\mu}^{(\beta)} = \frac{\varphi(0)}{1 - \Phi(0)} = \sqrt{\frac{2}{\pi}}. \tag{4.3.57}$$

So in the worst case, the limit is an unbiased estimator. △

4.4 An Alternative Space Estimator

The class of estimators that were considered in the previous section had been motivated by Graversen et al. (2001). However, combatting Goal 2 from Section 2.1 by adding a constant to

the estimator from the probabilistic space problem may seem rather simple since the constant only depends on the loss function and not the observed path. Thus, as discussed in Section 2.2 we can consider the alternative statistical space problem from Definition 2.2.4.

For a stopping time with $\mathbb{E}\sqrt{\tau} < \infty$ we can utilize Wald's identity and $S - B \overset{d}{=} |B|$ to rewrite (2.2.7) to get

$$\inf_{\tau} \mathbb{E}\left[S_1 - S_\tau + f(\tau)\right] = \sqrt{\frac{2}{\pi}} - \sup_{\tau} \mathbb{E}\left[S_\tau - f(\tau)\right] = \sqrt{\frac{2}{\pi}} - \sup_{\tau} \mathbb{E}\left[|B_\tau| - f(\tau)\right]. \tag{4.4.1}$$

For simplicity, we consider a linear time penalty i.e. $f(t) = ct$ for some $c > 0$. For a stopping time with $\mathbb{E}[\tau] < \infty$ we have $\mathbb{E}\left[B_\tau^2\right] = \mathbb{E}[\tau]$ hence we can consider the following equivalent optimal stopping problem:

$$W = \sup_{\tau} \mathbb{E}\left[|B_\tau| - c\,|B_\tau|^2\right]. \tag{4.4.2}$$

A similar optimal stopping problem was solved in Graversen and Peskir (1997). Specifically, the authors considered the case where the supremum is taken over all stopping times. Denoting the gain function by $G(x) = x - cx^2$ we then have a one-dimensional time-homogeneous optimal stopping problem with infinite horizon. Noting that G has unique supremum in $(2c)^{-1}$ and since $\lim_{t\to\infty} |B_t| = \infty$ P_x-a.s. and B is continuous then the stopping time

$$\tau_* = \inf\left\{ t > 0 \;\middle|\; |B_t| = \frac{1}{2c} \right\} \tag{4.4.3}$$

is P_x-a.s. finite for any $x \in \mathbb{R}_+$. Thus, τ_* is optimal in the infinite horizon case of (4.4.2) and the value function is $(4c)^{-1}$.

We are interested in the finite horizon case, specifically with $|B_0| = 0$. Since the gain is time homogeneous then the stopping set from the infinite horizon problem is a subset of the stopping set in the finite horizon problem. Thus using the Markovian approach we can consider the optimal stopping problem

$$V(t,x) = \sup_{\tau \le 1-t} \mathbb{E}_x\left[|B_\tau| - c\,|B_\tau|^2\right] \tag{4.4.4}$$

where $|B|_0 = x$ under P_x for $(t,x) \in [0,1] \times [0,(2c)^{-1}]$. For convenience, we set $\gamma = (2c)^{-1}$ in the following.

Theorem 4.4.1. *Consider the family of optimal stopping problems*

$$V = \sup_{\tau \in \mathcal{T}} \mathbb{E}\left[S_\tau - c\tau\right] \tag{4.4.5}$$

for $c > 0$. There exists a continuous decreasing function $b : [0,1] \to [0,\gamma]$ with $b(1) = 0$ such that the stopping set D is given by

$$D = \{(t,x) \in [0,1] \times [0,\gamma] \mid x \ge b(t)\} \tag{4.4.6}$$

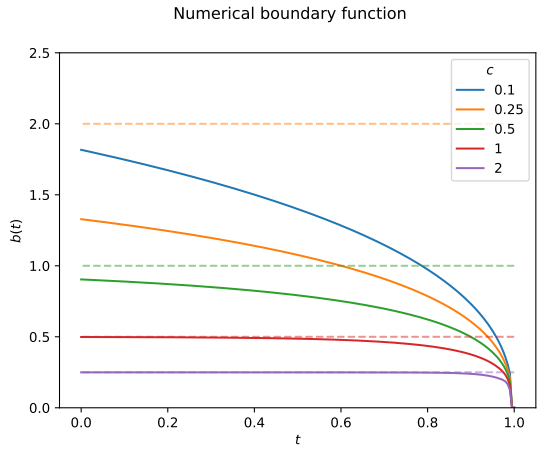


Figure 4.4.1: Numerical boundary for range of c parameters. Dashed lines are infinite horizon boundaries. The blue dashed line is at 5.

and the first hitting time τ_D is optimal. Furthermore, the value function from (4.4.4) is given by

$$\begin{aligned} V(t,x) &= \mathbb{E}_{t,x}\left[|B_1| - c\,|B_1|^2\right] - \int_0^{1-t} \mathbb{E}_{t,x}\left[\mathbb{L}_{|B|}G(|B_{t+s}|)I(|B_{t+s}| > b(t+s))\right] \mathrm{d}s \\ &= \int_0^\infty \left(y - cy^2\right) p(1-t;x,y)\,\mathrm{d}y + c\int_0^{1-t}\int_{b(t+s)}^\infty p(s;x,y)\,\mathrm{d}y\,\mathrm{d}s \end{aligned} \tag{4.4.7}$$

for $(t,x) \in [0,1]\times[0,\gamma]$ and specifically $V = V(0,0)$. The boundary function b is continuous, decreasing and dominated by γ, in particular it is characterized by

$$b(t) - cb^2(t) = \int_0^\infty \left(y - cy^2\right) p(1-t;b(t),y)\,\mathrm{d}y + c\int_0^{1-t}\int_{b(t+s)}^\infty p(s;b(t),y)\,\mathrm{d}y\,\mathrm{d}s \tag{4.4.8}$$

for $t \in [0,1]$. Furthermore, $V = V(0,0)$.

Proof. See Appendix B.3. □

The boundary function given by the nonlinear Volterra equation in (4.4.8) has been calculated for a range of c parameters and depicted in Figure 4.4.1. We have also depicted the infinite horizon boundary as dashed lines, which are larger than b as proved in Theorem 4.4.1.

Remark 4.4.2. One immediate concern about estimating S_1 by S_τ is that $S_\tau \leq S_1$ for any stopping time $\tau \in \mathcal{T}$. However, one should note that we can consider an estimator on the form aS_τ for $a > 0$ and get

$$\mathbb{E}\left[aS_1 - c'\tau\right] = a\mathbb{E}\left[S_1 - c\tau\right] \tag{4.4.9}$$

and since $c = c'/a > 0$ then Theorem 4.4.1 still applies. △

Chapter 5

Comparison of Estimators

In the present chapter, we will compare the estimators found in the previous chapter. The first section is devoted to a discussion of bias and variance for estimators with explicit rules in the linear time, entropy time, and polynomial probabilistic space formulations of optimal prediction. The second section will make an admissibility comparison of estimators showing that a trivial extension of the solution to the time and probabilistic space problems can be inadmissible in the statistical space problem with LINEX loss.

5.1 Mean Squared Error Comparison

We consider the family of estimators arising in the time problems with an explicit solution, i.e. estimators on the form

$$\tilde{\tau}_z = \inf\{t \in [0,1] \mid S_t - B_t \geq z\sqrt{1-t}\} \tag{5.1.1}$$

for $z \in \mathbb{R}_+$. As we saw in the proof of Corollary 4.2.2 the mean of the estimator is given by

$$\mathbb{E}\left[\tilde{\tau}_z\right] = \frac{z^2}{1+z^2}. \tag{5.1.2}$$

Moreover, for the second moment, we note as in Graversen et al. (2001), that a simple application of Itô's lemma implies $(B_t^4 - 6tB_t^2 + 3t^2)_{t\geq 0}$ is a mean zero martingale hence using $S - B \stackrel{d}{=} |B|$ yields

$$3\mathbb{E}\left[\tilde{\tau}_z^2\right] = \mathbb{E}\left[-\left|B_{\tilde{\tau}_z}\right|^4 + 6\tilde{\tau}_z\left|B_{\tilde{\tau}_z}\right|^2\right] \tag{5.1.3}$$

$$= -(z^4+6z^2)\mathbb{E}\left[\tilde{\tau}_z^2\right] + (2z^4+6z^2)\frac{z^2}{1+z^2} - z^4. \tag{5.1.4}$$

Thus, we get

$$\mathbb{E}\left[\tilde{\tau}_z^2\right] = \frac{z^6 + 5z^4}{(1+z^2)(3+6z^2+z^4)} \tag{5.1.5}$$

and the variance is

$$\mathrm{Var}\left(\tilde{\tau}_z^2\right) = \frac{2z^4}{(1+z^2)^2(3+6z^2+z^4)}. \tag{5.1.6}$$

In a classical statistical framework, one is typically interested in minimizing the bias and variance of the estimator. In particular, we look at minimizing the mean squared error

$$\mathrm{MSE}(\tilde{\tau}_z) = \mathrm{Var}(\tilde{\tau}_z) + \left(\mathbb{E}\left[\tilde{\tau}_z\right] - \frac{1}{2}\right)^2 = \frac{z^6 + 3z^4 - 3z^2 + 3}{4(1+z^2)(3+6z^2+z^4)} \tag{5.1.7}$$

which has minimum for $z \in \mathbb{R}_+$ in $z \approx 0.943$. Table 5.1.1 contains the relevant quantities computed for estimators on the form (5.1.1). We see that the unbiased estimator has a lower variance than the linear loss estimator. Furthermore, the best estimator according to an MSE criterion stems from a weighted linear loss where we put a higher penalty on overestimation relative to the unbiased estimator.

	z	$\mathbb{E}\left[\tilde{\tau}_z\right]$	$\mathrm{Var}(\tilde{\tau}_z)$	MSE	c_{wl}	c_{ent}
Linear	1.123	0.5577	0.05118	0.05450	1	1.243
Entropy	1.263	0.6146	0.04999	0.06312	0.7657	1
Unbiased	1	0.5	0.05	0.05	1.269	1.514
Minimal MSE	0.9430	0.4707	0.04855	0.04941	1.420	1.664

Table 5.1.1: Computed quantities for explicit estimators where c_{wl} and c_{ent} are the weights in the weighted linear and entropy loss functions respectively.

5.2 Admissibility Comparison

It should be clear that the comparison under mean squared error is a simple comparison that only takes into account the first two moments of the estimator. However, if we were to make a similar comparison in the statistical space branch, i.e. with estimators on the form $\tilde{S}_1 = B_{\tilde{\tau}} + \tilde{\mu}$ where $\tilde{\tau}$ is an optimal stopping time and $\tilde{\mu}$ is the optimal offset for a particular loss function then $\tilde{\tau}$ controls the variance while $\tilde{\mu}$ controls the bias. Such an analysis would lead to $\tilde{\tau} = 0$ and $\tilde{\mu} = \sqrt{2/\pi}$ having a MSE of 0.

Instead, we make an admissibility comparison as discussed in Section 2.1. In the following, we make use of the numerical procedure from Appendix A.2 to calculate the risks.

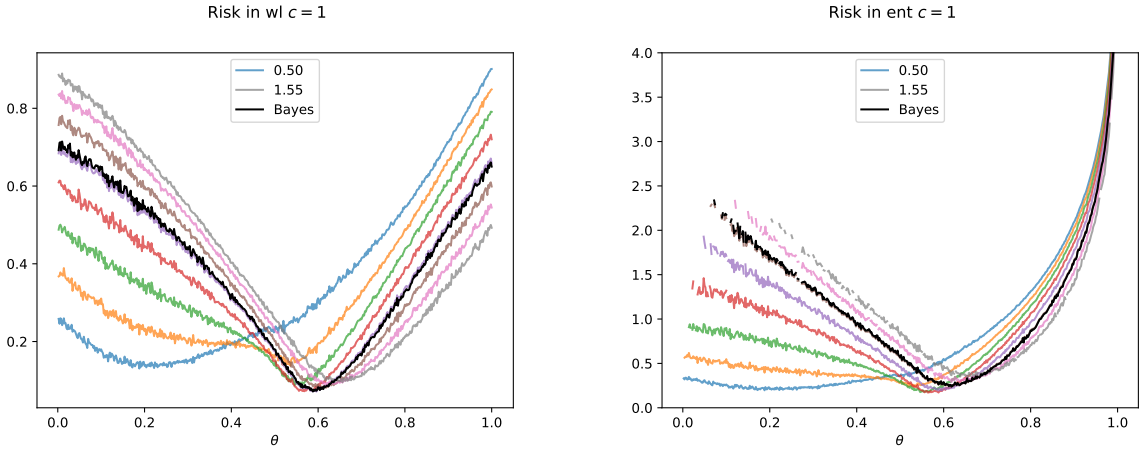


Figure 5.2.1: Risks of estimators on the explicit form (5.1.1) for $z \in [0.5, 1.55]$. Missing points in entropy loss are when there is a path for which $\tau(j) = 1$.

We start by considering the time problems. The risk under a weighted linear loss is given by

$$R_{wl}(\theta, \tau) = \mathbb{E}\left[c(\tau - \theta)^+ + (\tau - \theta)^- \mid \theta\right] \tag{5.2.1}$$

$$= \theta\left(P(\tau \leq \theta \mid \theta) - cP(\tau > \theta \mid \theta)\right) + \mathbb{E}\left[\tau\left(cI(\tau > \theta) - I(\tau \leq \theta)\right) \mid \theta\right]. \tag{5.2.2}$$

Similarly, for the entropy loss case, we have the risk

$$R_{ent}(\theta, \tau) = \log(1 - \theta)\left(cP(\tau > \theta \mid \theta) - P(\tau \leq \theta \mid \theta)\right) \tag{5.2.3}$$

$$+ \mathbb{E}\left[\log(1 - \tau)\left(I(\tau \leq \theta) - cI(\tau > \theta)\right) \mid \theta\right]. \tag{5.2.4}$$

We have plotted the risks of estimators $\tilde{\tau}_z$ on the form (5.1.1) under unit weights for a range of different $z \in [0.5, 1.55]$ in Figure 5.2.1. We see that in both cases all the estimators have risks that intersect, i.e. using the admissibility criterion no one estimator is uniformly better than any other. It should be noted that the same behavior happens with non-unit weights in the underlying loss function. This is somewhat expected since each of the estimators arrives from a similar loss function that penalizes overestimation differently. So a Bayes estimator under a weighted linear loss with a small weight $\tilde{c}$ will have low risk under a weighted linear loss with large weight $\bar{c}$ when the parameter θ is large.

Now consider the statistical space problem, specifically, we will consider the risk under a LINEX loss. The risk of an estimator $\hat{S}$ is given by

$$R_L(S_1, \hat{S}) = \mathbb{E}\left[e^{\alpha(S_1 - \hat{S})} \,\middle|\, S_1\right] - \alpha(\mathbb{E}[S_1 - \hat{S} \mid S_1]) - 1. \tag{5.2.5}$$

We consider three estimators; $\hat{S}_B^{(\alpha)}$ being the Bayes estimator from Section 4.3.2, $\hat{S}_A^{(a,c)} = aS_{\bar{\tau}_c}$ being the alternative estimator from Remark 4.4.2, and finally $\hat{S}_T^{(\mu,z)} = \mu + B_{\tilde{\tau}_z}$ being a mean-adjusted estimator from the time and probabilistic space formulation. In Figure 5.2.2 we have

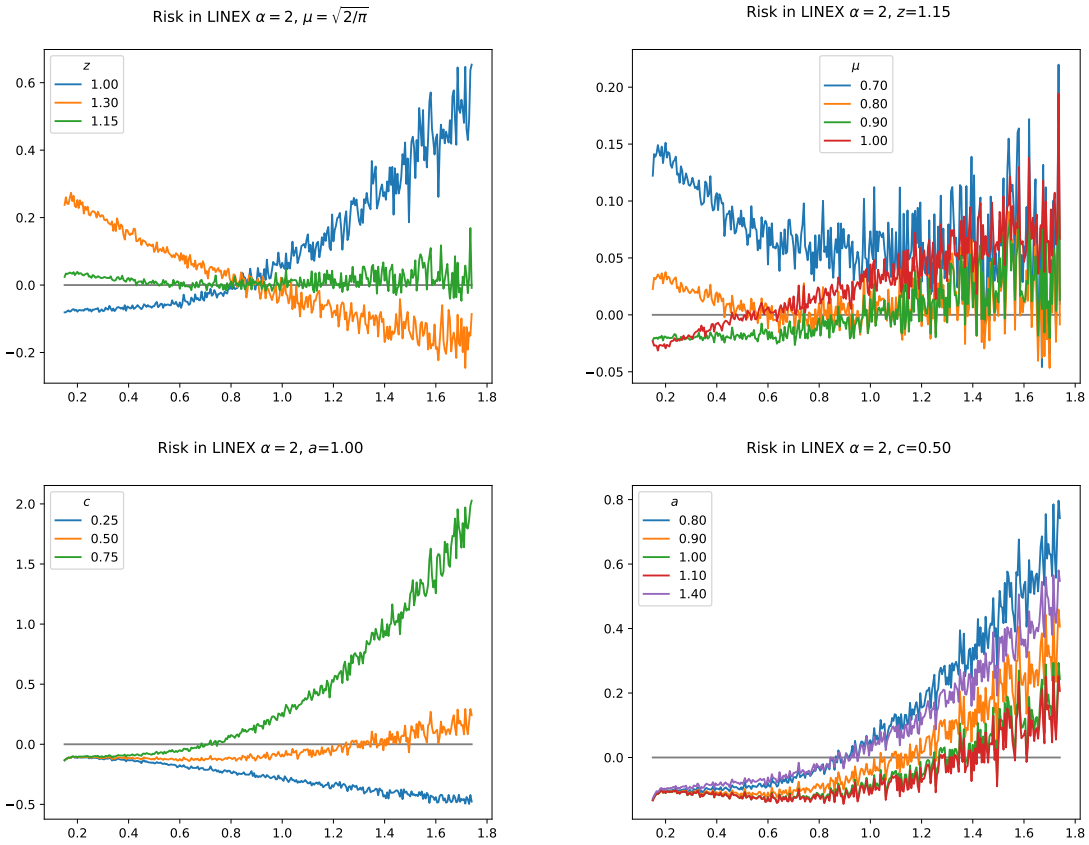


Figure 5.2.2: LINEX risks of estimators less LINEX risk of Bayes estimator with $\alpha = 2$. Top left; $\hat{S}_T^{(\sqrt{2/\pi},z)}$, top right; $\hat{S}_T^{(\mu,1.15)}$, bottom left; $\hat{S}_A^{(1,c)}$, and bottom right; $\hat{S}_A^{(a,0.5)}$.

plotted the difference between the risk of the Bayes estimator and the other estimators, e.g. $R_L(S_1, \hat{S}_T^{(\mu,z)}) - R_L(S_1, \hat{S}_B^{(\alpha)})$ when varying the parameters of the estimators.

We note from the bottom left panel that allowing the alternative estimator renders the $\hat{S}_B$ inadmissible, which is expected since the domination happens when the time penalty in Theorem 4.4.1 is low and thus $\overline{\tau}$ will be large hence $S_{\overline{\tau}}$ will be close to S_1. From the bottom right panel, we see that multiplying the S process by a constant a only seems to have a positive effect for α slightly larger than 1, which could be expected from the discussion in Section 4.4. The top right panel indicates that taking the stopping rule (5.1.1) is inadmissible under a LINEX loss with $\alpha = 2$ if the offset μ is smaller than the mean of S_1. Finally, the top left panel indicates that taking some optimal stopping rules from the time formulating and making the estimators unbiased by setting $\mu = \mathbb{E}[S_1]$ can be numerically close to inadmissible, e.g. for $z \approx 1.15$.

Chapter 6

Conclusion and Further Research

In this thesis, we have studied a range of optimal prediction problems, specifically we have considered problems from the time branch and the statistical space branch.

For the time problems we gave a Bayesian formulation and showed that the equivalence between the linear time problem and the squared probabilistic space problem could be extended to an equivalence between the weighted linear time problem and the squared probabilistic problem with a time penalty. Furthermore, we investigated the mixed probability and linear loss and saw that this problem has some unique intricacies in that the boundary seems to be non-monotone with connections back to the deterministic formulation of the problem. We also saw that the techniques from the linear time problems could be used to find explicit stopping rules for a range of logarithmic time problems.

We then went on to consider statistical space problems and saw that the only problem that had been solved yet was easy, particularly by the nature of the quadratic function. Furthermore, we extended the view to look at the celebrated asymmetric loss function, LINEX. We were able to solve the problem of exponential loss on underestimation while exploring that the converse case is more intricate. Finally, we took into account a different set of estimators by considering the running maximum with a time penalty.

We ended the thesis by making comparisons of the estimators. We found that the time estimator which minimizes variance and bias in the set of explicit optimal rules corresponding to a high penalty for overestimation. For the space problem, we saw that one should not be completely indifferent about which optimal stopping rule to use.

For future research, it would be interesting to consider the squared time problem and the weighted square loss in both the time and statistical space branches. Furthermore, extensions of the alternative space problem by considering other loss functions might also be fruitful.

Appendix A

Numerical Procedures

The present appendix will deal with the numerical procedures needed for nontrivial calculations in Chapters 4 and 5. The aim is to present the approach to solving free-boundary equations and calculating conditional expectations numerically. The routine has been implemented in Python 3.10.6 using the `SciPy` package for numerical optimization and integration. Furthermore, we have used the `NumPy` package for simulation and vectorization when needed. The implementation can be found on

https://github.com/Karl99Kristian/BayesianOptimalPrediction.

A.1 Nonlinear Volterra Equations of the Second Kind

There are many approaches to solving Volterra integral equations, however, efficient methods either require linearity or highly specialized algorithms for a particular nonlinear system as discussed in Press et al. (1992). One such specialized algorithm is the one proposed by Andersen et al. (2016) for the nonlinear Volterra equation arising from pricing of American options in a Black-Scholes market. Their approach utilizes a linearity in b of the gain function which yields a fixed point system that can reduce the time consumption from squared to linear in the time discretization. However, this linearity is not present in our cases hence we resort to the following general approach.

Consider the following nonlinear Volterra equation of the second kind:

$$f(t, b(t)) = \int_0^{1-t} h(t, s, b(t), b(t+s))\,\mathrm{d}s \tag{A.1.1}$$

where we want to solve for the unknown function $b : [0, 1] \to \mathbb{R}_+$. Assuming that the terminal boundary is some known value $b(1) \in \mathbb{R}_+$ we can solve for b iteratively. To solve the equation

we apply a low-order quadrature, namely a trapezoidal scheme, over a time grid $(t_i)_{i=0,\dots,n}$ for $n \in \mathbb{N}$ with $t_0 = 0$ and $t_n = 1$. For simplicity, we take the grid to be equidistant. We can then iterate backward in time by solving

$$f(t_i, b(t_i)) = \sum_{k=i+1}^{n} \frac{h(t_i, t_{k-1} - t_i, b(t_i), b(t_{k-1})) + h(t_i, t_k - t_i, b(t_i), b(t_k))}{2} \Delta t_k \tag{A.1.2}$$

for $i = 0, \dots, n-1$ with an application of the Newton-Raphson algorithm and an initial guess of $b(t_{i+1})$.

For the particular equations arriving in the thesis the f function is the difference between the expected value of the Mayer term at the terminal time and the Mayer term at $(t, b(t))$ corresponding to line (3.5.10). The h function is a sum of two space integrals with respect to the transition distribution of a reflected Brownian motion, e.g.

$$h(t, s, b(t), b(t+s)) = \int_{b(t+s)}^{\infty} \overline{M}(t+s, y) p(s; b(t), y) \, \mathrm{d}y - \int_{0}^{b(t+s)} L(t+s, y) p(s; b(t), y) \, \mathrm{d}y \tag{A.1.3}$$

where p is given by (4.1.10), $\overline{M}$ is the infinitesimal generator applied to the Mayer term, and L is the Lagrangian term corresponding to line (3.5.11) and (3.5.12). To ensure accuracy we calculate these space integrals by high-precision numerical integration quadrature, namely using the `qagse` routine (see Piessens et al., 2012). In the trapezoidal scheme, we have to be careful when evaluating h in $s = 0$ and $s = 1 - t$. We let $p(0, b(t), y) = \delta_{(b(t)=y)}$ be the Dirac measure such that

$$h(t, 0, b(t), b(t)) = \overline{M}(t, b(t)) - L(t, b(t)). \tag{A.1.4}$$

At $s = 1 - t$ we integrate $\overline{M}$ at maturity and let the Lagrangian term vanish, i.e.

$$h(t, 1-t, b(t), b(1)) = \int_{b(1)}^{\infty} \overline{M}(1, y) p(1-t; b(t), y) \, \mathrm{d}y. \tag{A.1.5}$$

Furthermore, at $i = n - 1$, the discretization can introduce problems, e.g. in (4.3.41) it can be shown that $b(t_{n-1}) = 0$ which is not the case since the boundary is strictly increasing. However, for simplicity, we set $b(t_{n-1}) = b(t_n)$ for all free-boundary equations.

Accuracy

To showcase the accuracy of the algorithm we vary the number of grid points n and perform the algorithm on an equation with a known analytical solution. Consider the optimal stopping problem from Theorem 4.2.5. By the considerations in Section 3.5 we get the free-boundary

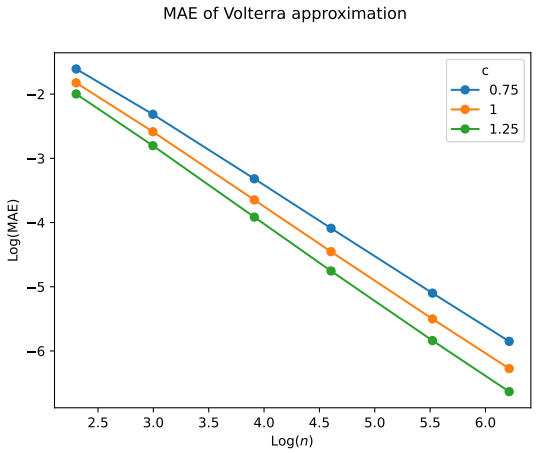


Figure A.1.1: Mean absolute error of approximate solution to (A.1.6) on log-log scale.

equation

$$\begin{aligned} 0 =&(1-t)\left(2\Phi\left(\frac{b(t)}{\sqrt{1-t}}\right)-2\right) \\ &+\int_0^{1-t}\int_{b(t+s)}^{\infty}\left(2\Phi\left(\frac{y}{\sqrt{1-(t+s)}}\right)-2\right)\cdot\frac{1}{\sqrt{2\pi s}}\left(e^{-\frac{(b(t)-y)^2}{2s}}+e^{-\frac{(b(t)+y)^2}{2s}}\right)\mathrm{d}y\,\mathrm{d}s \\ &-c\int_0^{1-t}\int_0^{b(t+s)}\left(2\Phi\left(\frac{y}{\sqrt{1-(t+s)}}\right)-1\right)\cdot\frac{1}{\sqrt{2\pi s}}\left(e^{-\frac{(b(t)-y)^2}{2s}}+e^{-\frac{(b(t)+y)^2}{2s}}\right)\mathrm{d}y\,\mathrm{d}s \end{aligned} \tag{A.1.6}$$

since for $M(t,x)=(1-t)\left(2\Phi\left(\frac{x}{\sqrt{1-t}}\right)-2\right)$ we have $M(1,|B|_1)=0$ and

$$\left(\frac{\partial}{\partial t}+\frac{1}{2}\frac{\partial^2}{\partial x^2}\right)M(t,x)=2\Phi\left(\frac{x}{\sqrt{1-t}}\right)-2. \tag{A.1.7}$$

We have implemented the algorithm from above to get an estimate $\tilde{b}_n$ of the boundary b on the grid. To measure the accuracy we calculate the mean absolute error by

$$\mathrm{MAE}_n=\frac{1}{n}\sum_{i=0}^{n}\left|\tilde{b}_n(t_i)-b(t_i)\right| \tag{A.1.8}$$

which has been plotted on log-log scale in Figure A.1.1. A regression yields an accuracy approximately of order $\mathcal{O}(n^{-1.1})$, so taking $n\approx 200$ is adequate for the analysis in this thesis.

A.2 Conditional Simulation of Estimators

Since no explicit expression exists for the moments of the estimators given the parameter we resort to Monte Carlo simulation in the calculation of risk in Section 5.2. The drawbacks of this approach are simulation and discretization error (see e.g. Glasserman, 2004), however, we will use it as a good first approach. Specifically, the simulation is prone to a large simulation error since variance reduction techniques such as importance sampling cannot easily be utilized since we are conditioning on the maximum and time of the maximum of the path.

The goal is to calculate

$$X_A = \mathbb{E}\left[g(B) \mid B \in A\right] \tag{A.2.1}$$

i.e. the expectation of a functional of the whole path $B = (B_t)_{t\in[0,1]}$ conditioned on the set A. Specifically A is

$$A_\theta = \left\{ B \;\middle|\; \theta = \inf\left\{ t \;\middle|\; B_t = \max_{t\in[0,1]} B_t \right\} \right\} \tag{A.2.2}$$

for $\theta \in [0,1]$ in the time problems and

$$A_s = \left\{ B \;\middle|\; s = \max_{t\in[0,1]} B_t \right\} \tag{A.2.3}$$

for $s \in \mathbb{R}_+$ in the space problems. To estimate X_A we will simulate paths by a simple random walk construction. Namely, we consider an equidistant grid $(t_i)_{i=0,\dots,n}$ and simulate $(Z_i)_{i=1,\dots,n}$ independent standard normal random variables and calculate the Brownian path $(B_{t_i})_{i=0,\dots,n}$ by $B_0 = 0$ and

$$B_{t_i} = B_{t_{i-1}} + \sqrt{\Delta t} Z_i \tag{A.2.4}$$

for $i = 1,\dots,n$ and $\Delta t = (n)^{-1}$. This is done N times, where we denote the j'th path by $B(j)$ for $1 \le j \le N$ and the value of the path at time t_i by $B_{t_i}(j)$.

For A_θ note that the discretization of θ lies on $(t_i)_{i=0,\dots,n}$ hence it is natural to consider the strata A'_{θ_l} defined by

$$A'_{\theta_l} = \left\{ B(j) \;\middle|\; \theta_l = \inf\left\{ t_i \;\middle|\; B_{t_i}(j) = \max_{t\in(t_i)_{i=0,\dots,n}} B_t(j) \right\} \right\} \tag{A.2.5}$$

for $\theta_l \in (t_i)_{i=0,\dots,n}$. For the spatial problems, however, the discretization of S_1 takes values in $\mathbb{R}_+$ and is not restricted to a discrete set of points. Thus, we consider a discretization of space $(s_k)_{k=0,\dots,m}$ such that $s_0 = 0$ and $s_m = \infty$ and $s_i < s_j$. We will partition the simulated paths as follows

$$A'_{s_k} = \left\{ B(j) \;\middle|\; s_k \le \max_{t\in(t_i)_{i=0,\dots,n}} B_t(j) < s_{k+1} \right\} \tag{A.2.6}$$

for $k = 0,\dots,m-1$. For a fine grid conditioning on A'_{s_k} will be close to conditioning on $S_1 = s_k$.

Recall the density of S_1. There is very little probability mass for large S_1, thus when simulating N paths we will expect to get only a few paths for which the maximum is large. Likewise in the time problem, θ has arc sine law yielding a small density around $1/2$. So if we were to simulate N paths and partition these into the strata and calculate the empirical mean we would expect a larger simulation error for the improbable strata. Instead, we implement an accept-reject algorithm to ensure N' paths in each stratum and then approximate X_A by

$$\hat{X}_{A'} = \frac{1}{N'} \sum_{B(j) \in A'} g(B(j)) \tag{A.2.7}$$

for A' being A'_{t_i} for $i = 0, \ldots, n$ or A'_{s_k} for $k = 0, \ldots, m$. It should be noted that the use of accept-reject sampling introduces dependence between the paths, however, the paths within a stratum are independent. Since we do not use all paths together this is not a concern.

For the g functional of the path $B(j)$ we remark that this will involve the calculation of the discretization of the stopping rule, namely

$$\tau(j) = \inf \left\{ t \in (t_i)_{i=0,\ldots,n} \;\middle|\; \max_{s \in (t_i^t)} B_s - B_t \geq b(t) \right\} \tag{A.2.8}$$

where $(t_i^t) \subseteq (t_i)_{i=0,\ldots,n}$ are the points on the grid which are less than or equal to t and b is the boundary function associated with the rule τ. It should be noted that $\tau(j)$ is subject to further discretization error since we only monitor the boundary at the grid points. So if the actual path were to cross the boundary only between the grid points the rule would not be executed. For further discussion on the simulation of first hitting times, we refer again to Glasserman (2004). Furthermore, we denote the discretization of θ and S_1 by $\theta(j)$ and $S(j)$. Thus, particular examples of what is calculated in Section 5.2 is

$$\mathbb{E}\left[\tau 1_{(\tau > \theta)} \mid \theta = t_i\right] \approx \frac{1}{N'} \sum_{B(j) \in A'_{t_i}} \tau(j) I(\tau(j) > t_i) \tag{A.2.9}$$

and

$$\mathbb{E}\left[e^{\alpha(S_1 - B_\tau)} \mid S_1 = s_k\right] \approx \frac{1}{N'} \sum_{B(j) \in A'_{s_k}} e^{\alpha(S(j) - B_{\tau(j)}(j))}. \tag{A.2.10}$$

Note in particular that in (A.2.10) we do not use S_k but $S(j)$ in the summand.

In Section 5.2 we have taken $n = 500$ and N' to be 700 in the time case while 1500 in the space case. Furthermore, we let (s_k) be equidistant with $\Delta s = 200^{-1}$ until a threshold of 1.75. We note that even though we used accept-reject sampling to get a larger amount of paths with large S_1 it can be seen in Figure 5.2.2, that the estimate still has a large variance. This is likely due to a larger variance in the path which results in a larger variance for the estimator of the risk.

Appendix B

Mathematical Results and Explorations

B.1 Proof of Corollary 4.2.4

We give details for the proof of Corollary 4.2.4 following the results from Graversen et al. (2001).

Proof of Corollary 4.2.4. The case $\alpha > 1/2$ follows directly from the equivalence to the weighted linear time problem in (4.2.54). For the case $\alpha \leq 1/2$ note that the result makes sense from the equivalence since $c \nearrow \infty$ if and only if $\alpha \searrow 1/2$. Formally we will show this from the original derivation of the squared space problem in Graversen et al. (2001), i.e. $\alpha = 1$. Recall the Itô-Clark representation of the maximum implies existence of an integrable and adapted process $H = (H_t)_{t\in[0,1]}$, in the sense that $\mathbb{E}\big[\int_0^1 H_t^2 \,\mathrm{d}t\big] < \infty$, such that

$$S_1 = \mathbb{E}\left[S_1\right] + \int_0^1 H_t \,\mathrm{d}B_t \tag{B.1.1}$$

where specifically

$$H_t = 2\left(1 - \Phi\left(\frac{S_t - B_t}{\sqrt{1-t}}\right)\right). \tag{B.1.2}$$

Note that we can write

$$\begin{aligned}\mathbb{E}\left[\left(B_\tau - \alpha S_1\right)^2\right] &= \mathbb{E}\left[B_\tau^2\right] + \alpha^2\mathbb{E}\left[S_1^2\right] - 2\alpha\mathbb{E}\left[B_\tau S_1\right] &\text{(B.1.3)}\\ &= \mathbb{E}\left[\tau\right] + \alpha^2 - 2\alpha\mathbb{E}\left[B_\tau S_1\right] &\text{(B.1.4)}\end{aligned}$$

by using $S_1 \overset{d}{=} |B_1|$ and Wald's identity twice after noting that τ is finite. Furthermore, nothing that $M_t := \int_0^t H_s \,\mathrm{d}B_s$ is a square-integrable martingale we get

$$\mathbb{E}\left[B_\tau S_1\right] = \mathbb{E}\left[B_\tau M_1\right] + \sqrt{\frac{2}{\pi}}\mathbb{E}\left[B_\tau\right] = \mathbb{E}\left[B_\tau M_\tau\right] = \mathbb{E}\left[\int_0^\tau H_t \,\mathrm{d}t\right] \tag{B.1.5}$$

by use of (B.1.1), the tower property, martingale property of M, and Itô's product rule. So in total

$$\mathbb{E}\left[(B_\tau - \alpha S_1)^2\right] = \mathbb{E}\left[\int_0^\tau (1 - 2\alpha H_t)\,\mathrm{d}t\right] + \alpha^2. \tag{B.1.6}$$

Denoting $F(z) := 1 - 2\alpha(2(1 - \Phi(z)))$ we note that $F(z) \geq 0$ for all $z \in \mathbb{R}_+$ when $\alpha \in [0, 1/2]$. Hence,

$$\mathbb{E}\left[\int_0^\tau (1 - 2\alpha H_t)\,\mathrm{d}t\right] \geq 0 \tag{B.1.7}$$

showing that instantaneous stopping is optimal. □

B.2 Exploration of the n'th Order Spatial Problem

We consider the $G_n(S_1 - (B_\tau + \mu))$ loss case for $\mu \neq 0$ from Section 4.3.1. In order to rewrite (4.3.4) we first note that for $q > 0$ we have

$$\begin{aligned}
\int_x^\infty z^q f_{1-t}(z)\,\mathrm{d}z =& 2(1-t)^{q/2} \int_{\frac{x}{\sqrt{1-t}}}^\infty z^q \varphi(z)\,\mathrm{d}z \\
=& 2(1-t)^{q/2} \left(\left(\frac{x}{\sqrt{1-t}}\right)^q \left(1 - \Phi\left(\frac{x}{\sqrt{1-t}}\right)\right) - \int_{\frac{x}{\sqrt{1-t}}}^\infty q z^{q-1} \left(1 - \Phi\left(z\right)\right) \mathrm{d}z \right) \\
=& x^q \left(1 - F_{1-t}(x)\right) + 2(1-t)^{q/2} \int_{\left(\frac{x}{\sqrt{1-t}}\right)^q}^\infty \left(1 - \Phi\left(z^{1/q}\right)\right) \mathrm{d}z
\end{aligned} \tag{B.2.1}$$

by integration by parts and two substitutions. So for $i \in \mathbb{N}$ writing

$$\begin{aligned}
H_i(\tau, S_\tau - B_\tau) :=& G_i(S_\tau - B_\tau) F_{1-\tau}(S_\tau - B_\tau) + \int_{S_\tau - B_\tau}^\infty G_i(z) f_{1-\tau}(z)\,\mathrm{d}z && \text{(B.2.2)} \\
=& G_i(S_\tau - B_\tau) + 2(1-\tau)^{i/2} \int_{\left(\frac{S_\tau - B_\tau}{\sqrt{1-\tau}}\right)^i}^\infty \left(1 - \Phi\left(z^{1/i}\right)\right) \mathrm{d}z && \text{(B.2.3)}
\end{aligned}$$

we can utilize the usual result of $S - B \overset{d}{=} |B|$ and (4.3.4) to get

$$\mathbb{E}\left[(S_1 - (B_\tau + \mu))^n\right] = \sum_{i=2}^n (-1)^{n-i} \binom{n}{i} \mu^{n-i} \mathbb{E}\left[H_i(\tau, |B_\tau|)\right] - n\mu^{n-1}\sqrt{\frac{2}{\pi}} + \mu^n. \tag{B.2.4}$$

Unfortunately, the dimensionality of the problem can not be reduced further since an application of the time-change from Lemma 3.3.1 will yield different killing rates for each term in the sum. Hence, the problem is two-dimensional, and considering $\mu > 0$ fixed the equivalent optimal stopping problem is thus

$$W_*(t, x) = \inf_{\tau < 1-t} \mathbb{E}_{t,x}\left[M(t + \tau, |B_{t+\tau}|)\right] \tag{B.2.5}$$

where $B_t = x$ under $P_{t,x}$ for $(t,x) \in [0,1] \times \mathbb{R}_+$ and

$$M(t,x) = \sum_{i=2}^{n} (-1)^{n-i} \binom{n}{i} \mu^{n-i} H_i(t,x). \tag{B.2.6}$$

Now we are ready to explore the shape of the continuation set and the stopping set. Note that the infinitesimal generator of the underlying two-dimensional process is $\mathbb{L}_Z = \frac{\partial}{\partial t} + \frac{1}{2}\frac{\partial^2}{\partial x^2}$ and a straightforward but lengthy calculation yields

$$K_i(t,x) := \mathbb{L}_Z H_i(t,x) = i \left[\frac{x^{i-1}}{\sqrt{1-t}} \varphi\left(\frac{x}{\sqrt{1-t}}\right) - (1-t)^{i/2-1} \int_{\left(\frac{x}{\sqrt{1-t}}\right)^i}^{\infty} \left(1 - \Phi\left(z^{1/i}\right)\right) \mathrm{d}z \right. \tag{B.2.7}$$

$$\left. + \frac{x^i}{1-t} \left(\Phi\left(\frac{x}{\sqrt{1-t}}\right) - 1 \right) \right] \tag{B.2.8}$$

$$+ i(i-1)x^{i-2} \left(\Phi\left(\frac{x}{\sqrt{1-t}}\right) - \frac{1}{2} \right). \tag{B.2.9}$$

Defining $K^{(n)} = \sum_{i=2}^{n} (-1)^{n-i} \binom{n}{i} \mu^{n-i} K_i$ we know that if $K^{(n)}(t,x) < 0$ for some $(t,x) \in [0,1] \times \mathbb{R}_+$ it is optimal to continue, i.e. $\{K^{(n)} < 0\} \subseteq C$. Likewise, we have $D \subseteq \{K^{(n)} \geq 0\}$. Plotting the $K^{(n)}$ function as done in Figure 4.3.1 for $n = 4$ reveals that the stopping set might consist of two disjoint sets. If this is the case then formulating a free-boundary problem and solving the resulting free-boundary equation will involve three unknown boundary functions. Furthermore, for $n = 6$ and $n = 8$, we see that it may be for higher order losses that it is optimal to stop at once since the stopping set is contained in a candidate stopping set which includes the origin.

B.3 Proof of Theorem 4.4.1

Some details which are not specific to this optimal stopping problem have been left out for brevity. We refer to Peskir and Shiryaev (2006) and the references therein for details. We first make a remark on the restriction of the state space.

Remark B.3.1. The restriction on $x \in [0, \gamma]$ in Theorem 4.4.1 is not necessary. It should be clear from the proof below that starting above γ will yield an increasing boundary being larger than γ for which many of the same arguments apply. However, we only need the solution for $(t,x) = (0,0)$, and hence we will not comment further on this situation. △

Proof of Theorem 4.4.1. 1. *Existence of optimal stopping time.* We first verify the existence of

an optimal stopping time in (4.4.4). For the integrability condition note that we have

$$\mathbb{E}_x\left[\sup_{0\le t\le 1}|G(|B_t|)|\right] \le \mathbb{E}_x\left[\sup_{0\le t\le 1}|B_t| - c\inf_{0\le t\le 1}|B_t|^2\right] \tag{B.3.1}$$

$$=\mathbb{E}\left[\sup_{0\le t\le 1}|x+B_t| - c\inf_{0\le t\le 1}|x+B_t|^2\right] \le \mathbb{E}_x\left[|x| + \sup_{0\le t\le 1}|B_t|\right] = x + \sqrt{\frac{2}{\pi}} \tag{B.3.2}$$

so the integrability condition (3.1.2) holds. Furthermore, G is continuous and the flow $x \mapsto \mathbb{E}_x[G(|B_\tau|)] = \mathbb{E}_0[G(|x+B_\tau|)]$ is continuous for any stopping time τ, so specifically it is lower semicontinuous (lsc). Since the supremum of lsc functions is lsc then $(t,x) \mapsto V(t,x)$ is lsc then by application of Corollary 3.1.4 yields existence of an optimal stopping time. Moreover, it is the first hitting time on the stopping set

$$\overline{D} = \{(t,x) \in [0,1]\times[0,\gamma] \mid V(t,x) = G(x)\}\,. \tag{B.3.3}$$

The continuation set is given by

$$C = \{(t,x) \in [0,1]\times[0,\gamma] \mid V(t,x) > G(x)\} \tag{B.3.4}$$

and since V is lsc and G is upper semicontinuous then C is open and $\overline{D}$ is closed. Thus, $\tau_{\overline{D}}$ is in fact a stopping time.

2. *Shape of D.* We now consider the structure of C and D. Note that since the gain function, G is independent of time, then taking supremum over a larger set results in a larger supremum, thus specifically

$$\sup_{\tau\le 1-t}\mathbb{E}_x\left[G(|B_\tau|)\right] \le \sup_{\tau}\mathbb{E}_x\left[G(|B_\tau|)\right] \tag{B.3.5}$$

where the second supremum is taken over all stopping times with respect to the filtration. Thus, since the infinite horizon problem yields

$$\sup_{\tau}\mathbb{E}_x\left[G(|B_\tau|)\right] = G(\gamma) \tag{B.3.6}$$

for $x \le \gamma$ and $|B|$ is continuous then $D \subseteq [0,1]\times[0,\gamma]$. Furthermore, rewriting V we get

$$V(t,x) = \sup_{\tau\le 1-t}\mathbb{E}[|x+B_\tau| - c\,|x+B_\tau|^2] \tag{B.3.7}$$

where $P = P_0$. It is thus seen that $x \mapsto V(t,x)$ is concave. From this, it follows that there exists a function b satisfying $0 \le b(t) \le \gamma$ for all $t \in [0,1]$ such that the continuation set is on the form

$$C = \{(t,x) \in [0,1]\times[0,\gamma] \mid x < b(t)\} \tag{B.3.8}$$

and the stopping set is the closure of

$$D = \{(t,x) \in [0,1]\times[0,\gamma] \mid x > b(t)\}\,. \tag{B.3.9}$$

As noted above the time-homogeneity of G implies $t \mapsto V(t,x)$ is decreasing since if $(t,x) \in C$ then considering some $0 \leq t' < t$ then

$$G(x) < \sup_{\tau \leq 1-t} \mathbb{E}_x \left[G(|B_\tau|) \right] \leq \sup_{\tau \leq 1-t'} \mathbb{E}_x \left[G(|B_\tau|) \right] \tag{B.3.10}$$

hence $(t',x) \in C$. Thus, we can also conclude that b is a decreasing function.

3. *Continuity of V.* We need to show continuity of $(t,x) \mapsto V(t,x)$. For this it is sufficient to prove that for all $(t_0,x_0) \in [0,1] \times [0,\gamma]$ it holds that the mapping $x \mapsto V(t_0,x)$ is continuous at x_0 and $t \mapsto V(t,x)$ is continuous at t_0 uniformly over $x \in [0,\gamma]$.

The continuity at x_0 follows from $x \mapsto V(t,x)$ being bounded and concave on $[0,\gamma]$. For the second part we fix some $0 \leq t_1 < t_2 \leq 1$ and $x \in [0,\gamma]$. Let τ_1 be the optimal stopping time for $V(t_1,x)$ and set $\tau_2 = \tau_1 \wedge (1-t_2)$, i.e. the minimum of the optimal stopping time for the earlier time and the remaining time as seen from t_2. In particular note that $\tau_2 \leq \tau_1$ and

$$0 \leq \tau_1 - \tau_2 = (t_2 - 1 + \tau_1) \wedge 0 \leq t_2 - t_1. \tag{B.3.11}$$

Since V is decreasing in t we have

$$\begin{aligned}
0 &\leq V(t_2,x) - V(t_1,x) && \text{(B.3.12)}\\
&= \sup_{\tau \leq 1-t_2} \left(\mathbb{E}_x \left(G(|B_\tau|) \right) \right) - \mathbb{E}_x \left[G(|B_{\tau_1}|) \right] && \text{(B.3.13)}\\
&\leq \mathbb{E}_x \left[G(|B_{\tau_2}|) \right] - \mathbb{E}_x \left[G(|B_{\tau_1}|) \right] && \text{(B.3.14)}\\
&= \mathbb{E}_x \left[|B_{\tau_2}| - |B_{\tau_1}| \right] + c\mathbb{E}_x \left[\tau_1 - \tau_2 \right] && \text{(B.3.15)}
\end{aligned}$$

by Wald's identity. Note that using the triangle inequalities we get

$$\begin{aligned}
\mathbb{E}_x \left[|B_{\tau_2}| - |B_{\tau_1}| \right] &\leq \mathbb{E}_x \left[\left| |B_{\tau_2}| - |B_{\tau_1}| \right| \right] && \text{(B.3.16)}\\
&\leq \mathbb{E}_x \left[|B_{\tau_1} - B_{\tau_2}| \right] && \text{(B.3.17)}\\
&= \mathbb{E} \left[|B_{\tau_1 - \tau_2}| \right] && \text{(B.3.18)}\\
&\leq \sqrt{\mathbb{E} \left[\tau_1 - \tau_2 \right]} && \text{(B.3.19)}
\end{aligned}$$

where the last inequality follows from Wald's inequality (i.e. an application of Wald's identity and Jensen's inequality). Collecting the inequalities we get that

$$|V(t_2,x) - V(t_1,x)| \leq \sqrt{t_2 - t_1} + c(t_2 - t_1) \tag{B.3.20}$$

which of course holds uniformly over $x \in [0,\gamma]$. So in total $(t,x) \mapsto V(t,x)$ is continuous hence instantaneous-stopping holds, i.e.

$$V(t,x)\big|_{x=b(t)-} = G(b(t)). \tag{B.3.21}$$

4. *Smooth-fit condition.* We need to show that $x \mapsto V(t,x)$ is C^1 at $b(t)$. Specifically, we need to consider the left-hand derivative. Consider a point $(t,x) \in [0,1] \times (0,\gamma)$ along the curve $x = b(t)$. Note that since $V \geq G$ and specifically $V(t,x) = G(x)$ since $(t,x) \in \overline{D}$ then

$$\frac{V(t,x) - V(t,x-\varepsilon)}{\varepsilon} \leq \frac{G(x) - G(x-\varepsilon)}{\varepsilon} \tag{B.3.22}$$

for any $\varepsilon > 0$ such that $x - \varepsilon > 0$. Hence

$$\frac{\partial^-}{\partial x} V(t,x) \leq G'(x) = 1 - 2cx. \tag{B.3.23}$$

Now for the converse let τ_ε bet the optimal stopping time for $V(t,x-\varepsilon)$. Then

$$\frac{V(t,x) - V(t,x-\varepsilon)}{\varepsilon} \geq \frac{\mathbb{E}_x\left[G(|B_{\tau_\varepsilon}|)\right] - \mathbb{E}_{x-\varepsilon}\left[G(|B_{\tau_\varepsilon}|)\right]}{\varepsilon} \tag{B.3.24}$$

since τ_ε is suboptimal at (t,x). Note that

$$\begin{aligned} &\mathbb{E}_x\left[G(|B_{\tau_\varepsilon}|)\right] - \mathbb{E}_{x-\varepsilon}\left[G(|B_{\tau_\varepsilon}|)\right] = \mathbb{E}\left[G(|x + B_{\tau_\varepsilon}|) - G(|x - \varepsilon + B_{\tau_\varepsilon}|)\right] \\ =&\mathbb{E}\left[|x + B_{\tau_\varepsilon}| - |x - \varepsilon + B_{\tau_\varepsilon}|\right] + c\mathbb{E}\left[|x - \varepsilon + B_{\tau_\varepsilon}|^2 - |x + B_{\tau_\varepsilon}|^2\right]. \end{aligned} \tag{B.3.25}$$

For the second expectation we get

$$\frac{c}{\varepsilon}\mathbb{E}\left[|x - \varepsilon + B_{\tau_\varepsilon}|^2 - |x + B_{\tau_\varepsilon}|^2\right] = -2cx + c\varepsilon + 2c\mathbb{E}\left[B_{\tau_\varepsilon}\right] = -2cx + c\varepsilon \to -2cx \tag{B.3.26}$$

for $\varepsilon \to 0$ since τ_ε is finite. For the first expectation in (B.3.25) we have

$$\frac{1}{\varepsilon}\mathbb{E}\left[|x + B_{\tau_\varepsilon}| - |x - \varepsilon + B_{\tau_\varepsilon}|\right] \tag{B.3.27}$$

$$=P\left(B_{\tau_\varepsilon} > -x + \varepsilon\right) + \mathbb{E}\left[\left(\frac{2}{\varepsilon}(B_{\tau_\varepsilon} + x) - 1\right) I\left(B_{\tau_\varepsilon} \in [-x, -x + \varepsilon]\right)\right] - P\left(B_{\tau_\varepsilon} < -x\right) \tag{B.3.28}$$

$$\geq P\left(B_{\tau_\varepsilon} \geq 0\right) - P\left(B_{\tau_\varepsilon} \in [-x, -x + \varepsilon]\right) - P\left(B_{\tau_\varepsilon} < -x\right) \tag{B.3.29}$$

since $x - \varepsilon > 0$. Noting that b is decreasing we can dominate τ_ε by $\sigma_\varepsilon := \inf\{s \geq 0 \mid |B_s + x + \varepsilon| \geq x\}$. Letting $\varepsilon \searrow 0$ then by $|B|$ being a regular diffusion we get $\tau_\varepsilon \to 0$ P_x-a.s. for $\varepsilon \to 0$. Going to P_0 we thus have that $\tau_\varepsilon \to 0$ P_0-a.s. when $\varepsilon \to 0$ so $B_{\tau_\varepsilon} \to 0$ P_0-a.s. Hence, by dominated convergence we have

$$\frac{1}{\varepsilon}\mathbb{E}\left[|x + B_{\tau_\varepsilon}| - |x - \varepsilon + B_{\tau_\varepsilon}|\right] \to 1 \tag{B.3.30}$$

i.e.

$$\frac{\partial^-}{\partial x} V(t,x) \geq 1 - 2cx. \tag{B.3.31}$$

So in total, we have the smooth-fit condition

$$\frac{\partial}{\partial x}V(t,x)\Big|_{x=b(t)-} = 1 - 2cx. \tag{B.3.32}$$

5. *Continuity of b.* We first show right continuity on $[0,1]$. Take some $t \in [0,1)$ and consider a monotone sequence $(t_n)_{n\in\mathbb{N}}$ with $t_n \searrow t$ for $n \to \infty$. Since b is decreasing $b(t+)$ exists and $(t_n, b(t_n)) \in D$ for all $n \in \mathbb{N}$ then $(t, b(t+)) \in \overline{D}$. Thus, we have $b(t+) \geq b(t)$. Hence, by using b is decreasing once again we get the right continuity of b.

For left continuity assume for contradiction that b has a discontinuity in $t_* \in (0,1)$, i.e. $b(t_*-) > b(t_*)$. We now fix a point $t' < t_*$ and $x' \in (b(t_*), b(t_*-))$ so that $(t_*, x') \in \overline{D}$. Consider the subset of C formed by the vertices $(t', b(t')), (t_*, b(t_*-)), (t_*, x')$ and (t', x'), denoted by R, see Figure B.3.1.

Note that G is C^2 and by the strong Markov property, V is $C^{1,2}$ on C, hence specifically they are so on R. We can then apply the second fundamental theorem of calculus twice to get

$$\begin{aligned}
V(t,x) - G(x) &= V(t,x) - V(t,b(t)) - (G(x) - G(b(t))) && \text{(B.3.33)}\\
&= \int_{b(t)}^{x} \left(\frac{\partial}{\partial x}V(t,u) - G'(u)\right) \mathrm{d}u && \text{(B.3.34)}\\
&= \int_{b(t)}^{x} \left(\frac{\partial}{\partial x}V(t,u) - \frac{\partial}{\partial x}V(t,b(t)) - (G'(u) - G'(b(t)))\right) \mathrm{d}u && \text{(B.3.35)}\\
&= \int_{b(t)}^{x}\int_{b(t)}^{u} \left(\frac{\partial^2}{\partial x^2}V(t,v) - G''(v)\right) \mathrm{d}v\,\mathrm{d}u && \text{(B.3.36)}
\end{aligned}$$

for any $(t,x) \in R$ where we have used the smooth- and continuous-fit conditions. Furthermore, since $t \mapsto V(t,x)$ is decreasing then

$$\frac{\partial}{\partial t}V(t,x) \leq 0. \tag{B.3.37}$$

Using the strong Markov property again yields that V solves

$$\frac{\partial}{\partial t}V(t,x) + \frac{1}{2}\frac{\partial^2}{\partial x^2}V(t,x) = 0 \tag{B.3.38}$$

on C hence also on R which implies that $\frac{\partial}{\partial t}V(t,x) \geq 0$. Using (B.3.36) we get

$$V(t,x) - G(x) \geq \int_{b(t)}^{x}\int_{b(t)}^{u} 2c\,\mathrm{d}v\,\mathrm{d}u > 0. \tag{B.3.39}$$

As such we have arrived at a contradiction since this implies $(t,x) \in C$ which contradicts $R \subset \overline{D}$. Thus, $b(t_*-) = b(t_*)$ and so b is continuous. Finally, note that $V(1,x) = G(x)$ for all $x \in [0,\gamma]$. Repeating the same argument for $t_* = 1$ yields $b(1-) = b(1) = 0$.

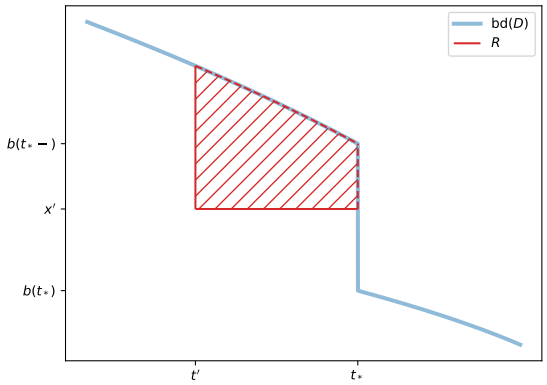


Figure B.3.1: Illustration of the set R under the contradictory assumption of b having a discontinuity at t_*. The dashed line means the boundary is not included, however smooth- and continuous-fit still holds.

Hence, we have shown sufficient conditions to utilize Section 3.5 to formulate free-boundary equations. Noting that

$$\mathbb{L}_{|B|}G(x) = -c \tag{B.3.40}$$

we get (4.4.7) and (4.4.8).

□

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